Technical Report – No. 2

Project: Carbonates Rock Physics 2: Petroelastic characterization of pre-salt carbonates for geophysical reservoir monitoring

Contract Nr. 24778-3

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1. Introduction

This technical report presents the current status of our ongoing research, detailing several key issues related to its development, including team recruitment, equipment acquisition, training, methodology, and preliminary results. The ROCKPHYS 2 project is divided into three main research areas: Rock Physics Laboratory Characterization (RPLC), Digital Rock Physics (DRP), and Integrated Data Analysis (IDA).

2. Project Status

This report covers the initial period of the project (April 2025 to September 2025). During this time, we utilized the dataset and outcrop samples collected during the ROCKPHYS 1 project for the analyses, and also conducted new experiments using outcrop analogs, as detailed in this report.

Equipment issues also occurred during this time, and maintenance procedures were implemented; however, some problems persist, limiting the full functionality of certain equipment.

The main Rock Physics System (Ergotech/Phoenix) has had its LVDTs (Linear Variable Displacement Transducers) replaced, allowing for axial deformation measurements to be performed again. Unfortunately, its cantilever failed, causing the interruption of radial deformation measurements. We have sent this part for repair in the UK.

The new Rock Physics System for sidewall cores (UENF/LENEP), which we started to call Corephys, is operational and has recently been used in some tests.

The Permo-Porosimeter Coreval 700 was recently repaired, which will enable future tests of porosity and permeability under confining pressure. It will also allow the evaluation of pore and bulk compressibilities.

3. Research Team

People	Education	Connection	Activity
Marco A. R. Ceia Orcid: 0000-0002-4366-9519	DSc. Reservoir and Exploration Eng.	Permanent Staff-UENF	Coordinator. Petrophysics and Rock Physics Measurement and Analysis

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Victor Hugo Santos Orcid: 0000-0001-5720-2744	DSc. Geology	Permanent Staff-UENF	Rock Characterization. Petrography.
Remilson B. Rosa	Lab. Technician	Permanent Staff-UENF	Plug preparation and assistance with rock physics measurements.
Débora de Mendonça Freire	Graduate Technician	Permanent Staff-UENF	Quotes, staff management, controlled chemicals monitoring, and reports.
Alexandre Telles	Graduate Technician	Permanent Staff-UENF	Planning and Implementation of Health, Safety, and Environment (HSE) procedures.
Lucas César Oliveira Orcid: 0000-0003-4147-0837	MSc. Reservoir and Exploration Eng.	Hired - FACC	Researcher 2 – Rock Physics data acquisition and analysis.
Sofia Bueno Orcid: 0000-0003-2086-2655	DSc Reservoir and Exploration Eng.	Hired - FACC	Researcher 13 - Low-frequency data acquisition supervisory. Velocity dispersion and elastic attenuation data analysis
Nicolau Azevedo Prates	Undergraduate student	Hired - FACC	Researcher 4 - Speed estimation support
Mateus dos Reis Oliveira	Undergraduate student	Hired - FACC	Researcher 5 – XRD and XRF measurement support
Herson Rocha 0000-0002-2494-6277	DSc Reservoir and Exploration Eng.	Hired - FACC	Researcher 3 - X-ray micro-CT data processing and analysis.

Yan Lucas Albuquerque Lopes	Undergraduate student	Hired - FACC	Researcher 8 – Code development support for MicroCT data analysis
Antonio Romero da Costa	D.Sc Physics	Hired - FACC	Researcher 9 – Rock Physics Modeling
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Marcus Vinicius Sousa Correa Orcid: 0000-0002-3005-8428	MSc Reservoir and Exploration Eng	Hired - FACC	Researcher 15 - Petrophysical and Rock Physics data Integration and management. Rock Physics Data Analysis.
José Fabiano Ferreira Viana	Technician	Hired - FACC	Lab Technician 1 - Equipment and system assembly, preparation, and maintenance.
Evandro Ribeiro Carvalho	Graduate Technician	Hired - FACC	Lab Technician 2 - Computer and network maintenance. Software installation.
Adriano Gatti Moreira	Technician	Hired - FACC	Lab Technician 3 – Rock Physics tests
Rafael Oliveira	Technician	Hired - FACC	Lab Technician 4 - Petrophysics tests. Laboratory organization. Sample cleaning.
Denize Saldanha	High School	Hired - FACC	Assistant Technician 5– Petrophysics data organization. Quotations. Reports formatting. Graph and Technical Image design
João Vitor Monteiro Mendonça da Silva	Technician	Hired - FACC	Lab Technician 6 - Software development support for petrophysical data analysis
Alessandro Roberto de Oliveira	Computer Science Graduate	Hired - FACC	Technician 7 – Design and maintenance of laboratory systems
Michele Fradique Leandro Damasceno	High School	Hired - FACC	Lab Technician 8 – 3D scanner operation for sample digitalization

Pedro Henrique Vasconcellos Mesquita	Technician	Hired - FACC	Lab Technician 9 – Code optimization/adaptation support
Irineu Lima Neto Orcid: 0000-0002-8972-3007	DSc. Reservoir and Exploration Eng.	Hired - FACC	Researcher – Digital Rock Physics.
Gilleade Silva	Technologist	Hired - FACC	Researcher 16. Software development for Rock Physics Lab Characterization. Optimization/Automatization of waveform analysis
Hugo Machado	Undergraduate student	Hired - FACC	Researcher 14 - Support Petrophysical data analysis

4. ROCK PHYSICS LAB CHARACTERIZATION

4.1. Pre-salt data

4.1.1 Attenuation and Dispersion.

The calibration and interpretation of seismic attenuation and velocity dispersion in reservoir rocks are crucial for understanding wave propagation under in situ conditions. The methodology initially presented in Tech Report 1 established a framework to connect laboratory-scale attenuation mechanisms to field-scale seismic responses through a multiscale calibration approach. The effects of rock heterogeneity, particularly at the microscale — such as squirt-flow mechanisms (Mavko, Mukerji, and Dvorkin, 1998) — can explain the attenuation observed in laboratory experiments (Jones, 1986). Continuous refinement of this calibration has been carried out using well seismic data to identify specific depths and corresponding core samples, enabling a detailed investigation of saturation and pore pressure effects. Within this process, fluid substitution in the water/oil transition zone and pressure substitution to in-situ stress conditions are employed to more accurately reproduce reservoir conditions. Additionally, VSP (Vertical Seismic Profiling) data at 20 Hz and interval velocity information serve as an intermediate frequency range, bridging the gap between sonic and ultrasonic measurements for dispersion model calibration.

Expanding upon this foundation, the current stage applies the same multiscale calibration methodology, integrating VSP-based dispersion analysis and characteristic frequency modeling. Figure 4.1.1.1 illustrates this multiscale calibration workflow. In Figure 4.1.1.1a, the core samples from well 1-BRSA-1045-SPS are overlaid on the VSI check-shot profile, showing interval velocities obtained at 20 Hz, which serve as the seismic-scale reference for comparison with other frequency domains. Figure 4.1.1.1b presents the experimental dataset acquired in the laboratory, including ultrasonic measurements at 1.3 MHz and low-frequency data from a permeoporosimeter Coreval 700, representing the asymptotic trend toward zero frequency. This permoporosimeter provides the bulk compressibility (Cb) of the samples through the combination of the pore volume measurements obtained at different pressures and the grain volume, which is considered unaltered within the studied pressure range. The inverse of Cb is the dry bulk modulus (Kdry). Considering that the Poisson ratio obtained using the ultrasonic velocities is the same in the low-frequency domain, one could use these two elastic moduli to compute the expected low-frequency velocity. The detailed methodology can be found in Technical Report #1 (Section 4.1).

The samples were first measured under dry conditions and later subjected to fluid substitution using the Gassmann model for representative water/oil saturations, allowing direct comparison with sonic log data.

From this multiscale dataset, a three-point nonlinear regression was fitted based on the Geertsma–Smit model, aiming to define a new characteristic dispersion function for the analyzed samples. This approach integrates high-

frequency laboratory measurements, low-frequency permeoporosimeter data, and well log records at the sonic scale, thereby establishing a continuous frequency-dependent framework for elastic parameters. The VSI check-shot point proved consistent with the characteristic frequency predicted by White, although a theoretically increasing velocity with frequency was expected. These results highlight that, for an accurate representation of the dispersion phenomenon, all frequency scales must be calibrated under equivalent experimental conditions, including pressure, saturation, and mineralogical composition, to ensure physical consistency across measurement domains.

a) b)

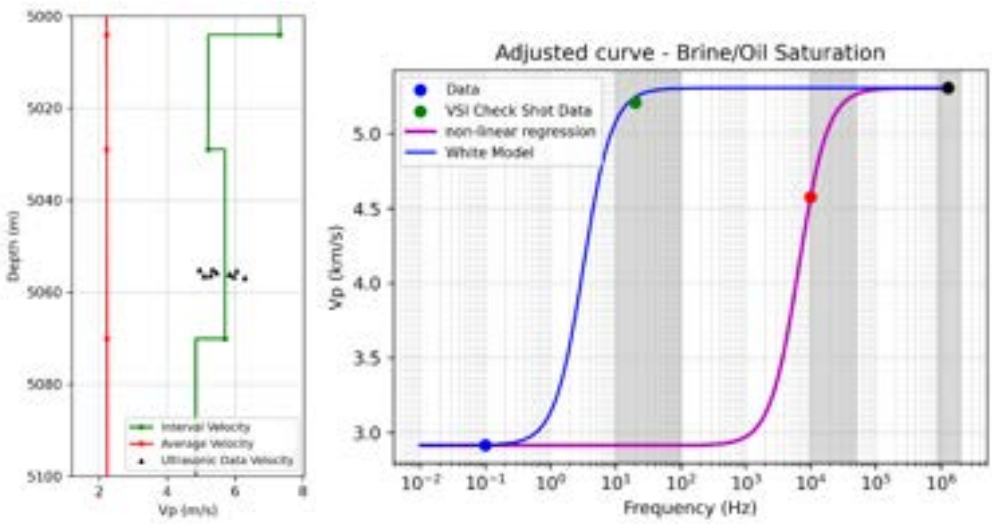


Figure 4.1.1.1 – a) Interval velocity profile indicating the studied samples. b) Nonlinear fit of the Geertsma-Smit model.

Spectral Ratio Analysis

Preliminary results from the spectral ratio technique were analyzed for samples from well 1-BRSA-1045-SPS, in which ultrasonic measurements were performed using P- and S-wave (S1 and S2) signals on dry samples under varying pressure conditions (0, 5, 6, 8.8, 11.5, 14.3, 17, 19.8, and 22.6 MPa). Each experimental rock sample was paired with an aluminum reference sample of the same length to ensure standardized comparisons and consistent experimental conditions. This pairing between experimental and reference samples is essential for correctly interpreting the effects of specific parameters, such as confining pressure and saturation, on the seismic response of rocks.

The analysis of the ultrasonic waveforms in the time domain enabled the determination of the arrival times of P, S1, and S2 waves for each sample. These times were obtained by identifying the first coherent break in the recorded signals, representing the propagation of the direct wavefront through the sample (Figure 4.1.1.2).

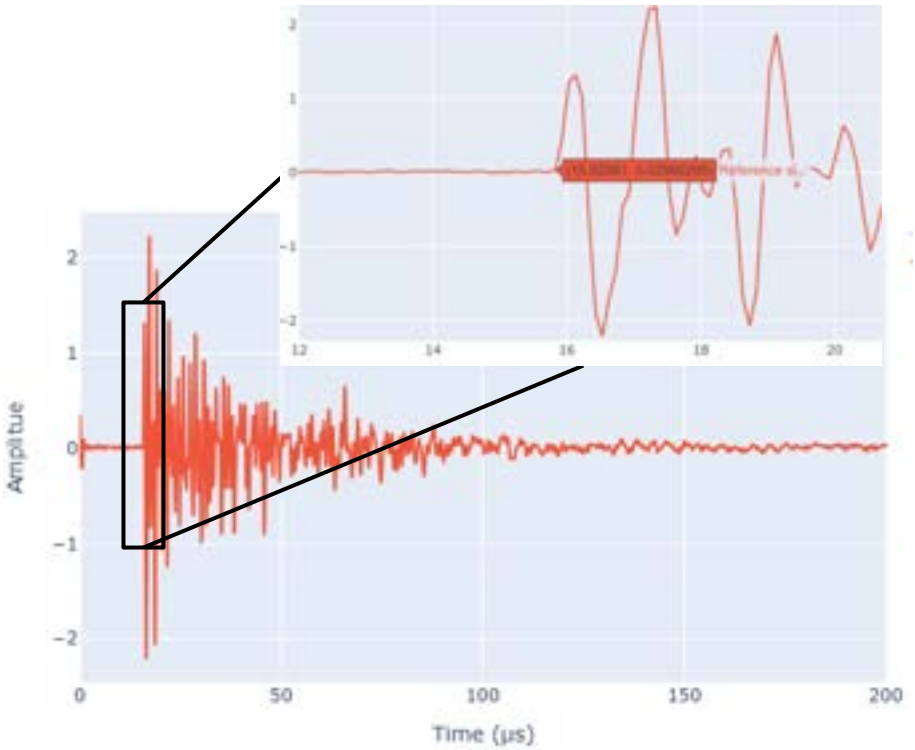


Figure 4.1.1.2 – Acquisition of the wave arrival time for an aluminum reference sample.

The reference sample had its arrival time determined under the maximum confining pressure applied during the experiment. This value was used as a comparative baseline for signal alignment and evaluation of the variations introduced by different testing conditions in the analyzed samples. To ensure the integrity of the analysis, a time window was applied to the recorded signal (Figure 4.1.1.3) to isolate the portion corresponding to the direct wave. This procedure aimed to reduce the interference of high-frequency noise and multiple reflections that could compromise the accuracy of the spectral analysis.

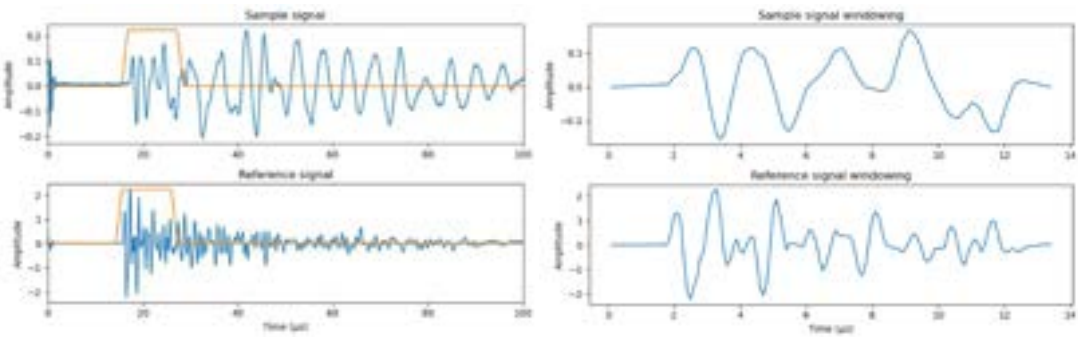


Figure 4.1.1.3 – Windowing of the sample signal and the reference sample signal.

The limits of the frequency band used in the spectral ratio technique were defined individually for each sample (Figure 4.1.1.4). This customized definition aimed to maximize precision in estimating the quality factor (Q), tailoring it to the specific spectral characteristics of each experiment.

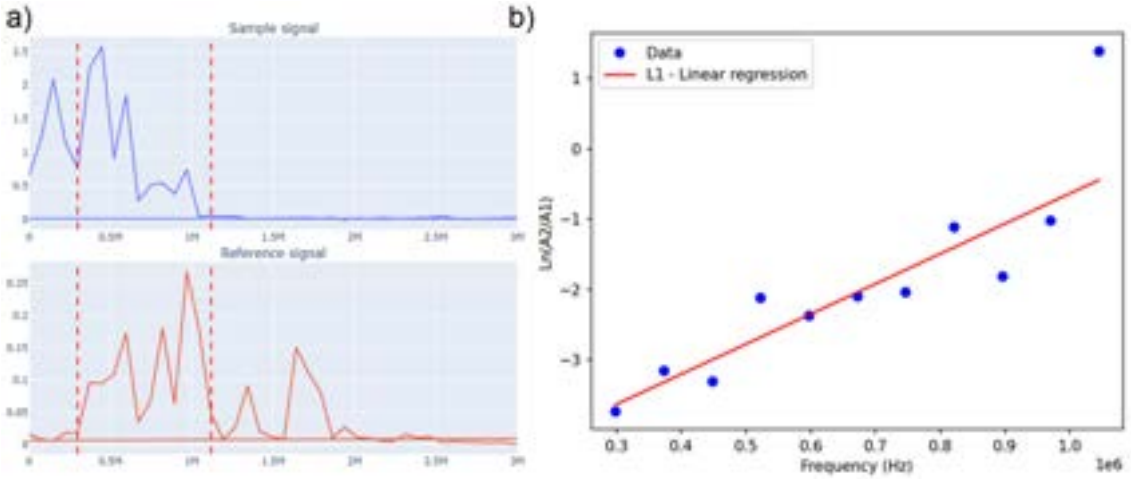


Figure 4.1.1.4 – a) Definition of the spectral frequency band. b) Normal L1 linear fitting.

Figure 4.1.1.5 presents a comparative analysis of five samples from the Barra Velha Formation, cut in different orientations relative to the bedding plane, focusing on the behavior of the quality factor (Q) under varying pressures and on the propagation of the P-wave (Figure 4.1.1.6). The samples cut in the horizontal direction (F7289H and F7296H), parallel to the bedding plane, showed interference effects in P-wave propagation associated with the presence of heterogeneities or microstructural complexities. These samples exhibited an increase in velocity with increasing confining pressure, indicating the progressive closure of microfractures. Directional anisotropy was observed, with the horizontal components of velocity (Vp_x and Vp_Y) exceeding the vertical component (Vpz) at higher pressures, suggesting more efficient closure of fractures oriented along the X and Y directions. The F7294H sample, in particular, showed an increase in the Q factor with pressure, attributed to the presence of pores with low sphericity and bimodal distribution (macropores and micropores), which tend to close under compression, reducing energy dissipation.

Samples cut in the vertical direction (F7290V and F7295V), perpendicular to the bedding plane, also exhibited clear elastic anisotropy, with higher velocities in the X and Y directions compared to Z, suggesting preferential closure of microfractures in horizontal planes. In both samples, variations in Q with increasing pressure were observed, attributed to possible mechanisms of reopening or reorientation of internal microfractures influenced by low pore sphericity. The F7295V sample showed higher sensitivity of Q to pressure variations, likely due to greater heterogeneity and a significant concentration of macropores with low aspect ratios.

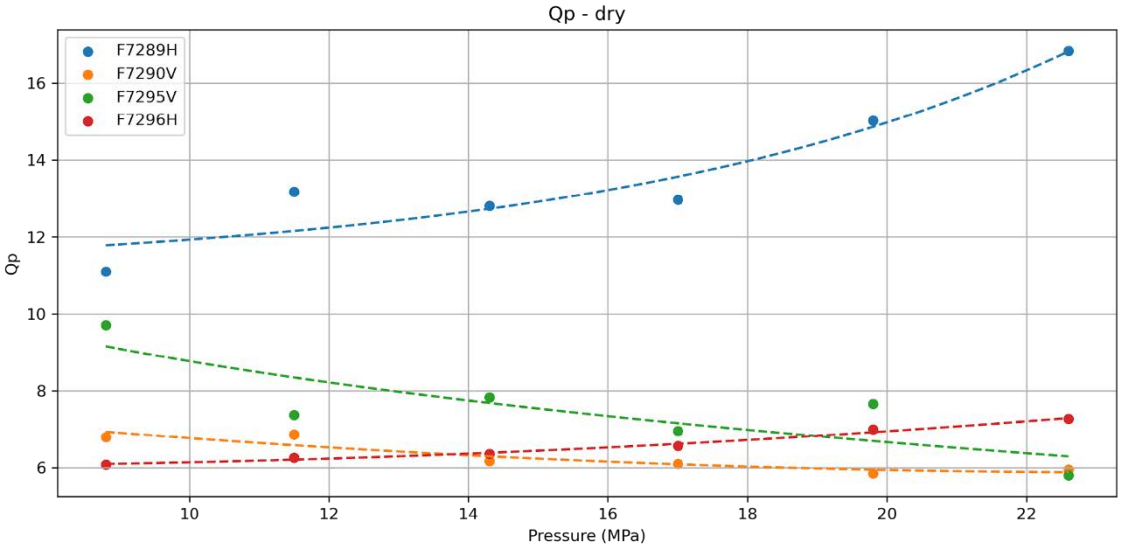


Figure 4.1.1.5 – Fitting curves of the estimated quality factor for P-wave (Qp) as a function of pressure for Barra Velha Formation samples under dry conditions.

Overall, the results suggest that sample orientation, pore aspect ratio distribution, and degree of fracturing have a probable influence on the mechanical and attenuative behavior of carbonate rocks under pressure, highlighting the anisotropic nature of these samples.

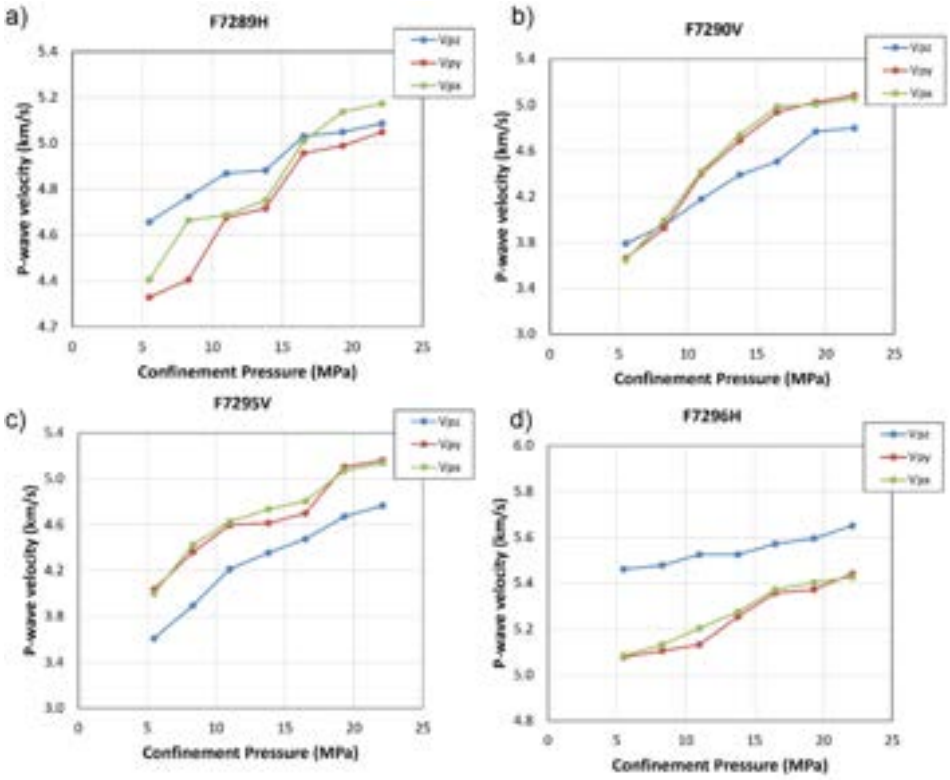


Figure 4.1.1.6 – P-wave velocity along the X, Y, and Z directions.

In addition to the P-wave analysis, the behavior of shear waves (S1 and S2) was evaluated for all samples, F7289H, F7296H, F7290V, and F7295V. The

mean value of the Q factors estimated for S1 and S2 was calculated to obtain Qs, as shown in Figure 4.1.1.7. In general, all samples followed the expected theoretical trend, where the quality factor increases with confining pressure. The horizontal samples showed a more pronounced increase in Qs, particularly sample F7289H, which exhibited the greatest growth, consistent with the behavior observed for the P-wave (Qp). The vertical samples (F7290V and F7295V) also presented an overall increasing trend, though less prominent, with smaller Qs variations between lower and higher pressures.

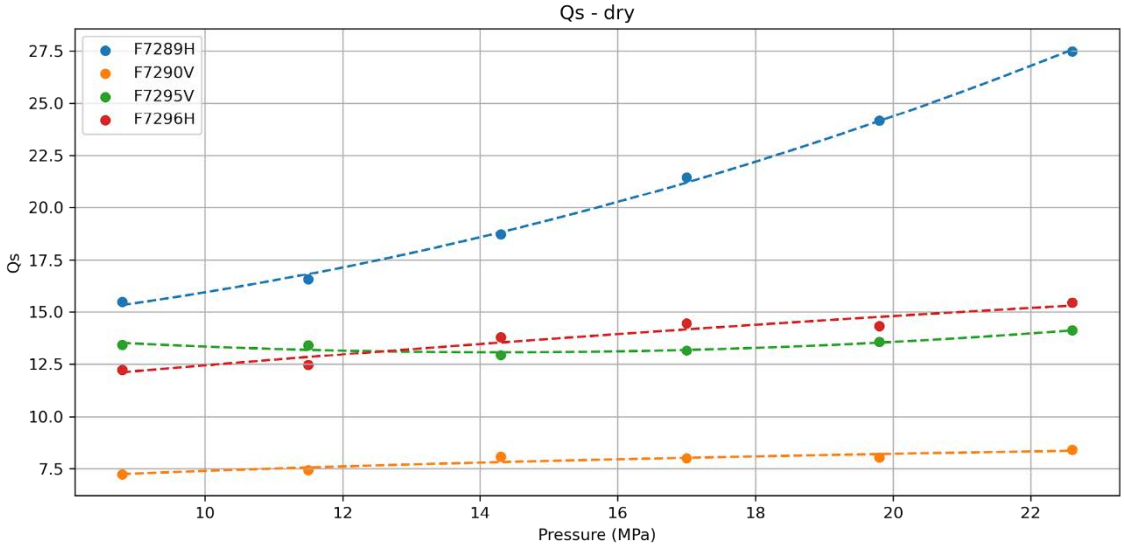


Figure 4.1.1.7 – Fitting curves of the estimated quality factor for S-wave (Qs) as a function of pressure for Barra Velha Formation samples under dry conditions.

These results confirm that S-waves respond similarly to P-waves under varying pressure conditions, revealing complementary insights into energy dissipation mechanisms and elastic anisotropy in carbonate rocks. The combined analysis reinforces the importance of considering multiple wave propagation modes for a comprehensive physical and seismic characterization of carbonate reservoirs.

After the estimation stage for the dry samples, measurements were also carried out on the samples subjected to saturation with BFW (Brazilian Formation Water), aiming to investigate the effect of fluid presence on the attenuation and elastic behavior of carbonate rocks under different pressure conditions. The experimental methodology followed the same parameters applied to the dry tests, ensuring reproducibility and direct comparability between the results obtained for the dry and saturated states. The quality factor (Q) analyses were conducted for both P-waves (Qp sat) and S-waves (Qs sat), as illustrated in Figures 4.1.1.8 and 4.1.1.9, respectively. As expected, the Q factor exhibited lower values under saturated conditions compared to the dry state for both P- and S-waves. This behavior is consistent with theoretical predictions, as fluid–matrix interactions

enhance internal damping within the rock, leading to increased energy dissipation.

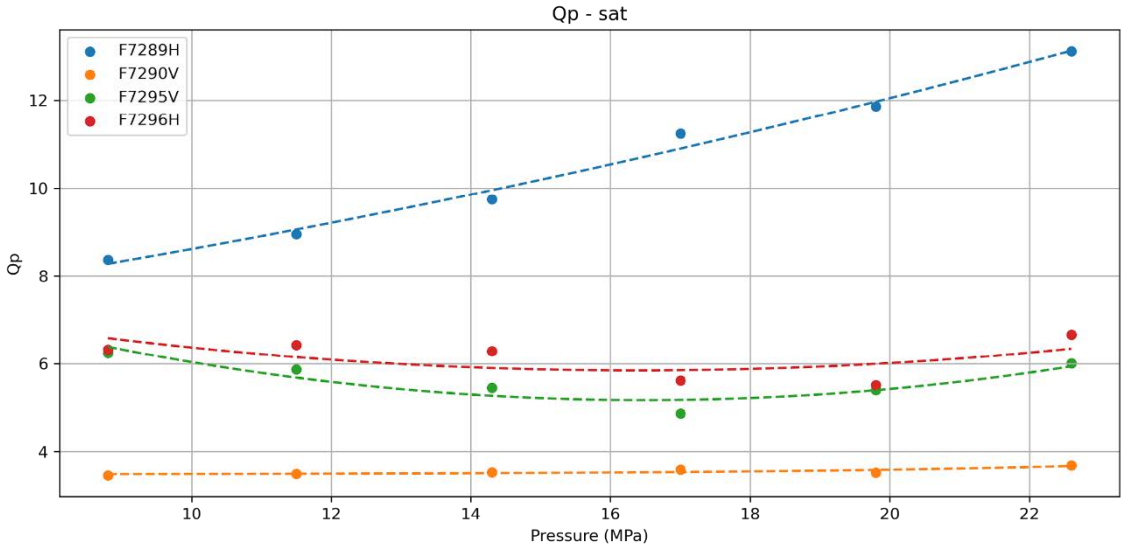


Figure 4.1.1.8 – Fitting curves of the estimated quality factor for P-wave (Qp) as a function of pressure for Barra Velha Formation samples saturated with BFW.

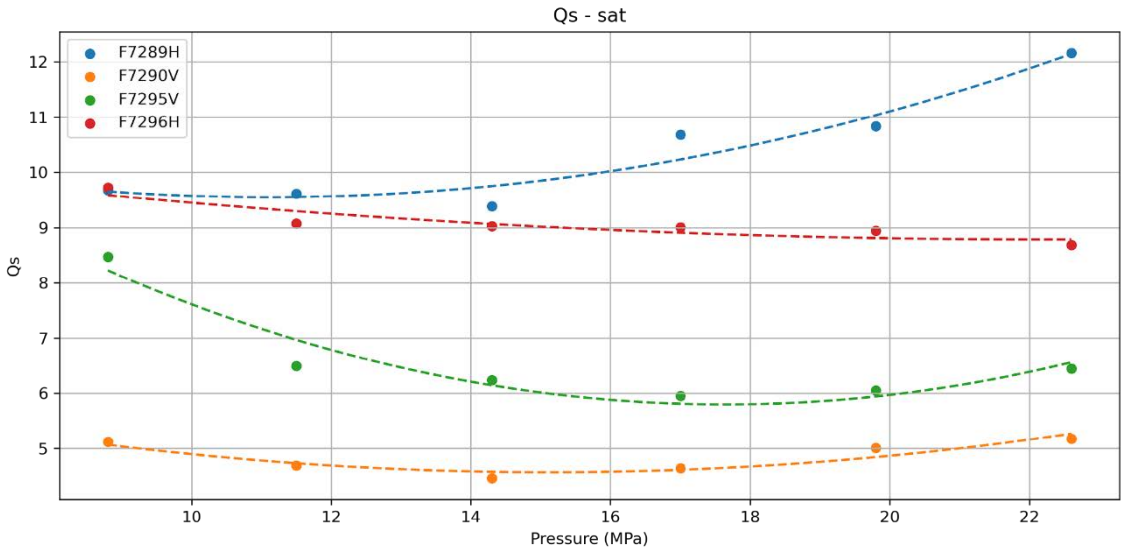


Figure 4.1.1.9 – Fitting curves of the estimated quality factor for S-wave (Qs) as a function of pressure for Barra Velha Formation samples saturated with BFW.

In the Qp sat analysis, sample F7289H remained the most prominent, showing the largest variation and the most pronounced increase in Qp with pressure, reflecting its sensitivity to compaction and the closure of pores and microfractures under confinement. Conversely, sample F7290V exhibited a stable behavior across all pressure levels, with very low variation in Qp, indicating a more homogeneous internal structure and reduced sensitivity to localized

compression effects. Samples F7295V and F7296H displayed similar trends, where Qp values decreased at intermediate pressures and then recovered at higher pressures, suggesting a non-linear elastic response possibly related to microfracture reorientation and fluid redistribution within the pore space.

In the Qs versus pressure analysis, a clear distinction was observed between horizontal and vertical samples. The horizontal samples (F7289H and F7296H) presented higher Qs values, while the vertical samples (F7290V and F7295V) exhibited lower values, indicating attenuation anisotropy related to microstructural orientation and fracture alignment. Sample F7289H, once again, showed the largest variation and a consistent increase in Qs with pressure, suggesting progressive closure of discontinuities and reduced internal damping. Sample F7296H started with Qs values similar to those of F7289H but then exhibited a slightly decreasing trend at higher pressures, with small variations, possibly indicating uniform saturation and lower local heterogeneity.

Among the vertical samples, F7290V maintained the lowest absolute Qs values, consistent with previous tests, reinforcing its higher energy dissipation capacity and lower relative stiffness. Similarly, sample F7295V showed a drop in Qs at intermediate pressures, followed by an increase at higher pressures, a behavior that may be associated with fluid redistribution and partial microfracture closure at elevated confinement levels.

4.2 Outcrop analogues

4.2.1 Pore Pressure experiments of the Coquinas (Morro do Chaves Fm.)

The propagation of seismic waves in sedimentary rocks is directly linked to their elastic properties, which are strongly influenced by effective pressure. In porous rocks, such as the carbonates, the pore pressure, that is, the pressure exerted by the fluid within the pore space, acts against the total stress applied on the rock frame, modifying its stiffness and consequently the P- and S-wave velocities.

In general, wave velocities increase with confining pressure due to a reduction in porosity, improved grain contact, and the closure of microcracks and fissures. This behavior follows a nonlinear relationship between velocity and pressure, typical of sedimentary formations. At low pressures, velocities rise sharply as pores and cracks close; at higher pressures, the increase stabilizes as the rock becomes compacted and discontinuities are minimized.

The parameter that effectively controls this behavior is the effective pressure, defined as:

$$p_{eff} = p_{conf} - \alpha\, p_{pore}, \qquad (4.2.1.1.1)$$

where p_{conf} is the total confining pressure, p_{pore} is the pore pressure, and α is Biot's coefficient describing the interaction between the solid matrix and the pore fluid. Variations in effective pressure directly influence the elastic response and anisotropy of the rock, making it a key factor in the seismic and petrophysical interpretation of deep reservoirs.

In this work, we repeated the pore pressure tests described in the Rockphys 1 (Phase 1), but this time using a Coquina sample (Morro do Chaves Fm.). This rock was studied during Phase 1, but had not undergone pore pressure tests.

The methodology applied in this study follows the experimental workflow described in the previous stage, now focused on understanding the variation of seismic velocities under different combinations of pore pressure and confining pressure in carbonate samples from the Morro do Chaves Formation (coquinas).

Empirical relationships often described by power-law or exponential functions are commonly used to represent the nonlinear dependency between velocity and pressure, allowing the development of calibrated models applicable to field-scale conditions. The theoretical framework is based on the Bowers model (1995), an empirical approach that relates velocity to effective pressure, especially in overpressured regimes. The general form of the model can be expressed as:

$$V = V_0 + A\left(p_{eff}\right)^B, \qquad (4.2.1.1.2)$$

where V_0 is the reference velocity at low pressure, and A and B are empirical constants fitted to laboratory or well-log data.

According to Bowers, when the measured pore pressure exceeds the expected normal hydrostatic pressure at a given depth, the rock is considered overpressured. Two main mechanisms can produce this condition: undercompaction, when rapid burial prevents pore fluid from escaping, thereby maintaining high pore pressure; and tectonic unloading, when stress release reduces the effective pressure without altering the burial depth.

Although practical, the Bowers equations have limitations, as they require local calibration with actual data to produce reliable predictions of velocity-pressure behavior.

Experimental Procedure

The experiments were conducted in a triaxial pressure vessel, which allowed for independent control of axial, confining, and pore pressures. The main steps of the procedure were as follows:

1. System calibration: The zero-time reference was obtained with the transmitter and receiver in direct contact (no sample), ensuring accurate travel-time measurements.

2. Sample preparation: Cylindrical samples were placed in the triaxial cell and fully saturated with BFW (Brine Formation Water).

3. Sensor installation: Axial P- and S-wave transducers were mounted for velocity acquisition along the sample axis.

4. Initial pressures:

    - Axial pressure: 6 MPa
    - Confining pressure: 5 MPa
    - A constant difference of 1 MPa was maintained to prevent horizontal fracturing.

5. Pressure loading:

    - Confining pressure varied from 5.3 MPa to 63.3 MPa
    - Axial pressure from 6 MPa to 64 MPa
    - Pore pressure increased from 0 MPa up to 53 MPa, with initial increments of 11.3 MPa.

6. Cyclic tests :

    - Cycles 1 and 2: Confinement between 63–40 MPa, pore pressure between 53–30 MPa
    - Cycles 3 and 4: Confinement between 63–30 MPa, pore pressure between 53–20 MPa

7. Unloading stage: Pressures were gradually reduced to evaluate elastic recovery and hysteresis effects.

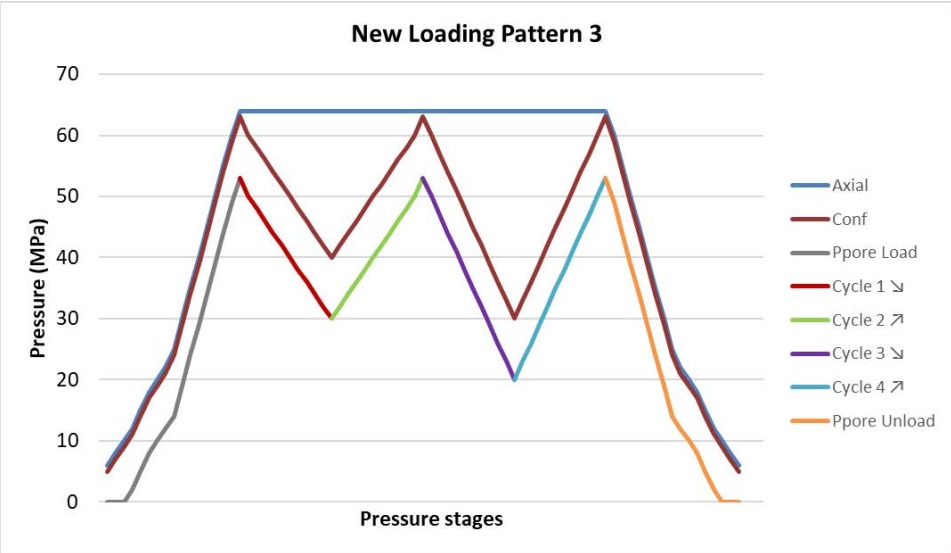


Figure 4.2.1.1.1 – Loading pattern model for pore pressure analysis.

Waveforms were recorded only when inlet and outlet gauges were equal, ensuring that the pressure inside the sample was fully equilibrated. This procedure prevents internal pressure gradients and guarantees that measured velocities represent the true elastic response under controlled pore pressure conditions, rather than transient fluid flow effects.

Results

The experimental results reveal a nonlinear relationship between seismic velocity and pore pressure, with clear evidence of elastic hysteresis. During loading (increasing pore pressure), both P- and S-wave velocities decrease, indicating a reduction in effective stiffness as the fluid pressure counteracts the confining stress. During unloading, velocities increase again, but not at the same rate, showing that the rock's response depends on the pressure path, a typical hysteretic behavior.

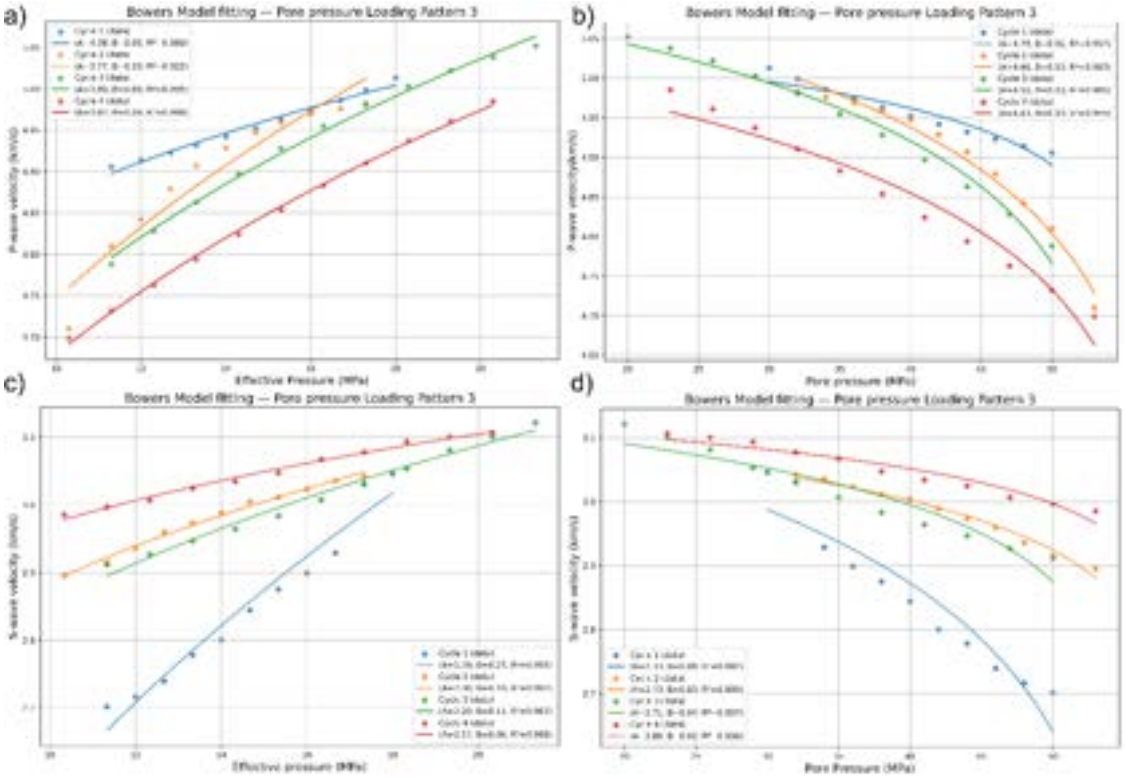


Figure 4.2.1.1.2 – Velocity versus effective pressure and velocity versus pore pressure for: a, b) P-wave and c, d) S-wave, including the Bowers model fitting for each analysis cycle along the Z-axis.

This phenomenon is consistent with rocks containing microcracks and heterogeneous pore structures: microcracks close rapidly at low effective pressures, producing steep velocity increases, and rounded pores respond more gradually, resulting in slower changes in velocity.

Furthermore, the measured velocities along the X, Y, and Z axes display anisotropy, suggesting heterogeneous texture or preferred orientation of fractures and laminations within the rock matrix.

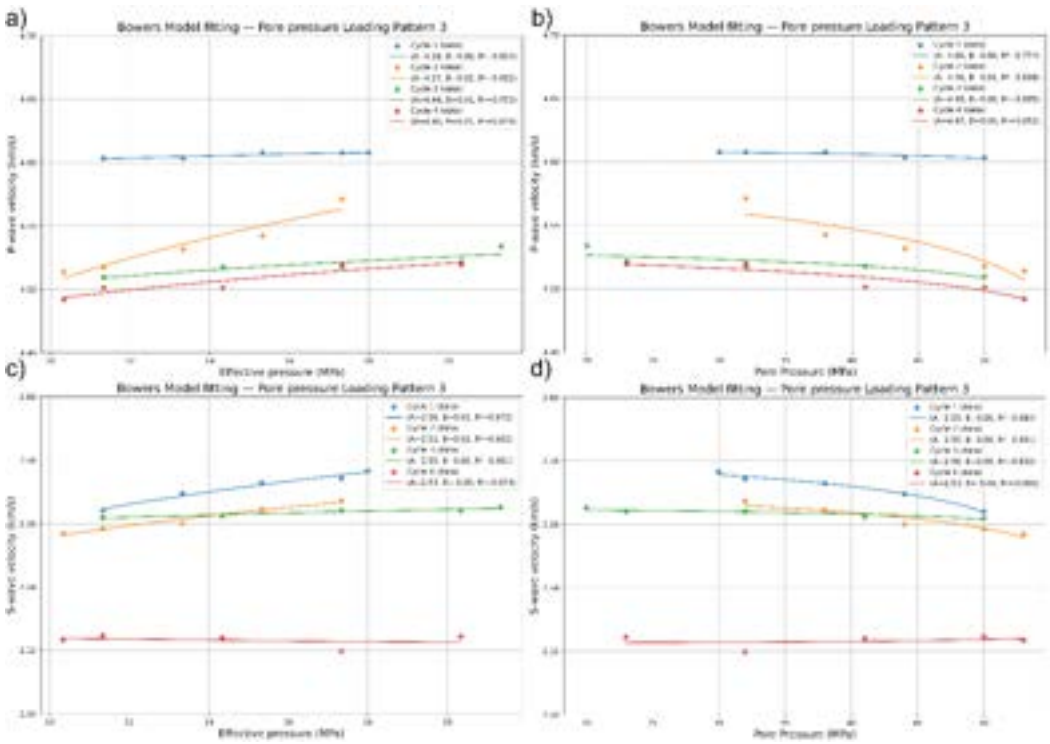


Figure 4.2.1.1.3 - Velocity versus effective pressure and velocity versus pore pressure for: a, b) P-wave and c, d) S-wave, including the Bowers model fitting for each analysis cycle along the X-axis.

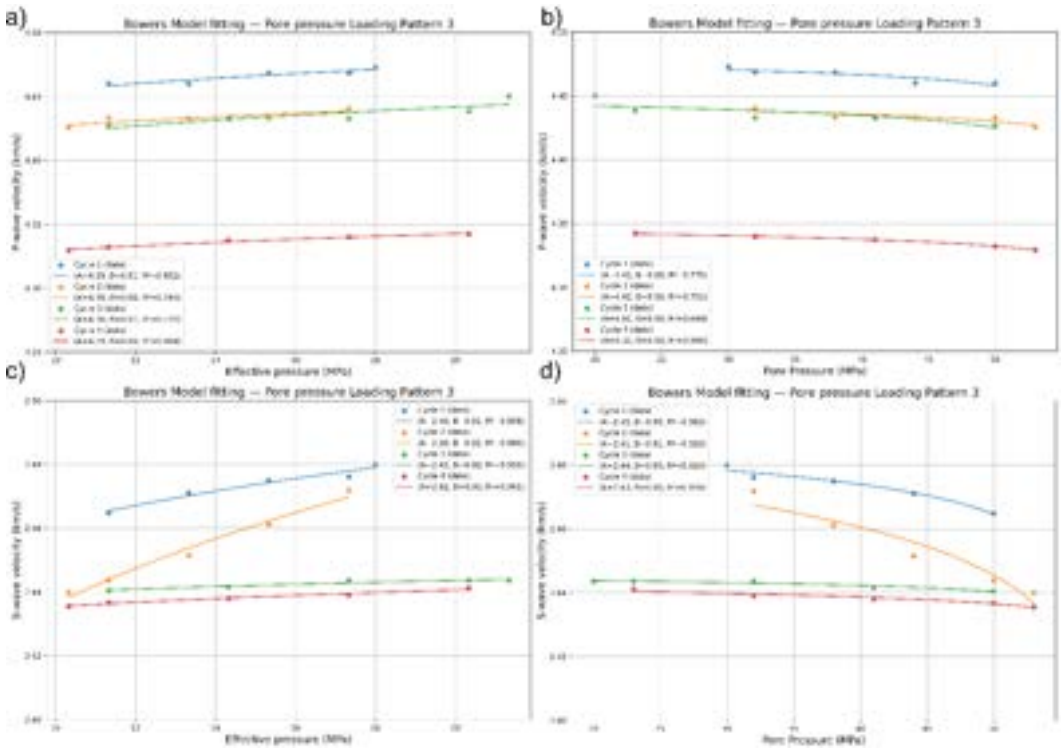


Figure 4.2.1.1.4 - Velocity versus effective pressure and velocity versus pore pressure for: a, b) P-wave and c, d) S-wave, including the Bowers model fitting for each analysis cycle along the Y-axis.

Seismic velocities decrease during pore pressure loading and recover during unloading, demonstrating elastic hysteresis. This path-dependent behavior reflects distinct mechanisms of pore and microcrack closure and reopening. The observed anisotropy among the principal axes confirms the heterogeneous nature of the samples, influenced by structural fabric and depositional features. Overall, these results enhance the understanding of how elastic velocities vary with pore pressure, contributing to more accurate velocity correction models for deep carbonate reservoirs.

Next steps:

- Complete the analysis of sample A18, currently under processing.
- Apply and compare different pore pressure models to refine the empirical fit.
- Evaluate the consistency of results among samples and identify trends related to anisotropy and rock stiffness under effective pressure control.

4.2.2 Fluid Substitution Experiments

FLUID SUBSTITUTION

One of the important challenges in rock physics analysis of logs, cores, and seismic data is predicting seismic velocities in rocks saturated with a single fluid based on velocities measured in dry conditions or from rocks saturated with another fluid. This challenge constitutes the fluid substitution problem, a fundamental tool in seismic rock physics that provides geoscientists with valuable insight into identifying the type of fluid present in a reservoir (Mavko et al., 2009; Huang et al., 2015). For this purpose, we describe an experimental methodology and present an initial result for fluid substitution tests.

METHODOLOGY

First, we characterize the petrophysical properties of the core plug sample, as shown in Table 4.2.2.1. Subsequently, the direct wave transmission test (ASTM D2845-08, 2008) was conducted to determine the P- and S-wave transit travel times. After that, the transmission test is performed in the Rock Physics Triaxial System equipment and with an Indiana Limestone sample in dry, saturated, and fluid substitution (partially saturated) conditions.

The velocity test was conducted at both dry and saturated conditions, with pressure ranges from 5 to 40 MPa. Next, the sample was 100% saturated with a BFW (Sw=1), which was introduced into the Rock Physics Triaxial System and applied pressure (5 to 40 MPa). Having reached the maximum pressure level, we utilized a fluid injection system coupled to triaxial system equipment to perform a

fluid injection process on the core plug sample. The fluid injection process utilizes an experimental apparatus (Figure 4.2.2.1), which consists of a variable displacement pump with a constant flow rate, a fluid transfer cell, a manometer, hydraulic connections and tubes, and a recipient for collecting effluent and pressure relief of the system. The fluid used for fluid substitution was a mixture of diesel oil and cyclohexane, with a ratio of 3 parts cyclohexane to 1 part diesel. We performed this measure in two steps.

#1: injection of a cyclohexane and diesel mixture to replace BFW, resulting in a final saturation proportion of the Sw=0.65 and So=0.35;

#2: injection of a BFW to replace the cyclohexane and diesel mixture, resulting in a final saturation of approximately Sw=0.86 and So=0.14. The register of wave transit travel times occurred during the load and unload confinement pressure of all experiments.

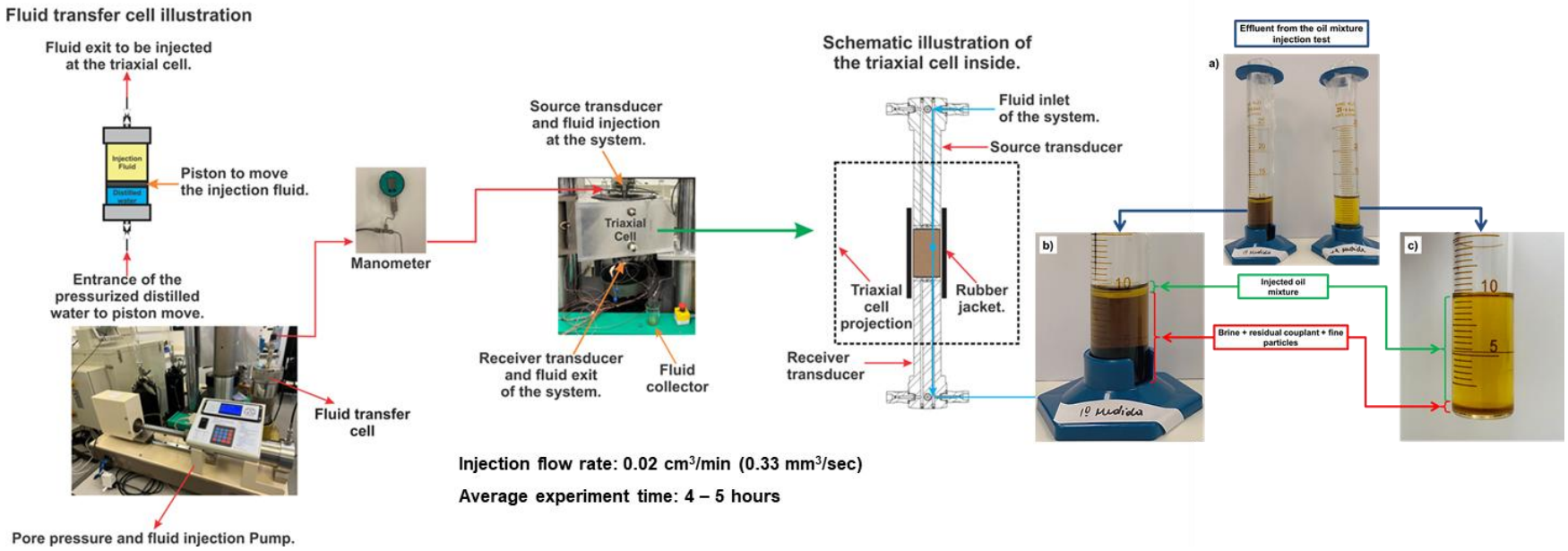
Table 4.2.2.1: Petrophysical properties of Indiana Limestone (ILS-70-3-L3) sample.

Sample	Indiana Limestone (ILS-70-3-L3)
Diameter (cm)	74.89
Length (cm)	37.98
Volume (cm^3)	84.844
Gas porosity – ϕ (%)	18.04
Dry bulk density – ρ_{dry} (g/cm^3)	2.2003
Grain density – ρ_{grain} (g/cm^3)	2.6845
Weight dry sample (g)	186.68
Pore volume (cm^3)	15.303

RESULTS

Figure 4.2.2.2 exhibits the initial results of the fluid substitution tests. The top Figure 4.2.2.2 shows the variation of P-wave velocity with effective pressure for different fluid saturation conditions. In all cases, velocity increases as effective pressure rises, consistent with the progressive closure of microfractures and the reduction of effective porosity, which enhances the rock's stiffness. The dry condition exhibits the lowest velocities, while the fully saturated state ($S_w = 1.00$) shows the highest values, with intermediate saturations (S_w = 0.86 and 0.65) falling between these limits. Figure 4.2.2.2 bottom presents the wiggle plot comparison of the P-wave forms and highlights the arrival times for the same saturation conditions at 40 axial and 39 confining pressures. The dry sample exhibits the longest arrival travel times, whereas increasing water saturation results in progressively shorter times, indicating higher velocities. Together, these results confirm the consistent influence of both effective pressure and fluid saturation on P-wave propagation in the tested sample.

Adaptation of the pore pressure system for fluid substitution



Injection flow rate: 0.02 cm^3/min (0.33 mm^3/sec)

Average experiment time: 4 – 5 hours

Figure 4.2.2.1: Fluid injection apparatus coupled in the Rock Physics Triaxial System.

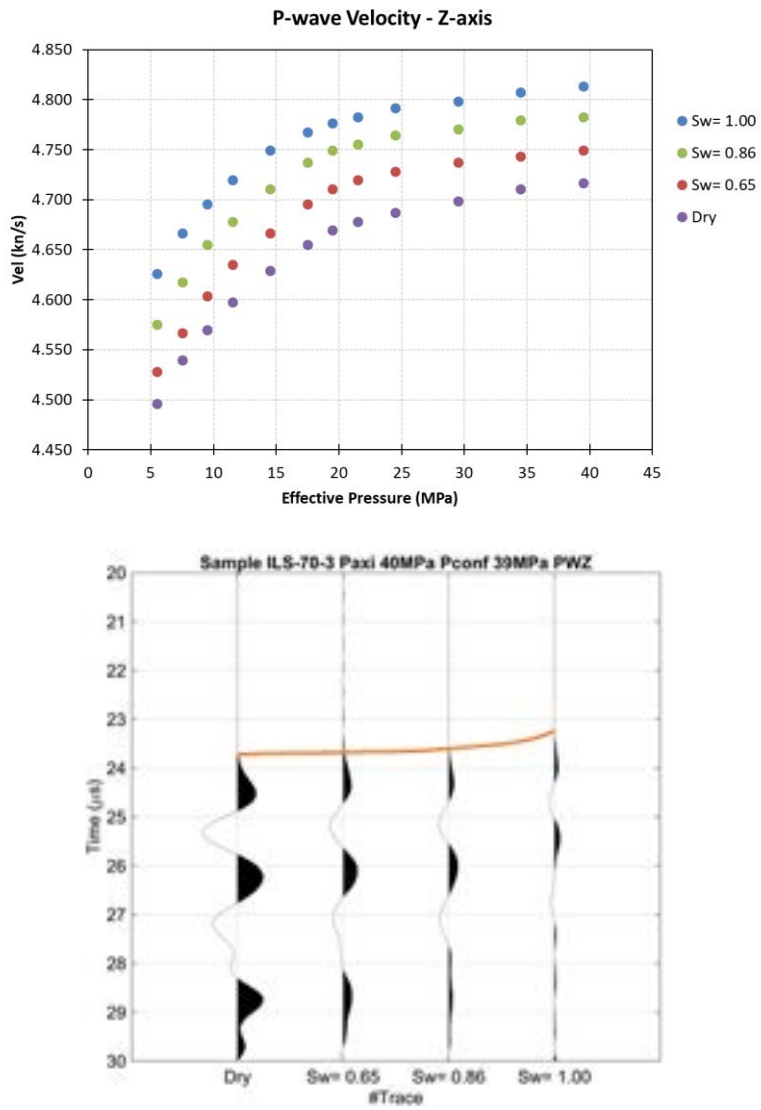


Figure 4.2.2.2: (Top) P-wave velocity in Indiana limestone (ILS-70-3) under different saturation and pressure conditions. (Bottom) Wiggle plot comparison of waveforms for the ILS-70-3 sample under different saturated conditions and the same pressure level.

4.2.3 Fluid properties

Pore fluids directly influence the seismic properties of rocks. The densities, bulk moduli, velocities, and viscosities of common pore fluids are usually simplified in some rock physics modeling. One of the primary objectives of geophysical exploration is to identify the pore fluids in hydrocarbon reservoirs at the subsurface level (Batzle and Wang, 1992). Moreover, the study of the physical and seismic properties of pore fluids is a crucial tool for reservoir flow

models, significantly influencing the decision-making process in oil production projects. Additionally, pore fluid property data are utilized to calibrate these models and reduce uncertainties in production forecasts (Oliver et al., 2008). For this purpose, we describe an experimental methodology in which the fluid properties were measured under pressure and temperature conditions close to those of the reservoir, to determine the P-wave velocity, bulk modulus, and density of fluids with different compositions. The fluids used were distilled water, Brazilian Formation Water (BFW; as reported in the latest Technical Reports), and NaCl brine.

FLUID BULK MODULUS AND DENSITY

The important characteristics of a fluid are the bulk compressibility, bulk modulus, and density, as outlined in Equations 4.2.3.1 to 4.2.3.3. To determine these properties, we use a pore pressure pump that comprises a variable displacement pump (Figure 4.2.3.1), which allows pressure (1 to 60 MPa) to be applied and volume variations to be monitored at a constant temperature (23 °C). The volume of the pump reservoir is 225,000 mm³ (225 mL), which is considered the initial volume for the tests. As the pressure increases, the fluid volume decreases, allowing the computation of these parameters and the calculation of the bulk compressibility (Equation 4.2.3.1) and bulk modulus (Equation 4.2.3.2) of distilled water, BFW, and NaCl brine (Figure 4.2.3.2). Considering that the fluid mass remains constant, it is possible to use the recorded volume variations to recalculate the fluid density (Equation 4.2.3.3) as a function of pressure (Figure 4.2.3.3).

$$C_{fluid} = -\frac{1}{V_0}\left(\frac{\partial V}{\partial P}\right)_{T=const} \tag{4.2.3.1}$$

$$K_{fluid} = \frac{1}{C_{fluid}} \tag{4.2.3.2}$$

$$\rho = \frac{m_0}{V(P)} \tag{4.2.3.3}$$

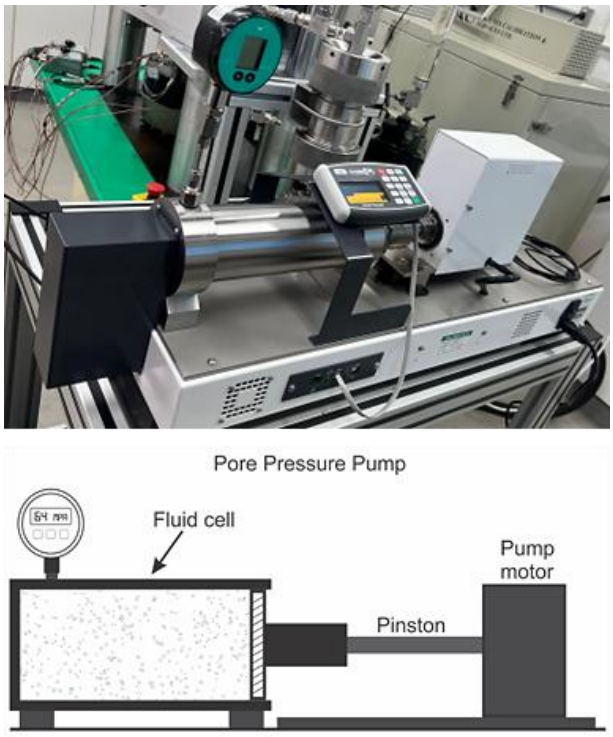


Figure 4.2.3.1: (Top) Pore pressure pump used in the compressibility test. (Below) Schematic drawing of a pore pressure pump.

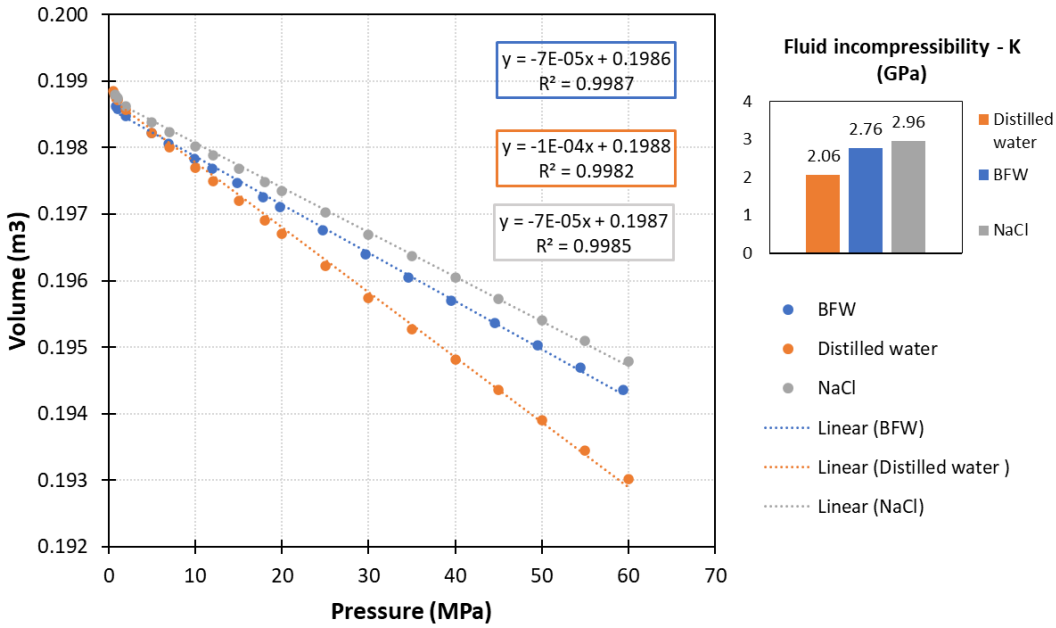


Figure 4.2.3.2: (Left) Results of the fluid compression test using a pore pressure pump. (Top right) Fluid incompressibility estimated by the fluid volume reduction test.

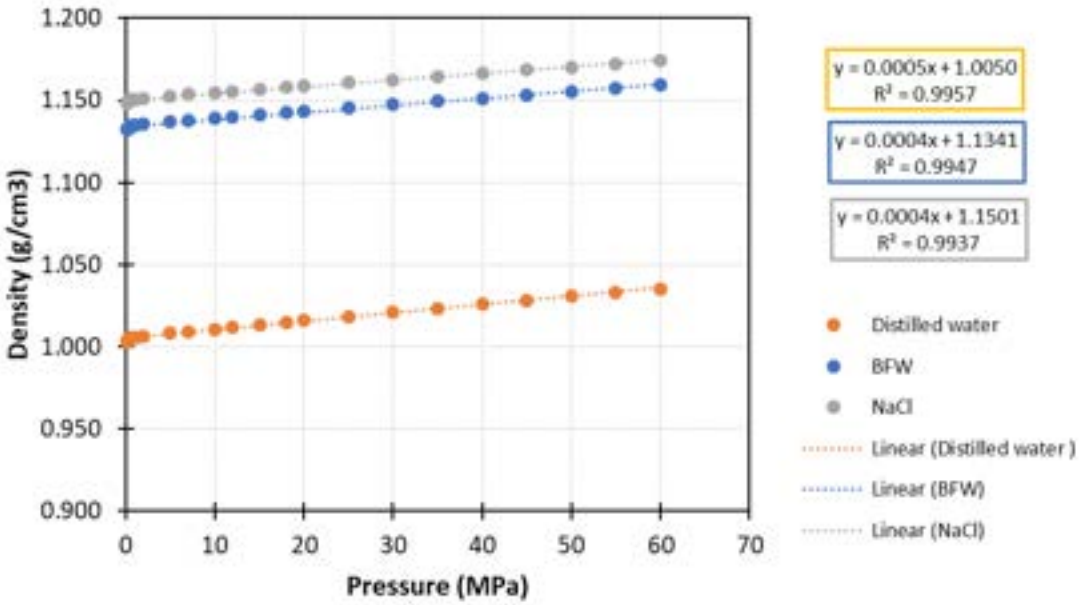


Figure 4.2.3.3: Results of fluid density varying with pressure.

FLUID VELOCITIES

The fluid velocity measurements were performed using the Triaxial Rock Physics System at LENEP. A hollow steel cylinder was positioned between the source and receiver transducers. Initially, the fluid was injected through the inlet of the bottom transducer to fill the internal volume of the cylinder. It was subsequently collected at the outlet of the top transducer and closed for the start of the pressure increment, and the P-wave travel time was registered to calculate wave velocity for each pressure level (1 to 53 MPa). Figure 4.2.3.4 shows a schematic drawing of the experimental setup used for fluid velocity measurements.

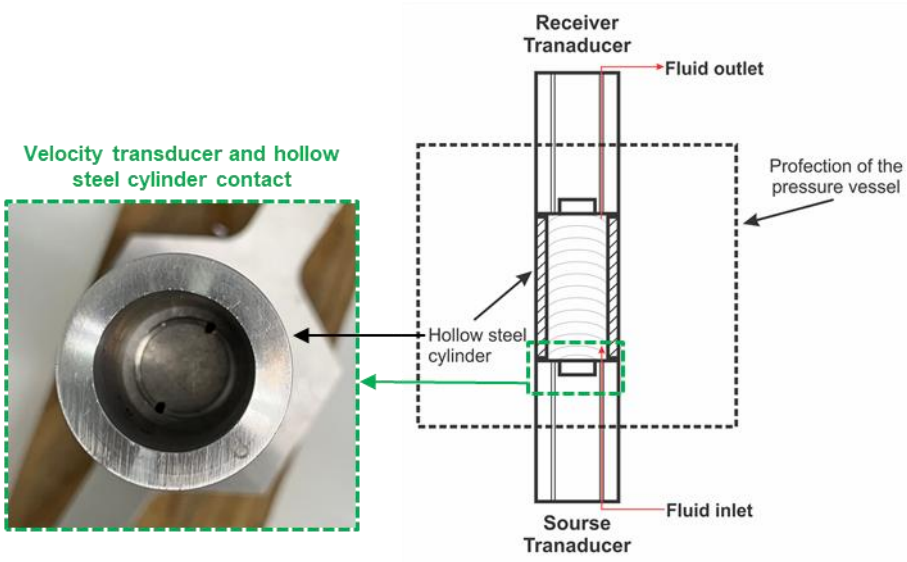


Figure 4.2.3.4: Experimental setup for fluid velocity measurement.

During the wave signal interpretation, two distinct waveforms were observed: the first arrival, attributed to the wave propagating through the stainless-steel cylinder wall, and the second, corresponding to the fluid arrival. Due to the overlap between these signals, it was necessary to apply a bandpass filter to suppress the response from stainless steel and facilitate signal processing. These data processing and analysis are illustrated in Figure 4.2.3.5.

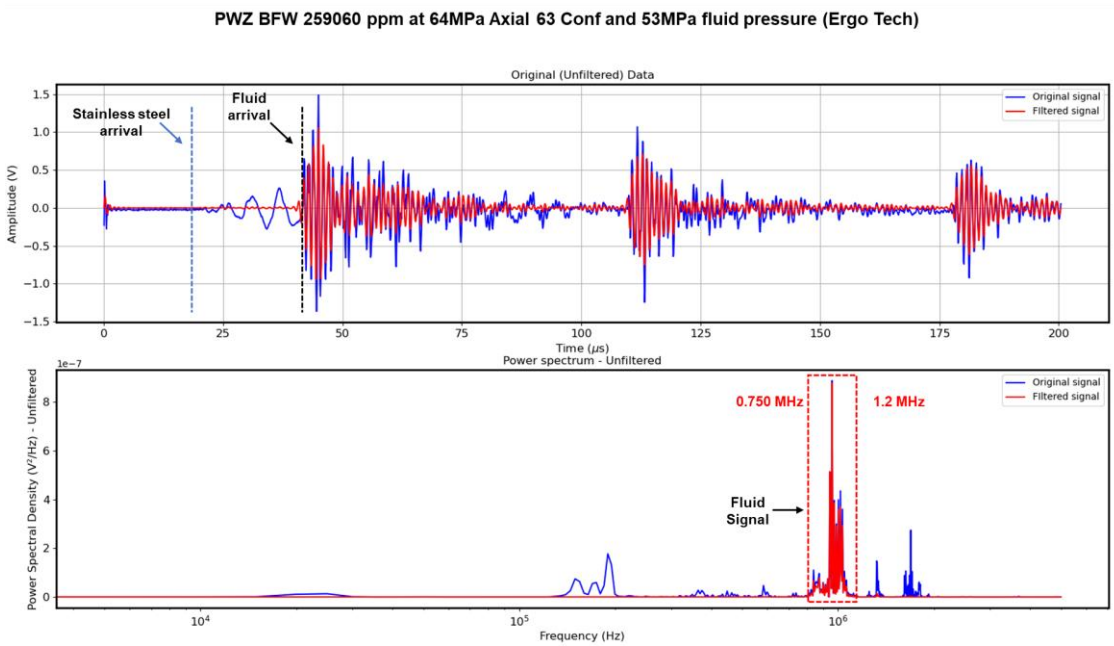


Figure 4.2.3.5: Data processing and analysis. (Top) Original and filtered P-wave signals. (Bottom) Power spectrum of the P-wave signal.

The velocity measurements were conducted in NaCl brine and BFW, both prepared with the same mass of dissolved total solids (259036 ppm), under pressures ranging from 1 to 53 MPa and temperatures between 23 and 100 °C. Figure 4.2.3.6 presents two figures for NaCl brine results: the first is a crossplot of velocity versus pressure at different pressure and temperature conditions, and the second is a wiggle plot showing the wave traces at the same pressure conditions and varying temperature. Note that, as the temperature increases, the P-wave velocity is affected, the wave propagation time increases, and the velocity decreases. Similarly, Figure 4.2.3.7 presents a result for BFW, displaying a crossplot of velocity versus pressure at varying temperatures. However, in the case of the BFW fluid, the velocity–pressure–temperature relationship displayed a distinct trend, characterized by a non-linear correlation. The effect of temperature on velocity was also different, with velocity increasing up to approximately 50 °C and then decreasing at higher temperatures. To better observe this behavior, velocity was plotted as a function of temperature, as shown in Figure 4.2.3.8, with each curve corresponding to a different pressure. It can be noted that temperature has a more substantial influence on velocity variation than pressure. In the BFW composition, the presence of multiple dissolved salts

suggests that temperature exerts a more significant effect due to molecular kinetic activity and volumetric fluid expansion.

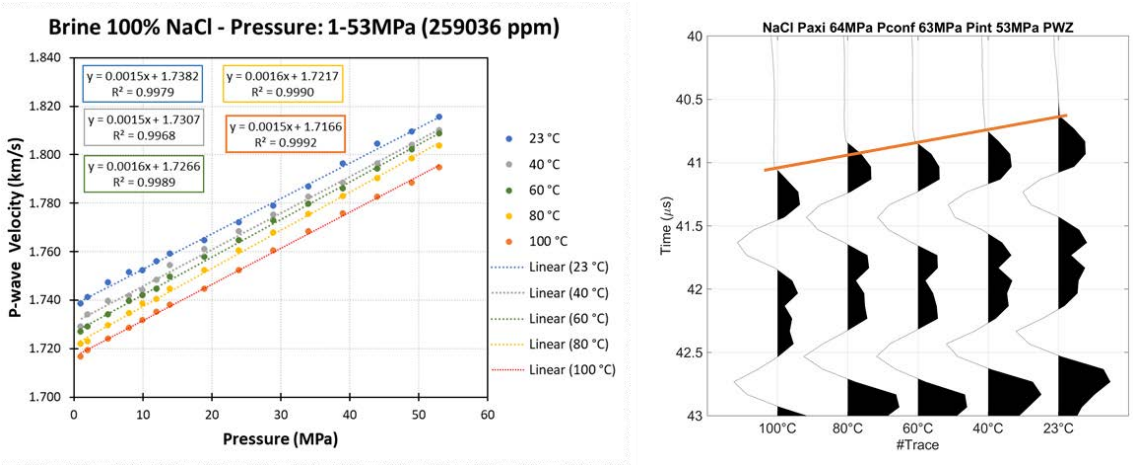


Figure 4.2.3.6: (Left) Fluid velocity analysis for sodium chloride brine (NaCl) with 259036 ppm, under varying pressure (1–53 MPa) and temperature (23, 40, 60, 80, and 100 °C). (Right) Wiggle plot comparison of waveforms for sodium chloride brine (NaCl) at 53 MPa of pressure and varying temperatures (23, 40, 60, 80, and 100 °C).

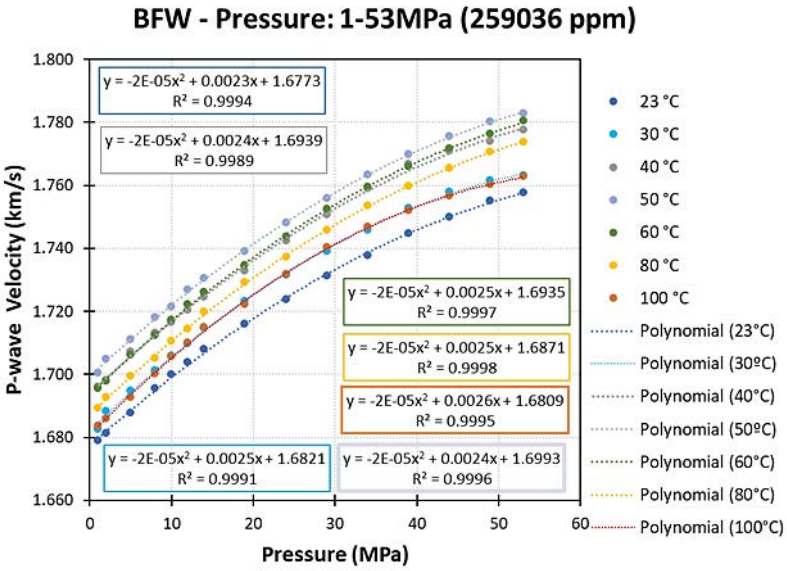


Figure 4.2.3.7: Fluid velocity analysis for BFW with 259036 ppm, under varying pressure (0.1–53 MPa) and temperature (23, 30, 40, 50, 60, 80, and 100 °C).

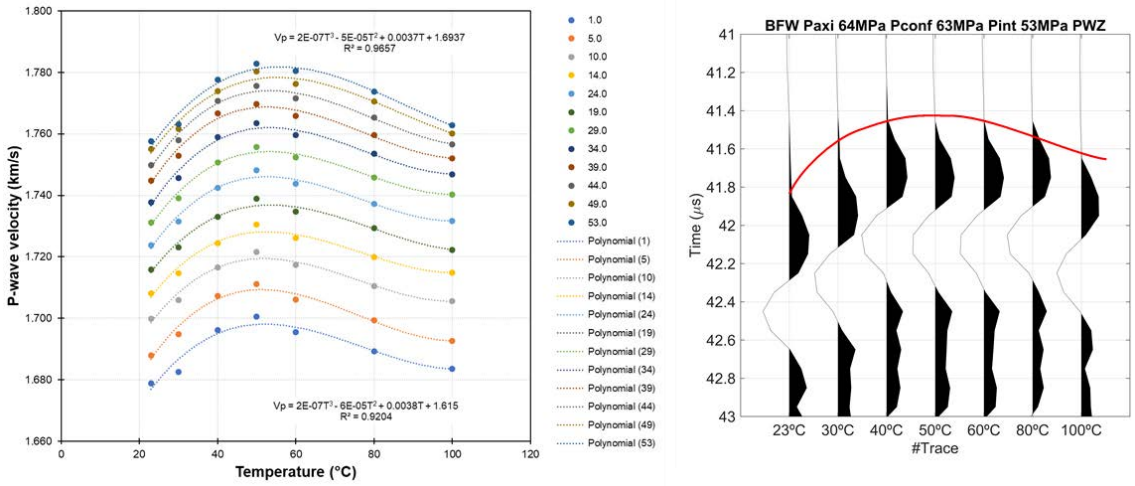


Figure 4.2.3.8: (Left) Fluid velocity analysis for BFW brine (259036 ppm) versus temperature with, under varying pressure (1 – 53 MPa) and temperature (23, 30,40, 50, 60, 80, and 100 °C). (Right) Wiggle plot comparison of waveforms for BFW brine at 53 MPa of pressure and varying temperatures (23, 30, 40, 50, 60, 80, and 100 °C).

FLUID VISCOSITY

A new Cannon-Fenske viscometer (Figure 4.2.3.9) was purchased to allow viscosity measurements to be carried out in our laboratory. With this new viscometer, it became possible to obtain reliable and repeatable viscosity data under controlled temperature conditions. this viscometer model is a simple capillary device used to measure the kinematic viscosity of Newtonian fluids. It works by measuring the time it takes for a liquid to flow through a thin glass capillary under the influence of gravity. Because the method assumes that viscosity does not change with flow rate, it's only suitable for fluids with constant viscosity, such as oils, alcohols, or brines.

We chose this equipment because the brine used in our tests behaves as a Newtonian fluid, meaning its viscosity stays stable regardless of shear conditions. The Cannon-Fenske allows us to obtain quick and reliable viscosity measurements under controlled temperature, which is important for checking fluid consistency and ensuring reproducibility in our lab experiments.

Figure 4.2.3.9 - Viscosimeter Cannon Fenske.

This BFW (Brazilian Formation Water) used in this study consists of a solution containing seven chemical reagents, formulated to reproduce the ionic composition and salinity of formation waters from Brazilian pre-salt reservoirs. This synthetic brine was prepared to simulate realistic reservoir conditions, ensuring that laboratory measurements accurately represent the physicochemical behavior of fluids encountered in subsurface environments.

The results obtained from the BFW are:

- T (average time) in sec – 547.33 s (the test was performed 3x)
- D (density) BFW – 1.13807 g/cm³ (measured using a 10 ml pipette)
- Cinematic viscosity – 15.101 cst
- Absolut viscosity – 17.186 cP

4.2.4 Porosity, Permeability, and Pore Compressibility – New samples

To improve the pore volume compressibility (PVC) model, new samples were tested in Coreval to provide information regarding porosity and permeability under pressure. Mineralogy was also determined using XRD equipment, allowing us to calculate the mineral compressibility (Cmineral). These new carbonate samples are from Kocurek Industries. PVC Measurements were obtained using the Unalmer-Swallwell method, based on nonlinear regression (power law function), between pore volume and confining pressure. The Coreval tests

surpassed 22.1 MPa, depending on the sample Unconfined Compressive Strength (UCS) provided by the Kocurek Industries site.

Table 4.2.4.1 – Kocurek carbonate samples porosity and permeability results tested in Coreval Equipment, Cmineral calculated from XRD results, and PVC values obtained from Unalmiser method.

Sample's name	Rock	Porosity (%)	k (mD)	PVC (1/GPa)	Load Pressure (MPa)	Cmineral
EQR-IL2-1A	Indiana Limestone	20.73	96.69	3.263	2.8	0.01409
EQR-IL2-1A	Indiana Limestone	20.60	97.32	1.632	5.5	0.01409
EQR-IL2-1A	Indiana Limestone	20.57	96.47	1.088	8.3	0.01409
EQR-IL2-1A	Indiana Limestone	20.53	95.73	0.816	11.0	0.01409
EQR-IL2-1A	Indiana Limestone	20.50	95.13	0.653	13.8	0.01409
EQR-IL2-1A	Indiana Limestone	20.47	94.32	0.544	16.5	0.01409
EQR-IL2-1A	Indiana Limestone	20.42	94.73	0.466	19.3	0.01409
EQR-IL2-1A	Indiana Limestone	20.40	93.21	0.408	22.1	0.01409
EQR-IL2-3A	Indiana Limestone	20.35	69.03	4.351	2.8	0.01409
EQR-IL2-3A	Indiana Limestone	20.25	68.23	2.176	5.5	0.01409
EQR-IL2-3A	Indiana Limestone	20.15	67.68	1.450	8.3	0.01409
EQR-IL2-3A	Indiana Limestone	20.11	67.08	1.088	11.0	0.01409
EQR-IL2-3A	Indiana Limestone	20.06	66.86	0.870	13.8	0.01409
EQR-IL2-3A	Indiana Limestone	20.02	66.58	0.725	16.5	0.01409
EQR-IL2-3A	Indiana Limestone	19.99	67.29	0.622	19.3	0.01409
EQR-IL2-3A	Indiana Limestone	19.97	66.77	0.544	22.1	0.01409
ILS 70-1 L3	Indiana Limestone	17.24	105.04	2.719	5.5	0.01409
ILS 70-1 L3	Indiana Limestone	17.11	103.86	1.813	8.3	0.01409
ILS 70-1 L3	Indiana Limestone	17.05	103.94	1.360	11.0	0.01409
ILS 70-1 L3	Indiana Limestone	17.08	104.61	1.088	13.8	0.01409

ILS 70-1 L3	Indiana Limestone	16.98	103.52	0.906	16.5	0.01409
ILS 70-1 L3	Indiana Limestone	16.93	103.67	0.777	19.3	0.01409
ILS 70-1 L3	Indiana Limestone	16.95	103.09	0.680	22.1	0.01409
SD-BL18-4	Silurian Dol	8.54	0.59	2.176	2.8	0.01410
SD-BL18-4	Silurian Dol	8.40	0.59	1.088	5.5	0.01410
SD-BL18-4	Silurian Dol	8.39	0.57	0.725	8.3	0.01410
SD-BL18-4	Silurian Dol	8.44	0.58	0.544	11.0	0.01410
SD-BL18-4	Silurian Dol	8.44	0.58	0.435	13.8	0.01410
SD-BL18-4	Silurian Dol	8.43	0.57	0.363	16.5	0.01410
SD-BL18-4	Silurian Dol	8.42	0.57	0.311	19.3	0.01410
SD-BL18-4	Silurian Dol	8.40	0.57	0.272	22.1	0.01410
SD-BL18-4	Silurian Dol	8.39	0.56	0.242	24.8	0.01410
SD-BL18-4	Silurian Dol	8.38	0.56	0.218	27.6	0.01410
SD-BL18-4	Silurian Dol	8.37	0.57	0.198	30.3	0.01410
SD-BL18-4	Silurian Dol	8.36	0.56	0.181	33.1	0.01410
SD-BL18-4	Silurian Dol	8.36	0.56	0.167	35.9	0.01410
SD-BL18-4	Silurian Dol	8.36	0.57	0.155	38.6	0.01410
SD-BL18-4	Silurian Dol	8.35	0.56	0.145	41.4	0.01410
AC-24-L3-01	Austin Chalk	27.29	7.07	1.450	2.8	0.01409
AC-24-L3-01	Austin Chalk	27.30	7.05	0.725	5.5	0.01409
AC-24-L3-01	Austin Chalk	27.28	6.97	0.483	8.3	0.01409
AC-24-L3-01	Austin Chalk	27.21	6.99	0.363	11.0	0.01409
AC-24-L3-01	Austin Chalk	27.15	6.95	0.290	13.8	0.01409
SD-BL-18-2	Silurian Dol	8.60	0.68	3.263	2.8	0.01410
SD-BL-18-2	Silurian Dol	8.55	0.64	1.632	5.5	0.01410
SD-BL-18-2	Silurian Dol	8.51	0.64	1.088	8.3	0.01410
SD-BL-18-2	Silurian Dol	8.50	0.64	0.816	11.0	0.01410
SD-BL-18-2	Silurian Dol	8.49	0.63	0.653	13.8	0.01410
SD-BL-18-2	Silurian Dol	8.48	0.63	0.544	16.5	0.01410
SD-BL-18-2	Silurian Dol	8.47	0.63	0.466	19.3	0.01410
SD-BL-18-2	Silurian Dol	8.46	0.63	0.4079	22.1	0.01410
SD-BL-18-2	Silurian Dol	8.44	0.63	0.363	24.8	0.01410
SD-BL-18-2	Silurian Dol	8.44	0.63	0.326	27.6	0.01410
SD-BL-18-2	Silurian Dol	8.42	0.62	0.297	30.3	0.01410
SD-BL-18-2	Silurian Dol	8.42	0.62	0.272	33.1	0.01410
SD-BL-18-2	Silurian Dol	8.41	0.62	0.251	35.9	0.01410
SD-BL-18-2	Silurian Dol	8.41	0.62	0.233	38.6	0.01410
SD-BL-18-2	Silurian Dol	8.40	0.62	0.218	41.4	0.01410
winscousin1	Winsconsin	4.07	0.04	33.721	2.8	0.01403
winscousin1	Winsconsin	3.91	0.03	16.861	5.5	0.01403

winscousin1	Winsconsin	3.79	0.02	11.240	8.3	0.01403
winscousin1	Winsconsin	3.71	0.02	8.430	11.0	0.01403
winscousin1	Winsconsin	3.65	0.01	6.744	13.8	0.01403
winscousin1	Winsconsin	3.60	0.01	5.620	16.5	0.01403
winscousin1	Winsconsin	3.53	0.01	4.817	19.3	0.01403
winscousin1	Winsconsin	3.49	0.01	4.215	22.1	0.01403
winscousin1	Winsconsin	3.47	0.01	3.747	24.8	0.01403
winscousin1	Winsconsin	3.44	0.03	3.372	27.6	0.01403
winscousin1	Winsconsin	3.38	0.03	3.066	30.3	0.01403
winscousin1	Winsconsin	3.35	0.01	2.810	33.1	0.01403
winscousin1	Winsconsin	3.32	0.03	2.594	35.9	0.01403
winscousin1	Winsconsin	3.29	0.03	2.409	38.6	0.01403
winscousin1	Winsconsin	3.25	0.03	2.248	41.4	0.01403
winscousin1	Winsconsin	3.29	0.03	2.108	44.1	0.01403
winscousin1	Winsconsin	3.20	0.01	1.984	46.9	0.01403
winscousin1	Winsconsin	3.21	0.01	1.873	49.6	0.01403
winscousin1	Winsconsin	3.18	0.03	1.775	52.4	0.01403
winscousin1	Winsconsin	3.14	0.03	1.686	55.2	0.01403
winscousin1	Winsconsin	3.13	0.03	1.606	57.9	0.01403
ILS 70-3	Indiana Limestone	16.51	134.04	2.176	2.8	0.01409
ILS 70-3	Indiana Limestone	16.52	132.48	1.088	5.5	0.01409
ILS 70-3	Indiana Limestone	16.44	129.40	0.725	8.3	0.01409
ILS 70-3	Indiana Limestone	16.42	133.22	0.544	11.0	0.01409
ILS 70-3	Indiana Limestone	16.38	129.52	0.435	13.8	0.01409
ILS 70-3	Indiana Limestone	16.39	130.17	0.363	16.5	0.01409
ILS 70-3	Indiana Limestone	16.36	136.91	0.311	19.3	0.01409
ILS 70-3	Indiana Limestone	16.35	132.63	0.272	22.1	0.01409
Wisconsin BL-18_4-F	Winsconsin	7.11	0.90	5.802	2.8	0.01403
Wisconsin BL-18_4-F	Winsconsin	7.02	0.87	2.901	5.5	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.99	0.86	1.934	8.3	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.95	0.85	1.450	11.0	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.93	0.84	1.160	13.8	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.92	0.83	0.967	16.5	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.91	0.83	0.829	19.3	0.01403

Wisconsin BL-18_4-F	Winsconsin	6.89	0.82	0.725	22.1	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.88	0.81	0.645	24.8	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.87	0.81	0.580	27.6	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.86	0.80	0.527	30.3	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.85	0.80	0.483	33.1	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.84	0.80	0.446	35.9	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.83	0.79	0.414	38.6	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.83	0.80	0.387	41.4	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.82	0.79	0.363	44.1	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.81	0.79	0.341	46.9	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.80	0.78	0.322	49.6	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.80	0.77	0.305	52.4	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.80	0.77	0.290	55.2	0.01403
Wisconsin BL-18_4-F	Winsconsin	6.79	0.77	0.276	57.9	0.01403

All information provided in Table 4.2.4.1 will be included in the two new PVC models, which will be explained in the next section.

4.2.5 Pore Compressibility Modeling

To improve the new compressibility model based on machine learning techniques, new carbonate samples were included, as described in section 4.2.4.

These samples are not Brazilian carbonates, which implies differences in petrophysical and geological characteristics. Since this new dataset introduces novel sample data into our model, the results consequently affected the previously determined hyperparameters, as the basic petrophysical parameters differ due to variations in lithology, location, depositional environment, and geological time scale.

As described in Technical Report #1, section 4.3.1.5.2, the Random Forest and Gradient Boosting algorithms provided the best machine learning performance. The inclusion of Kocurek samples slightly altered the results, leading to a decrease in R^2 and Mean Relative Error. Nevertheless, we are still

testing the Kocurek samples, and we plan to include ANP samples in these two updated models to further improve performance metrics such as R^2.

Additionally, we modified the data partitioning between training and testing sets. Previously, 80% of the data was used for training and 20% for testing. We now apply a 70/30 split, as this configuration has provided more precise and accurate results.

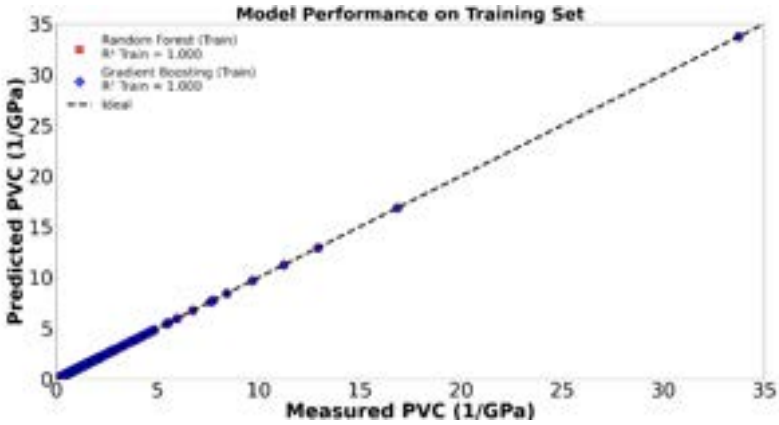


Figure 4.2.5.1 - Comparative performance of RF and GB predicting rock compressibility (PVC). The scatter plot shows the relationship between measured (x-axis) and predicted (y-axis) PVC values (1/GPa) for the training dataset (70% of total samples).

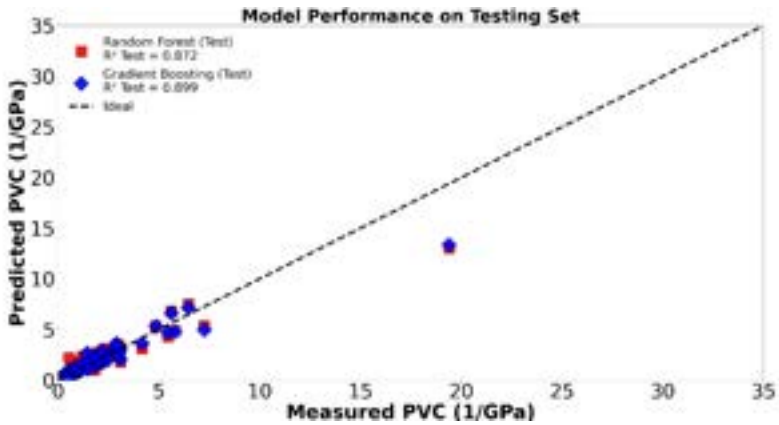


Figure 4.2.5.2 - Comparative performance of RF and GB predicting rock compressibility (PVC). The scatter plot shows the relationship between measured (x-axis) and predicted (y-axis) PVC values (1/GPa) for the testing dataset (30% of total samples).

Figure 4.2.5.2 shows the comparison between measured and predicted PVC values for the test set. Both models, RF and GB, performed well overall, with R^2 values of 0.872 and 0.899, respectively. Even so, some points deviate from the ideal 1:1 line, especially at higher PVC values, indicating that these data were more difficult to fit. This difference likely comes from the complexity of the dataset.

PVC depends on several interacting properties such as porosity, confining pressure, permeability, and mineral composition. When new carbonate samples from Kocurek were added, the dataset expanded to include a wider range of lithologies, thereby increasing the variability in the results. As a result, the models handled the general trend well but had more difficulty matching the extreme or more scattered measurements.

Table 4.2.5.1 - Best Hyperparameters from Random Forest tuned by the optimizer GridSearch.

Optimal value	Impact on model performance
bootstrap=False	Setting to False uses the full dataset for each tree, reducing randomness and slightly improving accuracy in smaller datasets.
max_depth=20	A depth of 20 allows enough complexity to capture relationships without overfitting.
max_features='sqrt'	Improves diversity among trees.
min_samples_leaf=1	A single sample per leaf allows detailed partitions; good when overfitting is controlled by depth.
min_samples_split=2	The default (2) allows trees to grow fully before pruning by other limits, like depth.
n_estimators=500	500 trees provide stable results and reduce variance without excessive computation time.

Table 4.2.5.2 - Best Hyperparameters from Gradient Boosting tuned by the optimizer GridSearch.

Optimal value	Impact on model performance
learning_rate=0.2	A moderate rate (0.2) ensures faster convergence while keeping model generalization.

max_depth=5	Shallow trees (depth 5) prevent overfitting and keep boosting steps general.
max_features='sqrt'	Using √(features) encourages diversity among weak learners.
min_samples_leaf=1	Ensures fine-grained splits where needed.
min_samples_split=2	Allows splits as soon as at least two samples exist, promoting detailed fitting when justified.
n_estimators=1000	A higher number of trees balances the smaller learning rate, achieving better accuracy.
subsample=1.0	Using all samples ensures consistency and stable learning, suitable when the variance is low.

In Tables 4.2.5.1 and 4.2.5.2, the best hyperparameters differ from those presented in Technical Report #1 (section 4.3.1.5.2) because new carbonate samples from Kocurek were included in the model development. The addition of these samples altered the data distribution, leading the GridSearch optimizer to identify a different set of optimal parameters.

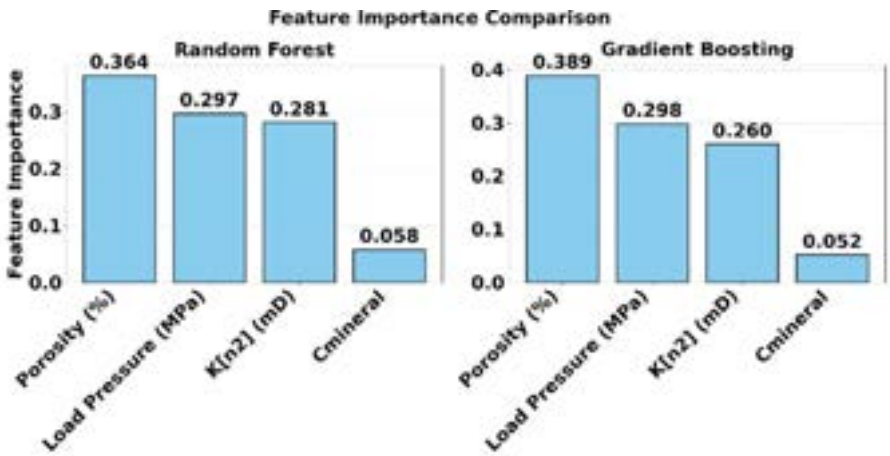


Figure 4.2.5.1 - Feature Importance Comparison between models. K[n2] is the gas permeability, and Cmineral is the mineral compressibility.

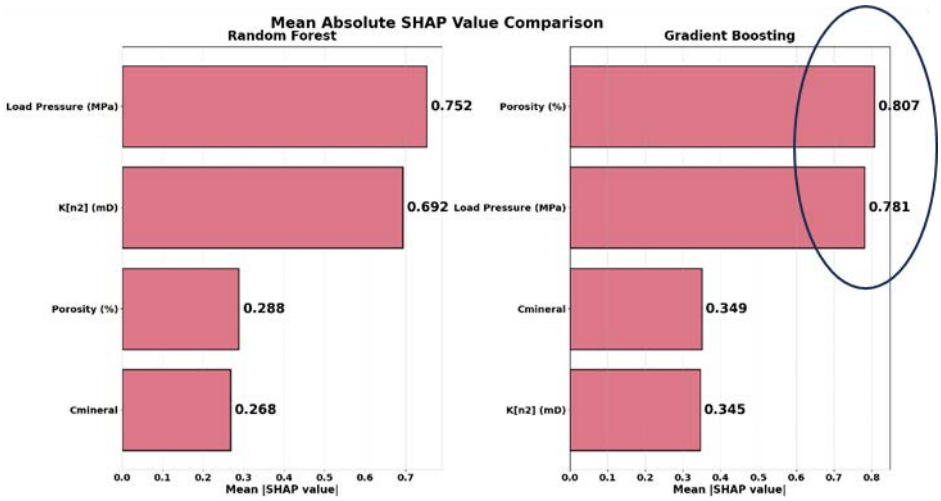


Figure 4.2.5.2 - Mean Absolute SHAP value comparison. K[n2] is the gas permeability, and Cmineral is the mineral compressibility.

For both Random Forest and Gradient Boosting models, Feature Importance measures how effectively each variable splits the data to reduce node impurity, reflecting its structural contribution to model construction. It indicates how the model learns from the data, but does not capture the direction or interaction effects of each variable. In contrast, Mean Absolute SHAP quantifies the average magnitude of a feature's influence on model predictions, showing how the model uses each variable to estimate pore compressibility. This tool is model-agnostic and computed using game theory, estimating the average marginal contribution of each feature to predictions across all possible feature combinations. It accounts for nonlinear interactions and provides a clearer view of the predictive impact of each input. In short, Feature Importance explains model learning, while Mean Absolute SHAP explains model prediction behavior.

The results show that Figure 4.2.5.4 compares the feature importance of input variables for the models. In both cases, porosity (%) is the most important factor, indicating that it is the primary factor influencing pore compressibility predictions. Load pressure (MPa) and permeability (k-mD) also make significant contributions, while mineral composition (Cmineral) has a minimal impact. The similar ranking across models suggests consistent variable influence and stable learning behavior between RF and GB. In Figure 4.2.5.5, for Random Forest, load pressure (MPa) and permeability (k-mD) are shown to have the strongest influence on predictions, indicating that variations in these parameters have the most significant impact on the model output. In contrast, for Gradient Boosting, porosity (%) and load pressure (MPa) are the dominant factors, indicating that this model places a greater emphasis on porosity when estimating pore compressibility. The differences between the two models suggest distinct ways

of weighting variable interactions, although both identify load pressure as a key factor driving the predictions.

5. DIGITAL ROCK PHYSICS

5.1. µCT simulations for Well #861

5.1.1. Introduction

This study complements the elastic anisotropy of horizontal and vertical core plug samples from Well #861 (3-BRSA-861-SPS), located in the Lapa Field of the Santos Basin, Brazil. The primary objective is to utilize a set of available µCT data volumes acquired from specimens taken from a reserved piece of the standard samples, which were performed at the MOGNO beamline facilities (SIRIUS/LNLS/CNPEM). In the last report (Tech Report 1 – Section 5.2), we discussed a methodology to quantify elastic anisotropy using triaxial P- and S-wave velocities and laboratory data measurements from vertical and horizontal standard samples (~38 mm in diameter). Thus, this study aims to perform digital image analysis of the µCT data volumes to characterize the pore system and mineral matrix, and to produce results that contribute to insights supporting the interpretation of anisotropy and Rock Physics modeling.

5.1.2. Digital Image Analysis of the µCT Data: focusing on the pore system

The available µCT data set comprises 10 specimens made from a piece of the core plug samples (Figure 5.1.1), respecting the field of view limitations of the MOGNO facilities and desired resolutions. Unfortunately, the apparatus could not be employed to image standard samples directly. A comparison between photos of samples at plug scale (~38 mm in diameter) and the new specimens prepared from pieces is not necessarily representative, because the new specimens were trimmed and sanded, causing difficulties in preserving visible megapores while trying to accommodate the required field of view in MOGNO's setup. However, micropore imaging tends to be best at high resolution. See the standard samples and texture description in Figure 5.1.2. Thus, we imaged a total of 9 specimens ~2.5 mm in diameter (voxel resolution 1.27 µm), and 1 specimen ~ 6 mm in diameter (F2870H) (voxel resolution 3.33 µm).

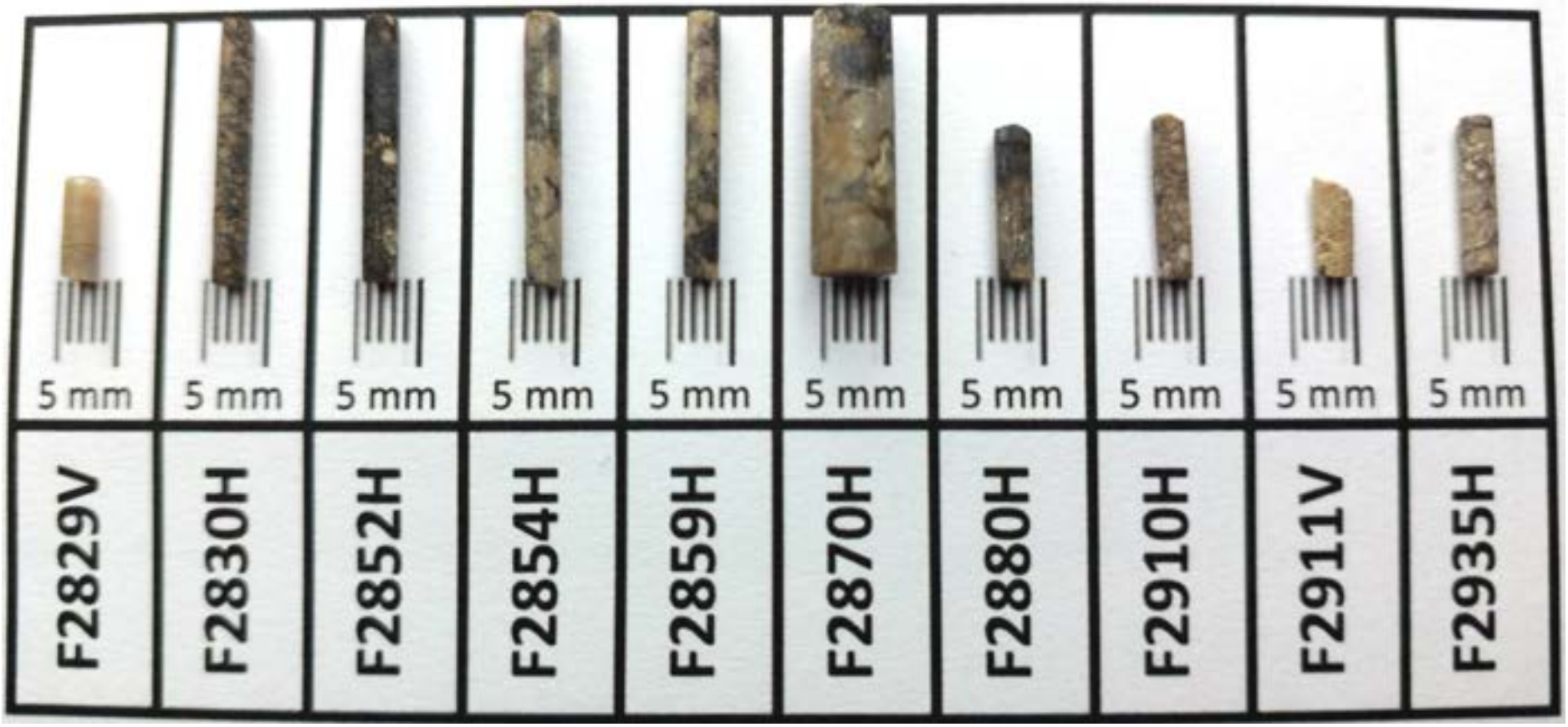


Figure 5.1.1 – Photo of specimens for MOGNO µCT acquisitions from core samples (standard samples of 1.5” in diameter). The specimens were trimmed and sanded from a reserved piece taken from each sample, exhibiting difficulties in preserving megapores. Detailing the data set of nine specimens ~2.5 mm in diameter (voxel resolution 1.27 µm), and one specimen ~ 6 mm in diam. (F2870H) (voxel resolution 3.33 µm).

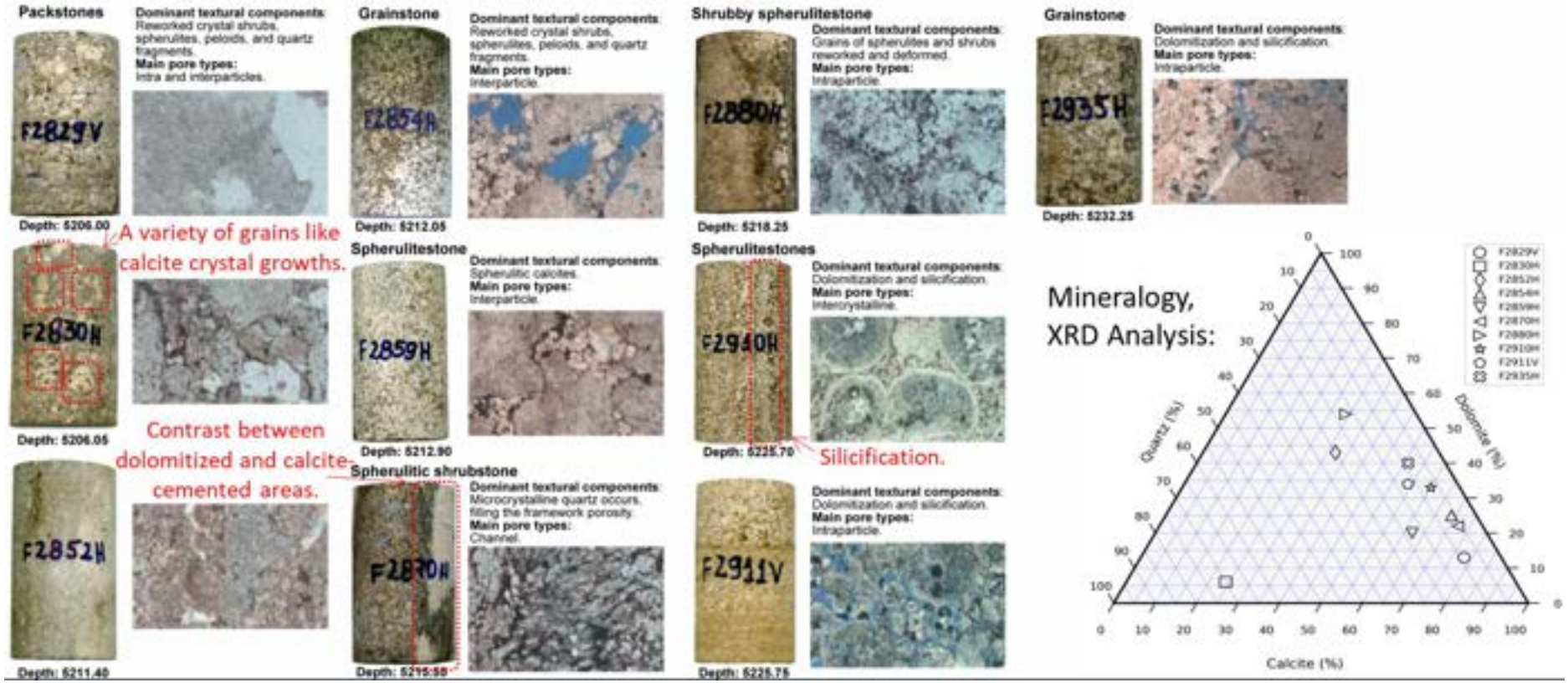


Figure 5.1.2 – Photos of the standard core samples (1.5” or ~38mm in diameter) in the data set, texture description, and laboratory mineralogy. Highlighting texture heterogeneity that may cause elastic anisotropy. Unfortunately, there is no µCT data volume of those entire standard samples (a contracted service suffered troubles in acquisitions, cancelling the service during the previous Rock Physics 1 project).

The adopted methodology for µCT image processing and pore system analysis is summarized in the steps:

1. Evaluate original data and convert to 8-bit grayscale, if necessary, to reduce file size for processing.
2. Isolate the specimen image within the full data volume for analysis, minimizing noise and enhancing the representative volume of interest.
3. Pre-processing data before segmentation: to evaluate brightness & contrast and suitable filters to reduce noise and artifacts; main gray level (brightness) adjustments: enhance contrast, flickering correction, gradient correction, Gaussian correction; primary image filters: Mean, Gaussian Blur; CT filter: Ring artifact removal.
4. Image segmentation: the main target is to perform a multithreshold (pores, solid1, solid2) by the automatic algorithm (Otsu) for pores, and solids of the matrix analysis guided by the interpreter. Additionally, consider evaluating a global multi-threshold to determine if it is suitable for identifying pores and solid phases.
5. Identify the pores and properties: pore segmentation by ellipsoid fits, aiming to compute geometric properties (area, perimeter, aspect ratio, etc.), considering a minimum of 2 pixels (voxels) for pore identification.

Steps 1 to 3 were started using the open-source software ImageJ and the licensed GeoDict 2024 software (LENEP/UENF). Steps 4 and 5 were performed using the GeoDict 2024 software. These software & applications are summarized for image processing using the specimen F2910H, with an image resolution of ~2.5 mm in diameter (MOGNO, voxel resolution 1.27 µm), as shown in Figure 5.1.3.

The identified pores and their computed geometric properties were evaluated in the results from step 5 for all specimens, examining the occurrence and pore contribution of the two available µCT resolutions, with a focus on the minimum and maximum pore areas in occurrence, as shown in Figure 5.1.4. We assumed the Anselmetti et al. (1998) classification for micropores with an area lower than 500 µm². The best image resolution (voxel resolution: 1.27 µm) is better capable of accessing microporosity, as expected.

In addition, Figure 5.1.5 displays a crossplot between the laboratory helium gas porosity from standard samples and the µCT image porosity from specimens, highlighting some limitations in pore system detection and specimen representativeness. For example, specimen F2829V showed very low image porosity and a matrix that appears to be non-representative of the standard sample. In general, we believe that a tiny portion of microporosity is accessible due to resolution limitations. However, meso-macropores and micropores can be quantified using µCT results. However, it is essential to consider their limitations.

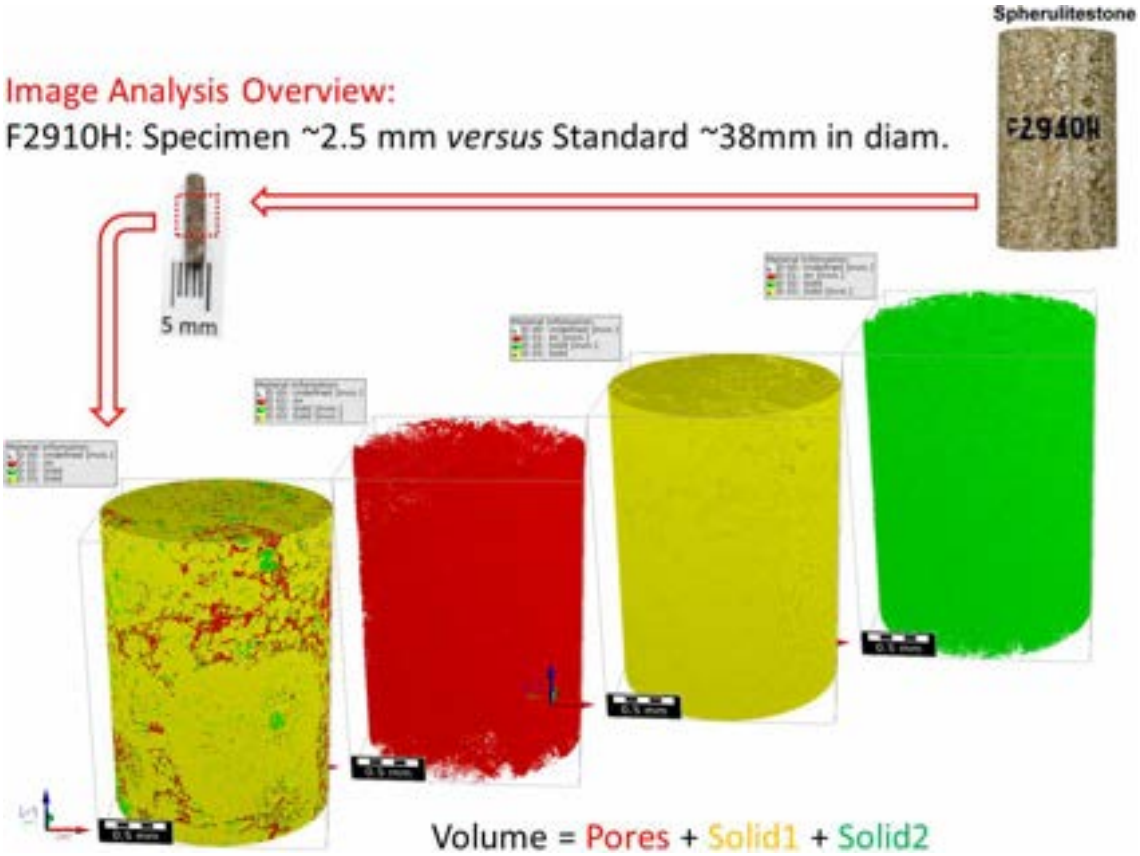


Figure 5.1.3 – Image analysis overview of the µCT data segmentation using a multi-thresholding strategy to identify the gray level phases: pore system, solid1, and solid2 (by the licensed GeoDict 2024 software). The same method was employed for all specimens in the dataset (see Figure 5.1.1).

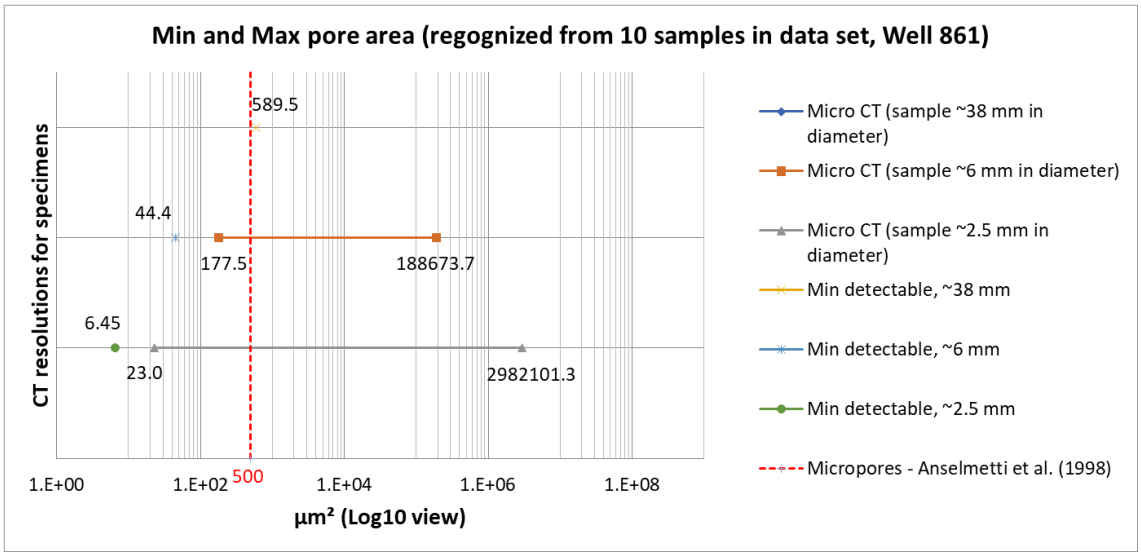


Figure 5.1.4 – Evaluating the recognized pores by the available µCT resolution, focusing on the minimum and maximum pore area in occurrence. We assumed the Anselmetti et al. (1998) classification for micropores with an area lower than 500 µm².

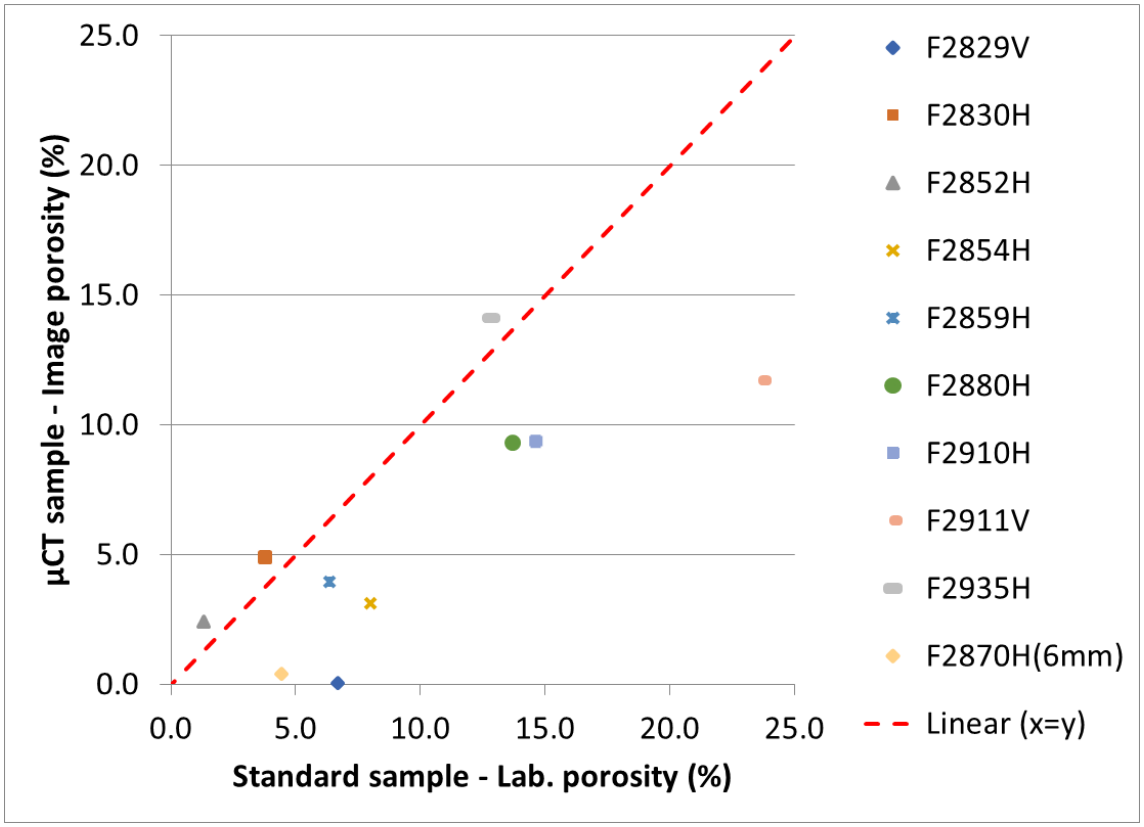


Figure 5.1.5 – Comparison between laboratory helium gas porosity from standard samples versus µCT image porosity in the results. Caution: The specimen F2829V showed a very low image porosity and matrix that appears to be non-representative of the standard sample.

The results include the image porosity contribution of all the identified pores for each specimen (fitted pore area and occurrence as shown in Figure 5.1.6), and the aspect ratio distribution in the volume (Figure 5.1.7), displaying a predominant range between 0.5 and 0.7 for most of the results, except the specimen F2829V (which is limited in representativeness).

The mean pore diameter is evaluated with respect to the pore aspect ratio intervals in Figure 5.1.8, and the results appear to be scattered for low aspect ratios (≤0.2). Additionally, the results in Figure 5.1.9 show the occurrence of minor, maximum, and mean pore diameters. The image resolution limits the identification of minor pore diameters. The maximum pore diameter result highlighted the specimen F2910H. This macropore may be related to the anisotropy (supposing that macropores occurrence may be associated with the high elastic anisotropy, whose results are retrieved from the previous Tech Report 1 – Section 5.2), comparing the maximum pore diameter computed for pore aspect ratio intervals and apparent anisotropy computed by the elastic laboratory measurements in Figure 5.1.10, denoting pore complexity resulting from textural anomalies that may cause anisotropy (see the high anisotropy of

sample F2910H), although limitations of specimen representativeness for standard samples. A low scattering of the identified pores is expected for homogeneous texture (very low anisotropy).

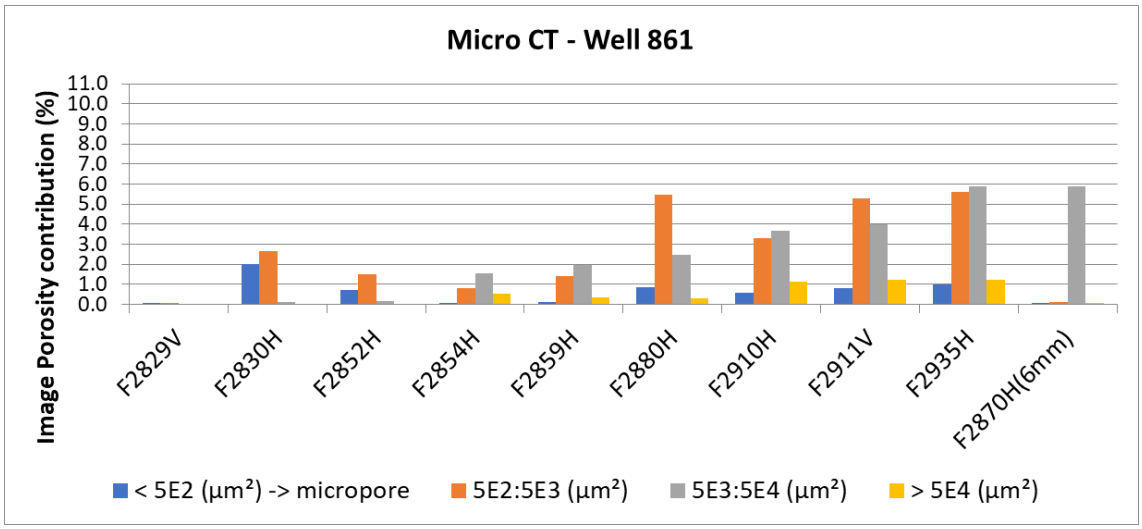


Figure 5.1.6 – Image porosity contribution of the identified pores fitted by the ellipsoids and area estimation in the µCT results. We assumed the Anselmetti et al. (1998) classification for micropores with an area lower than 500 µm².

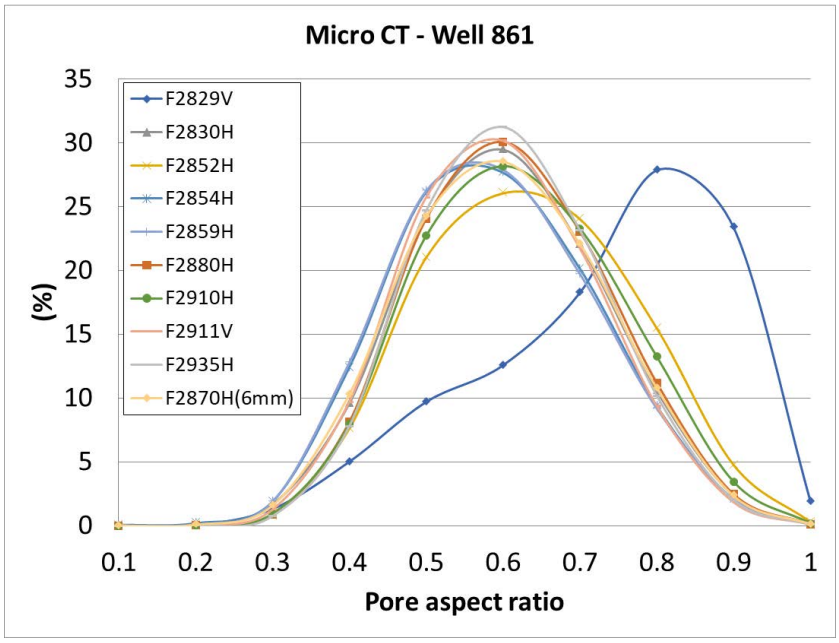


Figure 5.1.7 – Pore occurrence in the pore aspect ratio range for µCT volumes of specimens in the data set.

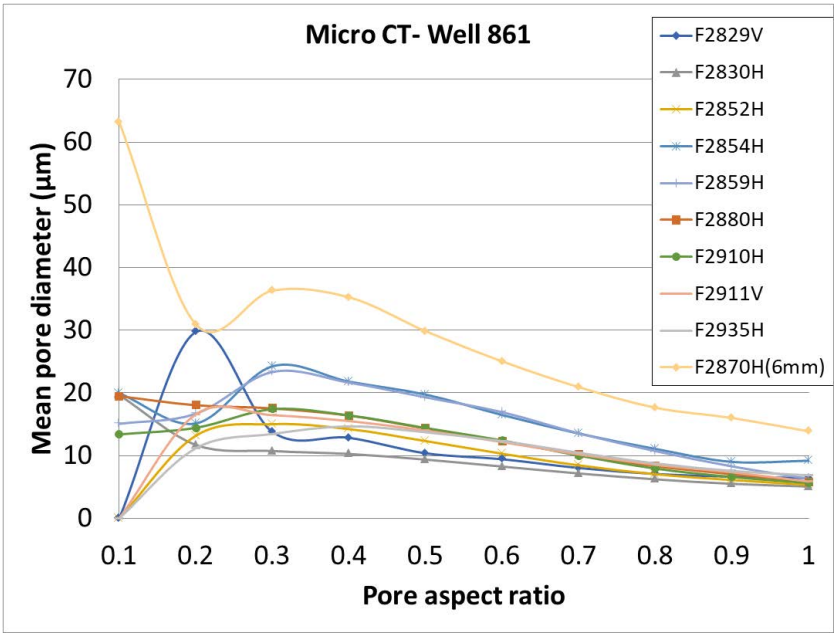


Figure 5.1.8 – Mean pore diameter computed for pore aspect ratio intervals.

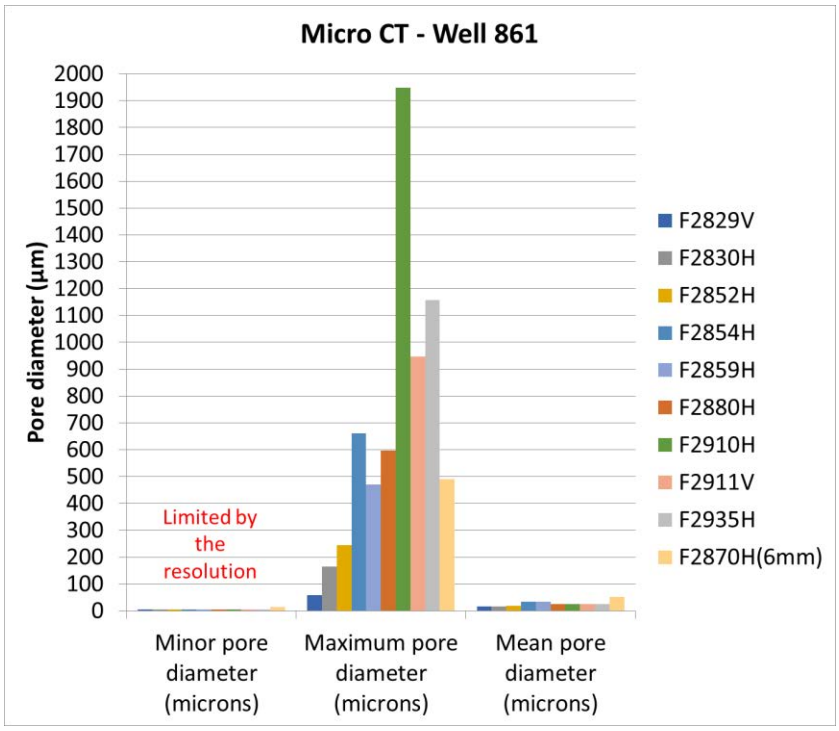
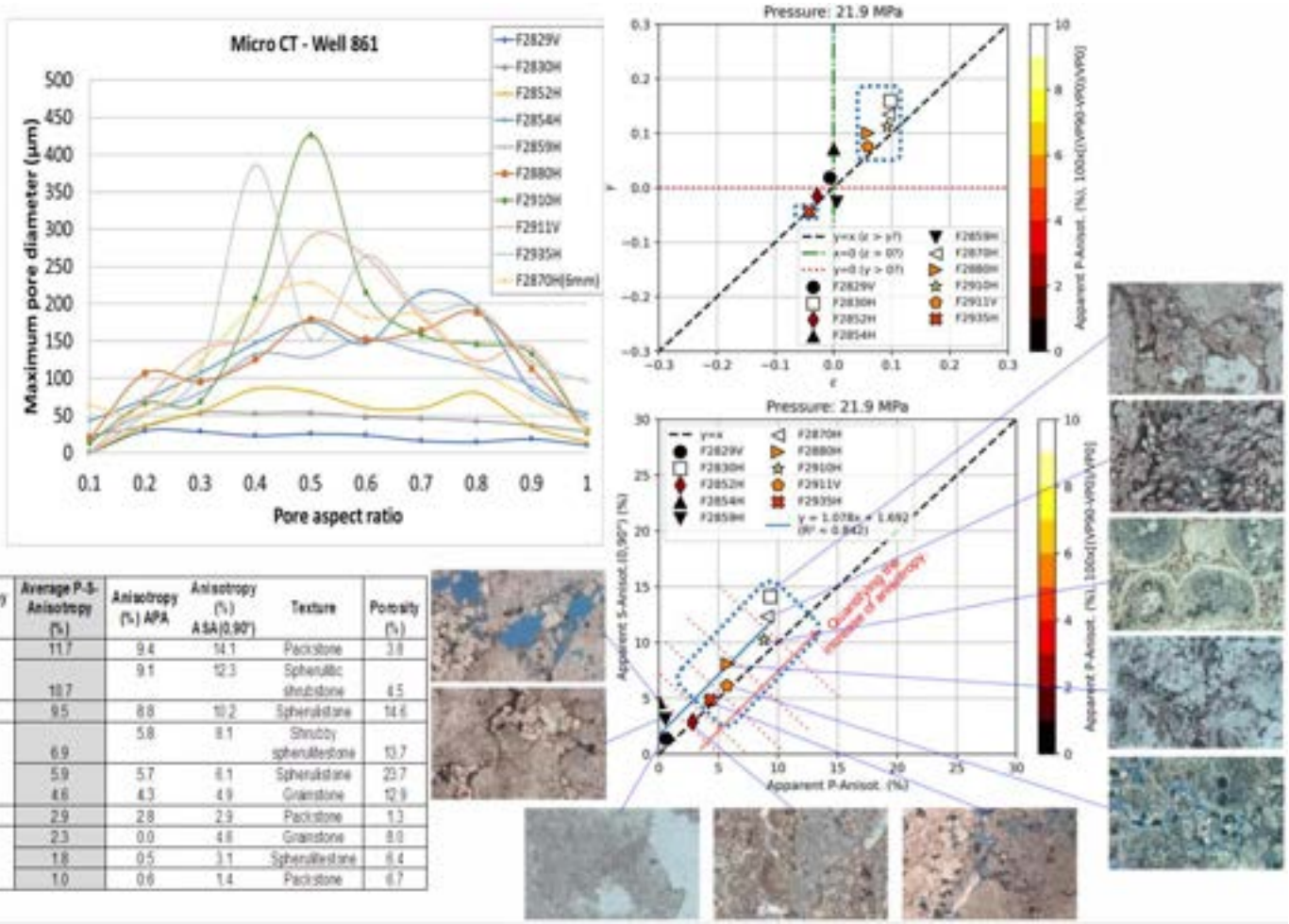


Figure 5.1.9 – Observing the minor, maximum, and mean pore diameter occurrence in the results. Unfortunately, the identified minor pore diameter is limited by the image resolution.



Sample	Anisotropy Group	Average P-S-Anisotropy (%)	Anisotropy (%) APA	Anisotropy (%) ASA(0,90°)	Texture	Porosity (%)
F2830H	I	11.7	9.4	14.1	Packstone	3.8
F2870H		10.7	9.1	12.3	Spherulitic shrubstone	4.5
F2910H	II	9.5	8.8	10.2	Spherulitstone	14.6
F2880H	III	6.9	5.8	8.1	Shrubby spherulitestone	10.7
F2911V		5.9	5.7	6.1	Spherulitstone	23.7
F2935H	IV	4.6	4.3	4.9	Grainstone	12.9
F2852H	V	2.9	2.8	2.9	Packstone	1.3
F2854H	VI	2.3	0.0	4.6	Grainstone	8.0
F2859H		1.8	0.5	3.1	Spherulitestone	6.4
F2829V		1.0	0.6	1.4	Packstone	6.7

Figure 5.1.10 – Crossplot of the maximum pore diameter computed for pore aspect ratio intervals from µCT of specimens denoting a possible heterogeneity (e.g., F2910H), and reviewing the apparent anisotropy and related results from the elastic laboratory measurements in standard samples.

5.1.3. Digital Image Analysis of the µCT Data: focusing on the mineral matrix

This study of the mineral matrix is related to step 4 in the methodology, as previously presented in Section 5.1.2, regarding image segmentation. We observed during the µCT gray 8-bit volume tests a good potential for multi-thresholding of the data, aiming to identify pores and two solid phases. Although pre-processing filters (step 3) employed in many ways as required for each specimen's volume may disrupt the gray tone band and make hard the thresholding, we decided to perform some multi-thresholding tests, and the best results are described: the "pore" phase was defined using the Otsu algorithm with low correction supervised by the interpreter, and the mineral matrix was divided into two phases named "solid 1" and "solid 2" exclusively by the interpreter.

Table 5.1.1 – Quantifying the µCT data volumes into three phases from an 8-bit gray level using the multi-thresholding interpretation. The "pore" phase was defined using the Otsu algorithm with low correction supervised by the interpreter, and the mineral matrix was divided into two phases named "solid 1" and "solid 2" exclusively by the interpreter. A darker shade of gray is expected for lower densities. It is essential to test minerals of each phase, assuming a mix of the three most important minerals identified by the laboratory XRD analysis (quartz, calcite, and dolomite).

Multi Threshold (dual) - CT specimen Well 861										
Phase	F2829V	F2830H	F2852H	F2854H	F2859H	F2880H	F2910H	F2911V	F2935H	F2870H(6mm)
Pore (%)	0.01	6.33	3.06	3.95	5.00	11.84	11.98	14.90	17.97	0.48
Solid 1 (%)	49.45	88.35	17.61	93.79	26.78	29.69	13.64	80.94	74.83	95.34
Solid 2 (%)	50.53	5.31	79.34	2.27	68.22	58.47	74.39	4.16	7.21	4.18
Sum (%)	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Test 1		Test 2	
Supposed occurrence		Supposed occurrence	
Quartz (Solid 1)	Calcite + Dolomite (Solid 2)	Quartz + Calcite (Solid 1)	Dolomite (Solid 2)

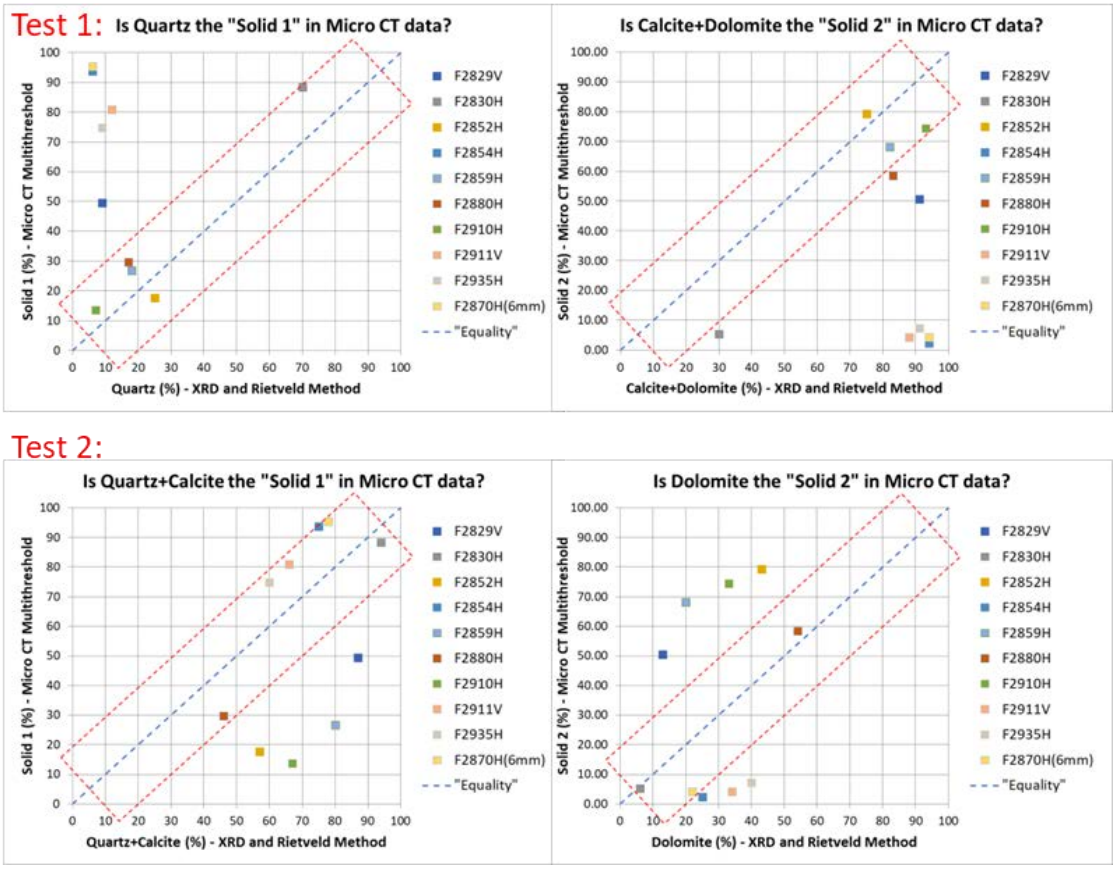


Figure 5.1.11 – Evaluating the mineral matrix divided into two phases: “solid 1” and “solid 2” from multi-thresholding tests of µCT data. The occurrence was attributed to multi-thresholding and compared with laboratory XRD results in tests 1 and 2, considering groups of quartz, calcite, and dolomite.

Thereafter, we evaluate the two mineral matrix phases, aiming to determine the predominance of mineralogy using laboratory XRD analysis results (quartz, calcite, and dolomite). A darker shade of gray is expected for lower densities, and it is essential for tests, assuming a mix of these three most important minerals. Table 5.1.1 summarizes the occurrence of each phase in the data volume, allowing us to test “solid 1” and “solid 2”, respectively, for two kinds of mineralogy groups:

- Test 1: quartz, and calcite + dolomite.
- Test 2: quartz + calcite, and dolomite.

The results of the tests are displayed in Figure 5.1.11, which shows crossplots between the multi-thresholding and laboratory XRD analysis, with an expected close match, as indicated by the ~20% difference (highlighted by the red dashed line area). Thus, the mineralogy can be successfully evaluated for each specimen by the:

Test 1: F2859H, F2852H, F2910H, F2911V.
Test 2: F2880H, F2830H, F2870H, F2935H, F2854H.
Mismatch: F2829V - it is limited in porosity and mineralogical representativeness.

5.1.4. A multiscale Rock Physics workflow for pore inclusion and next steps

The next step of this study aims to employ the workflow in Figure 5.1.12, proposed for a multiscale Rock Physics study. The main idea is to combine different scale results as input for a suitable Rock Physics modeling of the elastic properties and related P- and S-wave velocities (Vp and Vs, respectively). Thus, the input data can be originated from laboratory or well log scales, including the accounting mineral matrix, density, and pore type inclusions (macro-, meso-, and micropores, e.g., from NMR data logs). The µCT results should contribute to pore inclusion aspect ratios, except for micropores assuming a resolution limitation, which justifies a determination by fitting at minimum error, comparing the measured and estimated Vp. Many Rock Physics inclusion models (DEM, T-Matrix, etc.) could be evaluated in this way, trying to calibrate appropriate parameters for similar texture or hydraulic flow units (HFU) zones.

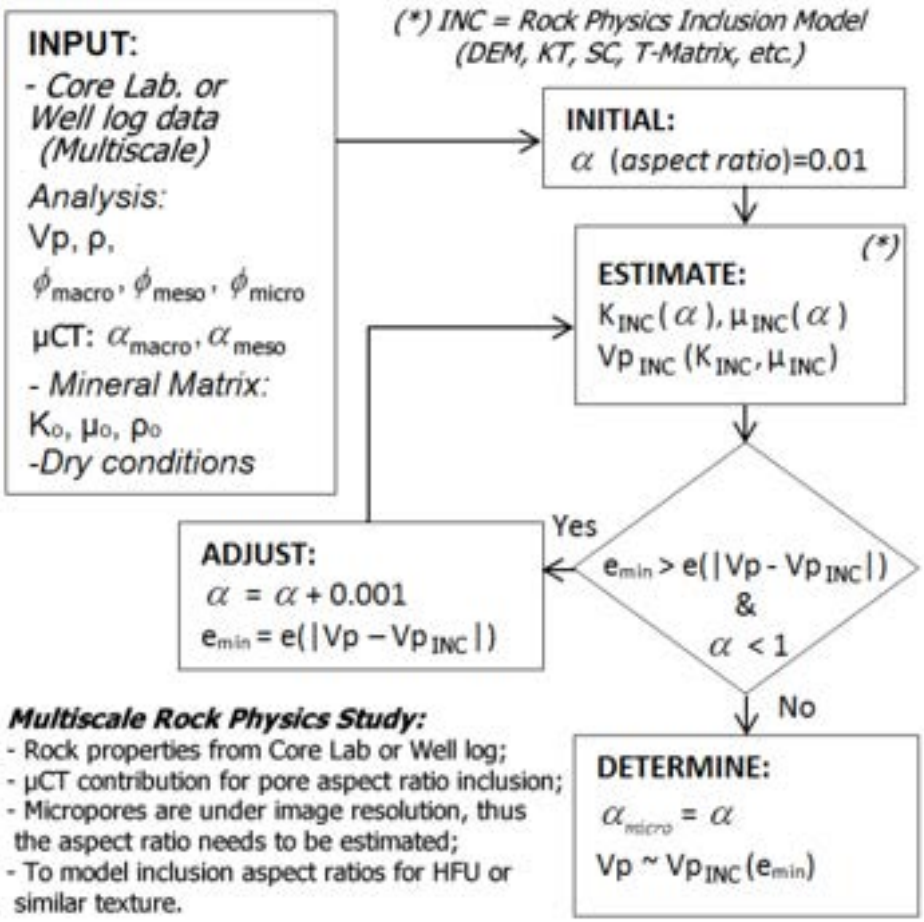


Figure 5.1.12 – A workflow for a multiscale Rock Physics study. The input data originate from laboratory or well log scales, including the accounting mineral matrix, density, and pore type inclusions (macro-, meso-, and micropores). The μCT results should contribute to pore inclusion aspect ratios, except for micropores, assuming a resolution limitation, which justifies a determination by fitting at minimum error, comparing the measured P-wave velocity with that estimated by a desired Rock Physics inclusion model (DEM, T-Matrix, etc.) for similar texture or hydraulic flow units (HFU).

We are currently working on Python scripts for the workflow in Figure 5.1.12, which predicts the setup of the desired Rock Physics inclusion model (DEM, T-Matrix, etc.). This effort is supported by the pyrockphys library, which will be utilized for future tests on relevant data sets of interest.

5.2. **Micro-CT image processing and elastic property characterization**

5.2.1 Introduction

Pre-salt layer carbonates are among the main exploration targets in Brazil. However, their microstructural complexity generates significant challenges for geophysical characterization (Vasquez et al., 2019). Elastic anisotropy, common in these rocks, directly affects seismic interpretation, especially when symmetry and medium heterogeneity deviate from ideal anisotropic behavior.

Elastic anisotropy is a critical property in carbonate reservoirs, influencing the interpretation of seismic velocity, stress distribution, and fracture behavior. In pre-salt carbonates, complex pore geometries and diagenetic alterations induce variable elastic responses, making laboratory characterization difficult (Mavko et al., 2009; Martínez & Schmitt, 2013). Digital rock physics (DRP), based on micro-computed tomography (μCT), provides a robust framework for connecting microstructural and elastic domains, allowing direct numerical simulations under controlled conditions (Vasquez et al., 2019; Lima Neto et al., 2023).

This study leverages the GeoDict software (ElastoDict module) and Python scripts to analyze carbonate samples from Well #1045 (1-BRSA-1045-SPS). The primary objectives are to:

I. Compute the effective stiffness matrix and directional P- and S-wave velocities from μCT-based digital rock models under a confining pressure of 22.1 MPa, in microtomography computer data with 12.14 μm of voxel resolution and 1.5" inches in diameter;
II. Extract the Thomsen anisotropy parameters (ε, δ, γ) to classify the magnitude of anisotropy (Thomsen, 1986);
III. Quantify deviations from elliptical anisotropy using non-ellipticity indicators (η, η′, η″), providing deeper insight into the anisotropic character (Thomsen, 1986; Alkhalifah & Tsvankin, 1995; Stovas & Ursin, 2007);
IV. Validate the applicability of transverse isotropy (VTI/HTI) symmetry models and correlate the numerical results with laboratory data.

By doing so, this work establishes a robust digital workflow to decipher the intricate elastic behavior of pre-salt carbonates, ultimately contributing to more accurate reservoir characterization.

5.2.2 Materials and Methods

5.2.2.1 Digital Rock Samples and Acquisition

Three digital carbonate samples - F7290V, F7292H, and F7293H - were obtained from μCT imaging of cores from Well #1045, with a voxel resolution of approximately 12.14 μm and a cylindrical diameter of 1.5 inches (Figure 5.2.1).

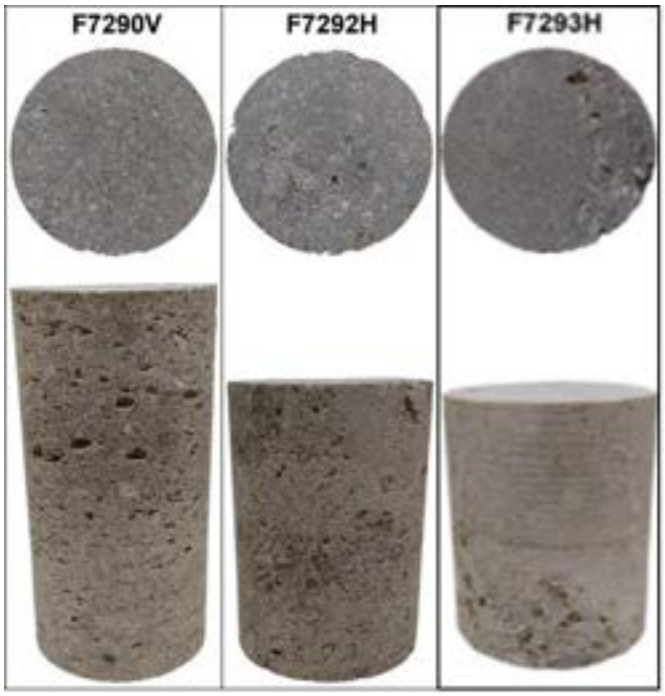


Figure 5.2.1. Views of carbonate plug samples F7290V, F7292H, and F7293H from Well #1045. The images display the top (cross-sectional) and lateral (longitudinal) views, highlighting textural and porosity contrasts among the specimens.

5.2.2.2 Computational Workflow

The workflow consisted of the following stages:

- Volume subdivision:

Each µCT dataset was divided into six subvolumes to optimize computational efficiency and reduce memory demand during simulations (Figure 5.2.2). The spatial coordinates of these subvolumes were extracted from the samples (Table 5.2.1).

Table 5.2.1. Definition of subvolumes from samples showing their spatial coordinates within the original image data.

Sample	Subvolume	X Coordinates (pixels)	Y Coordinates (pixels)	Z Coordinates (pixels)
F7292H (2200 × 2176 × 2400)	6	0 – 2200	0 – 725	0 – 1200
	5	0 – 2200	725 – 1450	0 – 1200
	4	0 – 2200	1450 – 2176	0 – 1200
	3	0 – 2200	0 – 725	1200 – 2400
	2	0 – 2200	725 – 1450	1200 – 2400
	1	0 – 2200	1450 – 2176	1200 – 2400
F7290V (2203 × 2115 × 1999)	6	0 – 2203	0 – 704	0 – 999
	5	0 – 2203	705 – 1409	0 – 999
	4	0 – 2203	1410 – 2115	0 – 999
	3	0 – 2203	0 – 704	1000 – 1999
	2	0 – 2203	705 – 1409	1000 – 1999
	1	0 – 2203	1410 – 2115	1000 – 1999
F7293H (2244 × 2160 × 2300)	6	0 – 2244	0 – 720	0 – 1150
	5	0 – 2244	720 – 1440	0 – 1150
	4	0 – 2244	1440– 2160	0 – 1150
	3	0 – 2244	0 – 720	1150 – 2300
	2	0 – 2244	720 – 1440	1150 – 2300
	1	0 – 2244	1440 – 2160	1150 – 2300

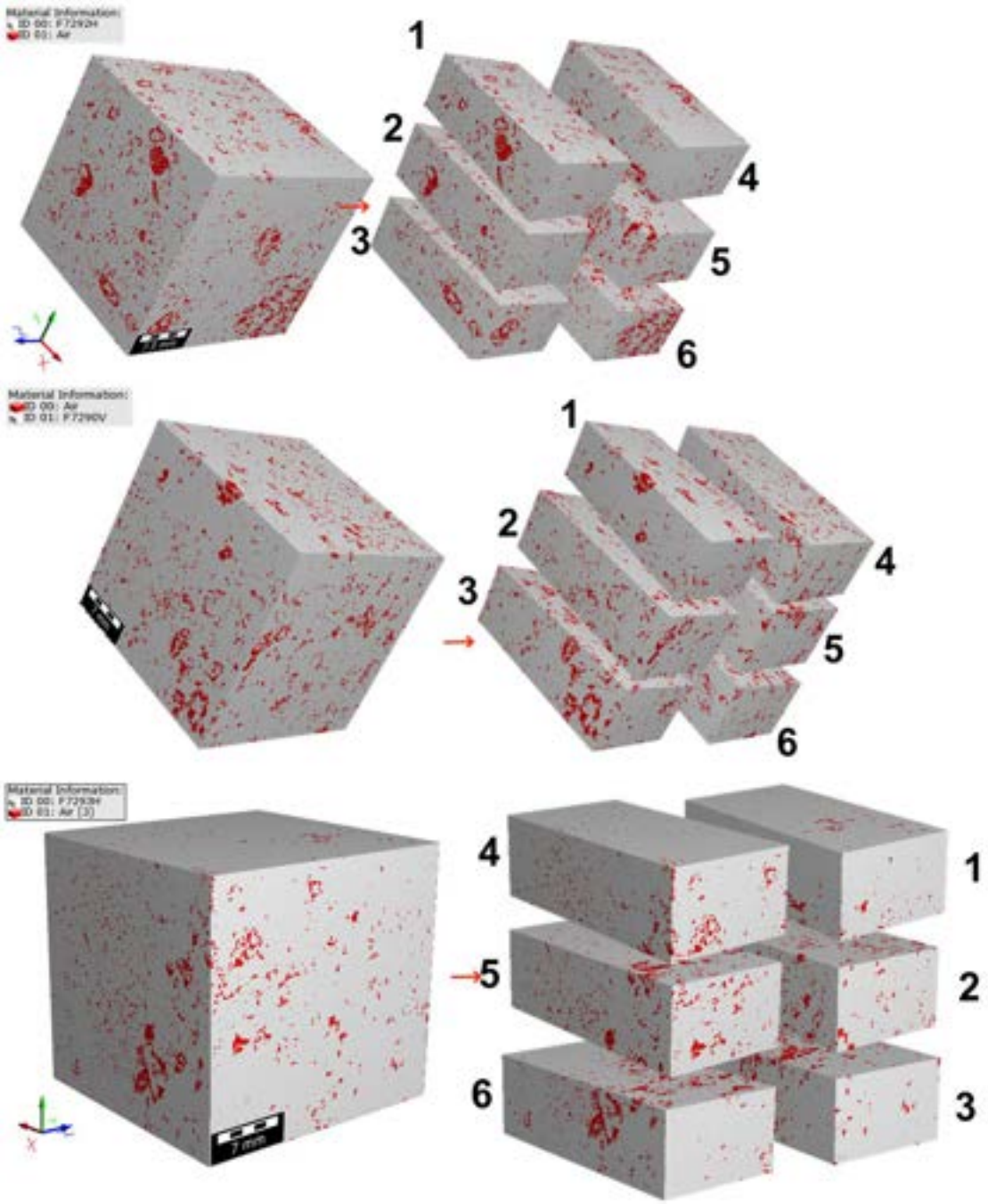


Figure 5.2.2. Three-dimensional µCT-based digital models of carbonate samples F7292H, F7290V, and F7293H from Well #1045, showing pore structures (in red) within the rock matrix (in gray). Each sample was segmented and subdivided into six subvolumes to optimize computational processing in GeoDict simulations.

- Elastic simulation in GeoDict (ElastoDict Module):

The Deformation and Effective Stiffness modules of GeoDict were employed to calculate directional stiffness matrices under dry conditions. Simulation time varied from 24 to 48 hours per subvolume.

- Stiffness Tensor Averaging:

Based on the input parameters, GeoDict solves six load cases or experiments (three compressions along the x, y, and z axes, and three shearing experiments), which are sequentially computed by the ElastoDict solver to predict the entries of the 6x6 effective elasticity tensor. Each of these six simulations is performed by applying load-case-specific displacements at the structure's boundaries and calculating the corresponding stresses (Rutka et al., 2006; Rutka & Wiegmann, 2006). Hooke's Law for the general anisotropic case is obtained by averaging the stresses over the structure and presented in Eq. 5.2.1:

$$\sigma_{ij} = \sum_{r,s=1}^{3} c_{ijrj}\varepsilon_{rs} \qquad r, s, i, j \in \{1,2,3\} \tag{5.2.1}$$

where σ is the stress tensor, c is the elasticity or the stiffness tensor, and ε is the deformation tensor.

In Voigt notation, the stresses and strains, as well as the elasticity tensor, are often expressed. In this manner, the stresses and strains are expressed as 6×1 vectors, and the elasticity tensor is reduced to a 6×6 matrix (Helnwein, 2001; Moakher, 2009). This results in a notation that is more understandable and succinct. It is advantageous to use the so-called reduced suffix notation (e.g., Chadwick et al., 2001) with the following index assignment due to certain symmetry properties, as seen in Table 5.2.1.

Table 5.2.2. Configuration of the index of the stiffness tensor according to its symmetry.

Tensor notation index ij	11	22	33	23 or 32	31 or 13	12 or 21
denoted also by	xx	yy	zz	yz	zx	xy
Voigt notation index I	1	2	3	4	5	6

Next, considering the symmetry of the elasticity tensor, have that $c_{ijkl} = c_{jikl} = c_{ijlk} = c_{klij}$, in which $i, j, k, l \epsilon \{1,2,3\}$. Thus, the stress tensor σ_{ij} can be written as σ_I in Voigt notation as follows:

$$\sigma = (\sigma_{11}\ \sigma_{22}\ \sigma_{33}\ \ \sigma_{23}\ \sigma_{31}\ \sigma_{12}\ ) \tag{5.2.2}$$

The strain tensor σ_I also can be written as ε_1 in Voigt notation, as follows:

$$\varepsilon = (\varepsilon_{11}\ \varepsilon_{22}\ \varepsilon_{33}\ \ \varepsilon_{23}\ \varepsilon_{31}\ \varepsilon_{12}\ ) \tag{5.2.3}$$

Consequently, using Voigt notation, the following coefficients can be applied to convert the symmetric fourth-order elasticity tensor c_{ijkl} into a symmetric matrix C_{IJ}. Therefore, the generalized Hooke's law can be written as a matrix-vector product, and Eq. 5.2.1 can be written as follows (Eq. 5.2.4) and in matrix form (Eq. 5.2.5).

$$\sigma_I = \sum_{j=1}^{6} C_{IJ}\varepsilon_j \qquad I = 1, \dots, 6 \tag{5.2.4}$$

$$\begin{aligned}&(\sigma_{xx}\ \sigma_{yy}\ \sigma_{zz}\ \ \sigma_{yz}\ \sigma_{xz}\ \sigma_{xy}\ ) \\ &= (C_{11}\ C_{12}\ C_{13}\ \ C_{14}\ C_{15}\ C_{16}\ \ C_{12}\ C_{22}\ C_{23}\ \ C_{24}\ C_{25}\ C_{26}\ \ C_{13}\ C_{23}\ (\end{aligned} \tag{5.2.5}$$

This elasticity or stiffness tensor represents the most general stress-strain relationship for a linear elastic anisotropic solid, such as a triclinic solid with no material symmetry (Nayfeh, 1995).

- Computation of Thomsen Parameters:

From the averaged stiffness tensors, Thomsen's anisotropy parameters (Thomsen, 1986) were derived as:

$$\varepsilon = \frac{C_{11} - C_{33}}{2C_{33}} \tag{5.2.6}$$

$$\delta = \frac{(C_{13} + C_{55})^2 - (C_{33} - C_{55})^2}{2C_{33}(C_{33} - C_{55})} \tag{5.2.7}$$

$$\gamma = \frac{C_{66} - C_{44}}{2C_{44}} \tag{5.2.8}$$

- Non-Ellipticity indicators:

According to Thomsem (1986), Alkhalifa (1995), and Stovas &Ursin (2007), Deviations from elliptic anisotropy were assessed through the parameters η,η′, and η″, providing sensitivity to complex pore and grain structures, based on the following equations:

$$\eta = \frac{\varepsilon - \delta}{1 + 2\delta} \tag{5.2.9}$$

$$\eta' = \varepsilon - \delta \tag{5.2.10}$$

$$\eta'' = \frac{\varepsilon - \delta}{(1 + 2\delta)^2}\left(1 + \frac{2\gamma^2\delta}{\gamma^2 - 1}\right) \quad (5.2.11)$$

- Angular velocity functions:

A script in Python automates deformation tensor manipulation and angular velocity computation, in vertical and horizontal directions, using the following equations:

$$V_P(\theta) \approx V_{P0}\,[1 + \delta\, sen^2\theta + \eta' sen^4\theta] \quad (5.2.12)$$

$$V_{SV}(\theta) \approx V_{S0}\left[1 + \left(\frac{V_{P0}}{V_{SV0}}\right)^2 \eta' sen^2\theta + \left(\frac{V_{P0}}{V_{SV0}}\right)^2 \eta' sen^4\theta\right] \quad (5.2.13)$$

$$V_{SH}(\theta) \approx V_{S0}\,[1 + \gamma\, sen^2\theta] \quad (5.2.14)$$

5.2.3 Results and Discussions

5.2.3.1 Stiffness and Deformation Analysis

The effective stiffness matrices derived from the GeoDict simulations revealed distinct behaviors consistent with the intrinsic heterogeneity of the carbonate matrix. For each subvolume, the directional deformation matrices confirmed anisotropic deformation responses. For F7292H, the values of C_{11} and C_{33} were close, indicating weak anisotropy, while F7290V exhibited greater contrast between directions, compatible with moderate anisotropy (Figure 5.2.3).

Matriz: Stiffness					
83.3452	26.8301	27.0183	-0.0171	0.1837	-0.0867
26.8301	83.2815	27.0513	-0.0045	0.0179	-0.0327
27.0183	27.0513	84.3237	-0.0192	0.1595	-0.0604
-0.0171	-0.0045	-0.0192	28.1518	-0.1621	-0.0502
0.1837	0.0179	0.1595	-0.1621	28.0923	-0.0784
-0.0867	-0.0326	-0.0604	-0.0502	-0.0784	27.8598

Matriz: Stiffness					
89.8315	25.4870	24.6895	0.0403	-0.5146	0.2220
25.4870	89.5749	24.6028	0.2391	-0.1595	0.2338
24.6895	24.6028	84.4273	0.2198	-0.3890	0.0424
0.0403	0.2391	0.2198	30.7649	0.0242	-0.2587
-0.5146	-0.1595	-0.3890	0.0242	30.8654	-0.0143
0.2220	0.2338	0.0424	-0.2587	-0.0143	31.9410

Figure 5.2.3. Comparison between stiffness matrices of samples F7292H (top) and F7290V (base), respectively.

To assess the directional elastic behavior of the digital carbonate samples, a comparative analysis of Deformation and Effective Stiffness simulations was performed for samples F7292H and F7290V along the X, Y, and Z axes. The simulations, carried out with the ElastoDict module of the GeoDict software, enabled quantification of displacement magnitudes and evaluation of anisotropic

responses under applied stress conditions (Figure 5.2.4). The results presented in Table 5.2.3 integrate both the maximum displacement values and the corresponding stiffness behavior, allowing for a detailed interpretation of how microstructural features influence elastic anisotropy. This approach provides a direct correlation between digital deformation patterns, stiffness variations, and the overall mechanical response of each carbonate sample.

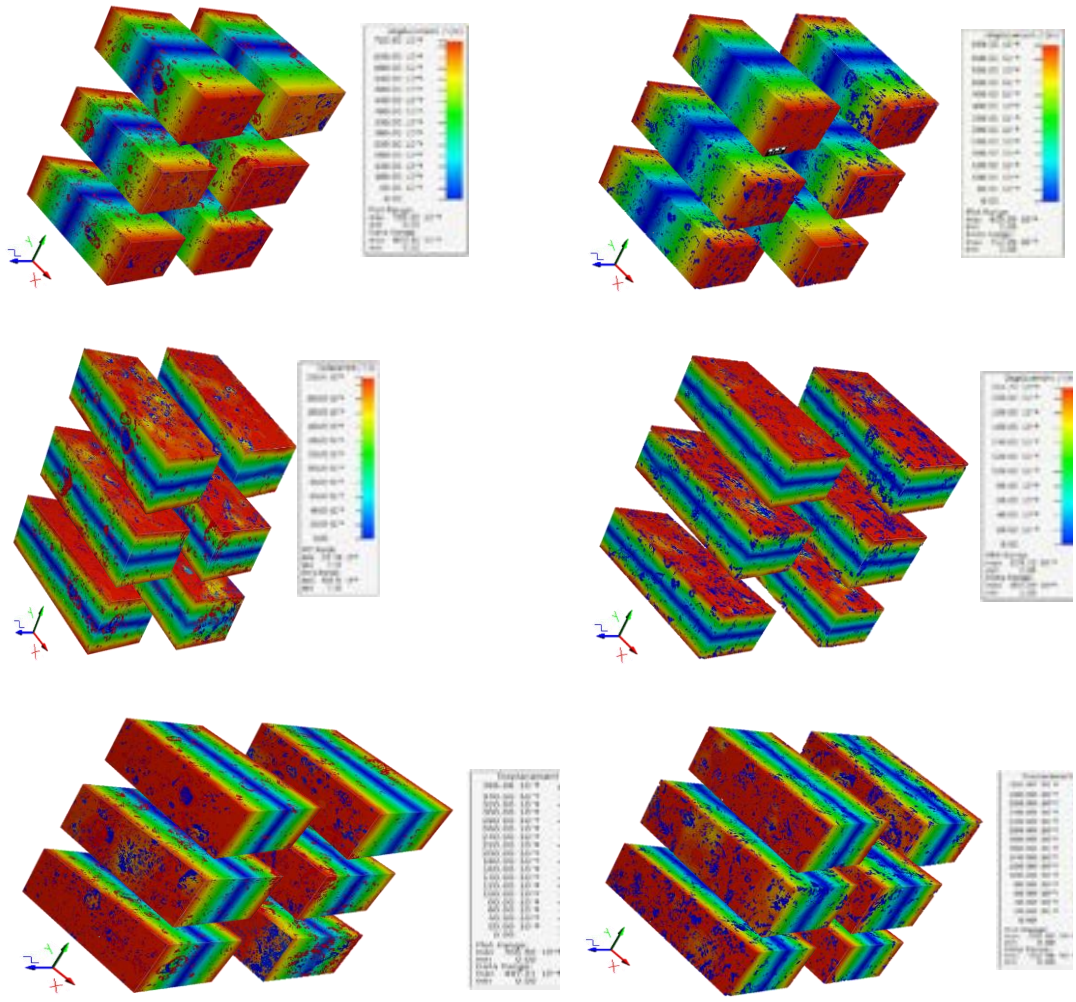


Figure 5.2.4. Comparison between the Displacement field and Deformation and Effective Stiffness for carbonate samples F7292H (left) and F7290V (right) obtained from GeoDict (ElastoDict) simulations along the X (top), Y (middle), and Z (bottom) axes. The color scale represents total displacement magnitude (in meters), with warmer tones indicating higher deformation.

Table 5.2.3. Comparison of Deformation and Effective Stiffness for Samples F7292H and F7290V.

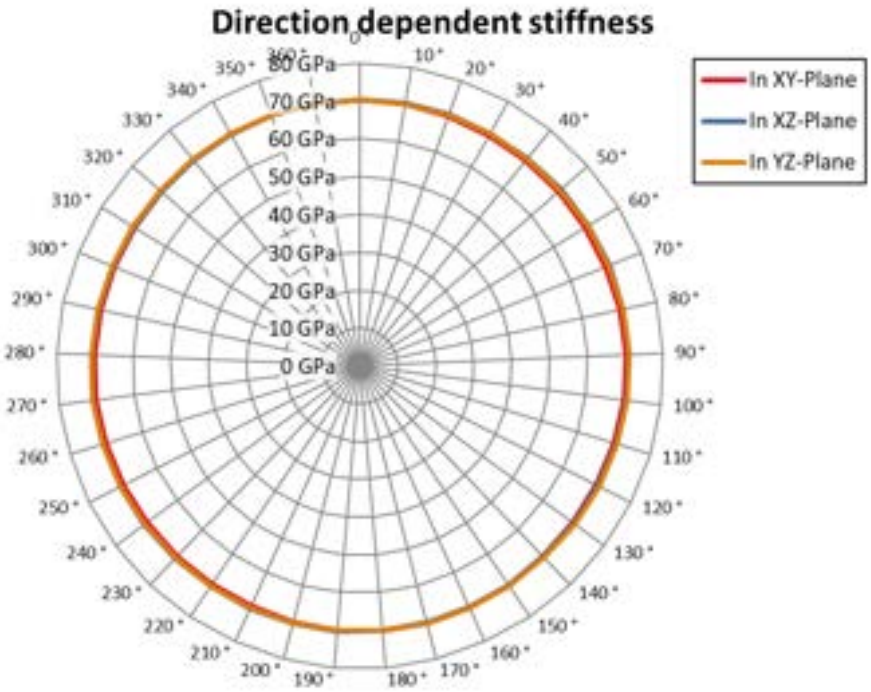
Axis	Sample	Maximum Displacement (m)	Deformation Characteristics	Effective Stiffness Response	Interpretation
X-axis	F7292H	7.26×10^{-7}	Displays a **uniform displacement field** with minor heterogeneities across subvolumes. Deformation concentrated near intergranular pores.	The **stiffness matrix shows lower anisotropy**, suggesting a more homogeneous response to horizontal stress.	Indicates **weak anisotropy**; the pore distribution causes minor heterogeneity but maintains an overall isotropic-like response.
	F7290V	6.60×10^{-7}	Exhibits **localized deformation zones** aligned with the pore network, producing uneven strain fields.	The **effective stiffness tensor reveals moderate anisotropy**, with noticeable variation in C_{11}, C_{12}, and C_{13} components.	The sample is **more anisotropic**, influenced by directional pore elongation and connected vugs along the X-axis.
Y-axis	F7292H	2.31×10^{-7}	Deformation pattern similar to the X-axis, though slightly more irregular due to microstructural alignment.	Minor reduction in stiffness values along the Y direction compared to X.	Reflects **weak transverse anisotropy**, consistent with a nearly VTI-type behavior.
	F7290V	2.15×10^{-7}	Shows **stronger asymmetry** in strain distribution, with distinct displacement gradients along structural features.	Effective stiffness significantly varies ($C_{22} < C_{11}$), confirming horizontal anisotropy.	Suggests **directional stiffness reduction**, typical of microstructural alignment and secondary porosity effects.
Z-axis	F7292H	3.67×10^{-7}	Deformation is predominantly vertical, with limited lateral expansion.	Stiffness slightly increases along the Z direction ($C_{33} > C_{11}$).	Indicates **slight vertical reinforcement**, typical of compacted carbonate textures.
	F7290V	3.03×10^{-7}	Shows **concentrated vertical strain** in	Effective stiffness tensor highlights	Confirms **weak-to-moderate VTI anisotropy**,

Axis	Sample	Maximum Displacement (m)	Deformation Characteristics	Effective Stiffness Response	Interpretation
			zones of high porosity, with visible non-linear displacement behavior.	**anisotropic behavior** ($C_{33} \gg C_{11}$, C_{22}).	consistent with experimental velocity anisotropy.

Figure 5.2.5 displays the polar diagrams of the deformation planes for samples F7292H and F7290V, illustrating the angular dependence of the elastic response obtained from GeoDict (ElastoDict) simulations. These diagrams show the magnitude of deformation as a function of propagation direction, providing a graphical representation of the anisotropic behavior of each carbonate sample.

Sample F7292H exhibits nearly circular contours at around 70 GPa pressure, indicating a weakly anisotropic response that is consistent with a more homogeneous pore distribution across the XY, XZ, and YZ planes. In contrast, sample F7290V exhibits slightly elongated lobes along specific orientations in the XZ and YZ planes, also under approximately 70 GPa of pressure. This pattern reveals directional mechanical anisotropy associated with preferential pore alignment and textural heterogeneity, while in the XY plane, the pressure measurement reached around 80 GPa.

The comparison of the two polar plots confirms that sample F7290V displays a higher degree of elastic anisotropy. These results underscore the strong correlation between digital deformation fields and the intrinsic microstructure that governs the elastic behavior of carbonate rocks.



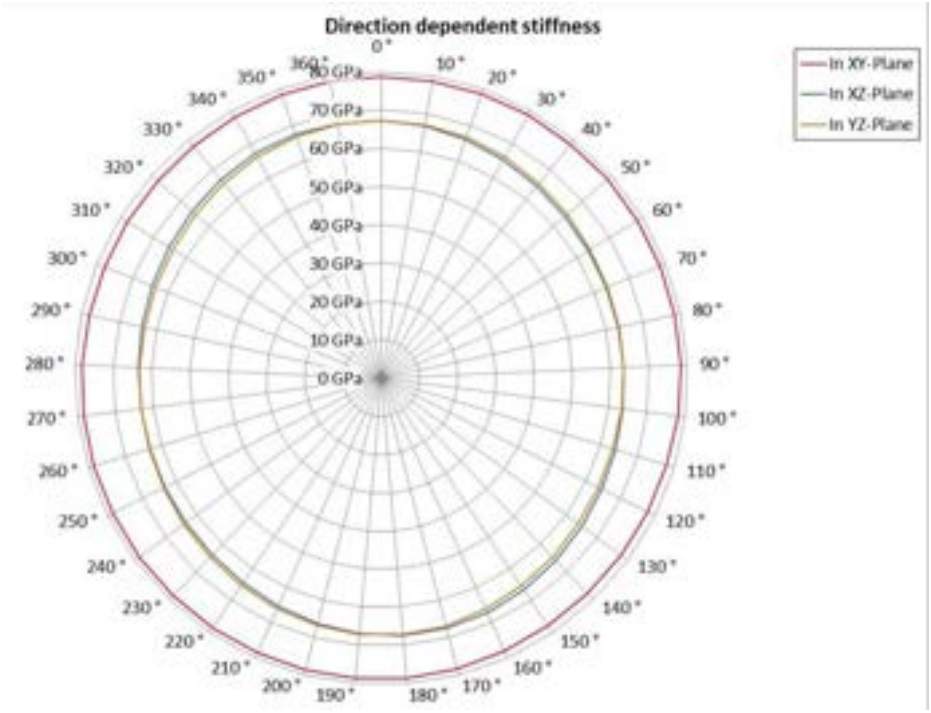


Figure 5.2.5. Polar diagrams of deformation planes for carbonate samples F7292H (top) and F7290V (base) obtained from GeoDict (ElastoDict) simulations.

The diagrams illustrate the angular variation of deformation magnitude, emphasizing the degree and orientation of elastic anisotropy.

Figure 5.2.6 shows the angular variation of compressional (V_P) and shear (V_{SV} and V_{SH}) wave velocities for samples F7292H and F7290V, which were calculated using the stiffness tensors derived from the digital simulations. The results, shown in Figures 6a and 6b, demonstrate how wave velocities change with propagation angle (θ) relative to the symmetry axis, allowing the characterization of the degree and nature of elastic anisotropy.

For sample F7292H (top), the compressional velocity (V_P) decreases slightly from 6,060 km/s at 0° to 6,025 km/s at 90°, indicating a weak P-wave anisotropy. The vertical shear velocity (V_{SV}) increases modestly from 3,500 km/s to 3,625 km/s, while the horizontal shear velocity (V_{SH}) remains nearly constant, ranging from 3,500 km/s to 3,482 km/s. These small variations reflect an almost isotropic elastic behavior, confirming that F7292H is a weakly anisotropic carbonate with minor structural directionality. In contrast, sample F7290V exhibits more pronounced velocity variations (base). The P-wave velocity (V_P) increases from 5,950 km/s at 0° to 6,171 km/s at 90°, while V_{SV} increases from 3,590 km/s to 3,798 km/s and V_{SH} from 3,590 km/s to 3,660 km/s. These changes, although moderate, indicate a weak-to-moderate vertical transverse isotropy (VTI), consistent with microstructural anisotropy observed in the μCT-based simulations.

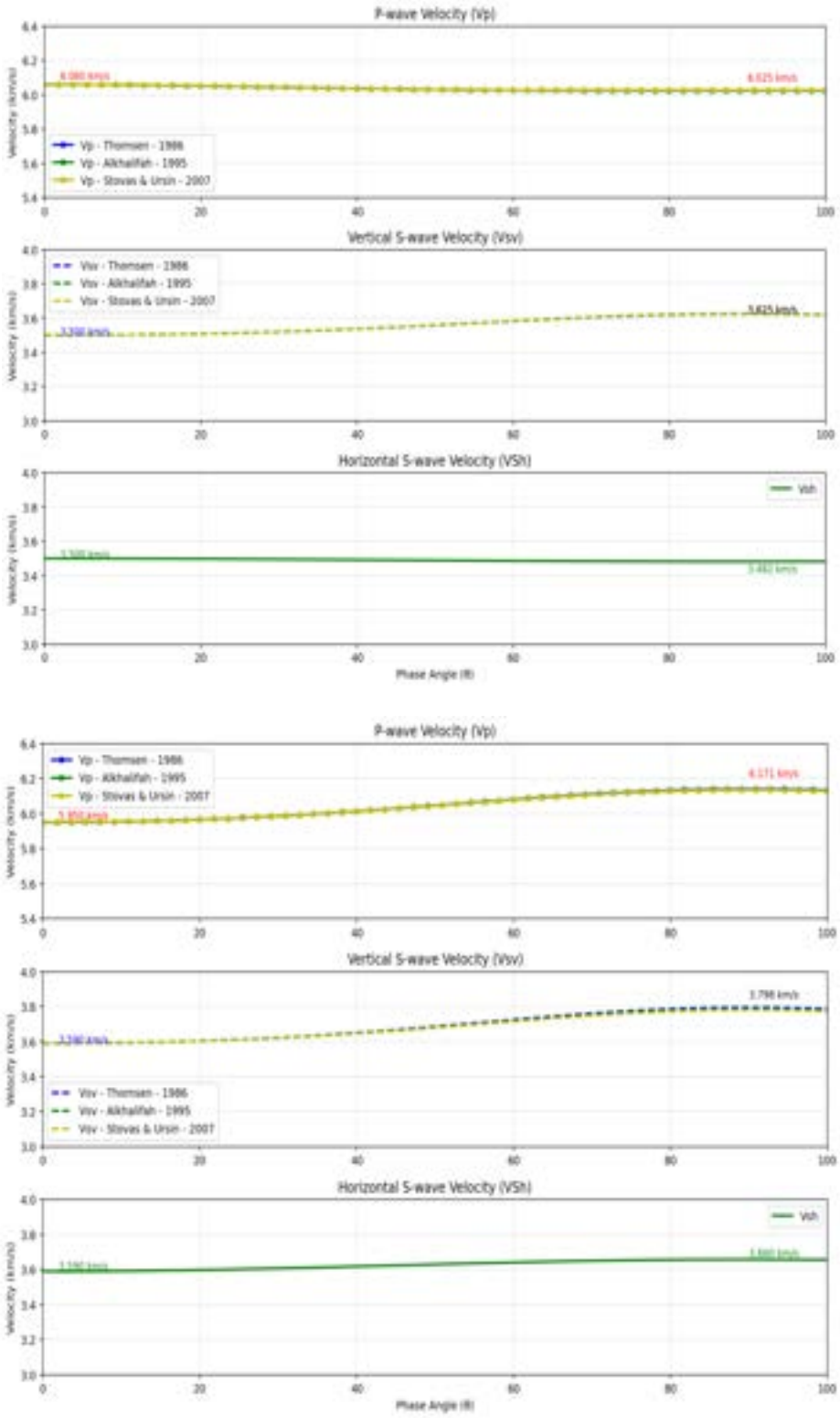


Figure 5.2.6. Angular variation of P-wave (V_P), vertical shear wave (V_{SV}), and horizontal shear wave (V_{SH}) velocities for carbonate samples F7292H (top) and F7290V (base). Theoretical trends are shown according to the formulations of Thomsen (1986), Alkhalifah (1995), and Stovas & Ursin (2007).

The comparison between laboratory-measured and numerically simulated velocities for samples F7292H and F7290V (Figure 5.2.7) demonstrates the strong consistency between experimental ultrasonic data and digital modeling results derived from the GeoDict (ElastoDict) simulations. The scatterplot compares compressional (V_P) and shear (V_{SV} and V_{SH}) wave velocities obtained in both conditions, with the red dashed line (y = x) representing perfect agreement.

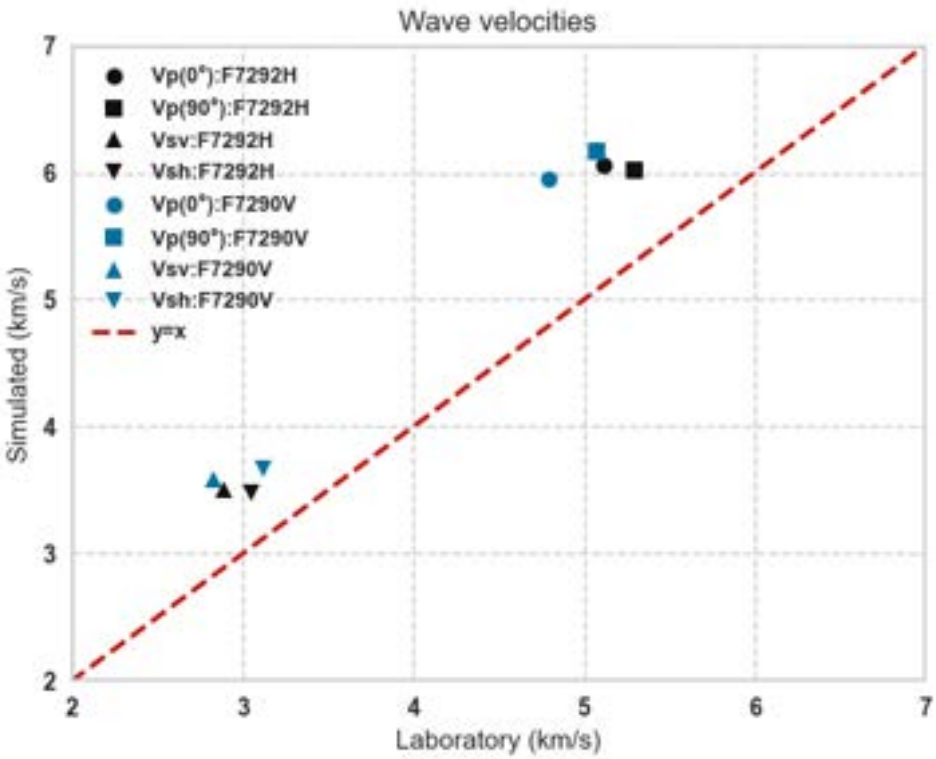


Figure 5.2.7. Comparison between laboratory-measured and simulated wave velocities for carbonate samples F7292H and F7290V. The red dashed line (y = x) indicates perfect correspondence between experimental and numerical data. The proximity of most data points to this line confirms the high reliability of the digital simulation results in reproducing real elastic properties and anisotropic trends.

For sample F7292H, both the P-wave and S-wave velocities are closely aligned with the reference line, indicating an excellent match between the simulated and measured data. The digital results reproduce the laboratory values with only minor deviations, confirming that the microstructural modeling accurately captures the elastic response of this weakly anisotropic carbonate.

In contrast, sample F7290V shows slightly higher simulated velocities compared to the laboratory data, particularly in the P-wave component. The simulated P-wave velocities range from 5,950 to 6,171 km/s, which is marginally above the experimental range of approximately 4,797 to 5,070 km/s. However, the S-wave

velocities, ranging from 3,590 to 3,660 km/s, remain in close agreement with the measured values. This slight overestimation in simulation may be attributed to the assumption of dry conditions in the numerical model, as well as potential differences in boundary constraints or micro-CT segmentation thresholds, which can increase the calculated stiffness.

The results indicate that the numerical simulations successfully reproduce the elastic anisotropy and magnitude of wave velocities observed experimentally, validating the use of micro-CT–based digital rock physics for estimating directional mechanical properties. The strong correlation between measured and simulated velocities confirms that the derived stiffness tensors are representative of the true physical response of the carbonate samples.

Table 5.2.4 shows that an analysis of a consistent and significant pattern emerges where the simulated results overestimate the actual laboratory measurements across every single data point. For sample F7292H, the compressional wave velocities are over-predicted by nearly one kilometer per second, resulting in relative errors of 18.45% and 13.79%, while its shear wave errors range from 14.05% to 21.28%. The overestimation is even more pronounced for sample F7290V, which shows the highest discrepancies in the set; its compressional wave simulations have relative errors exceeding 24% and 21%, and its vertically polarized shear wave shows a substantial 26.90% error. The data clearly indicate that the simulation model systematically generates higher velocities than were physically measured in the lab, with the performance varying based on the specific sample and the type of wave.

Table 5.2.4. Comparison between laboratory-measured and simulated wave velocities for samples F7292H and F7290V.

Sample	**Wave Type**	**Laboratory (Vlab) [km/s]**	**Simulated (Vsim) [km/s]**	**Absolute Error [km/s]**	**Relative error (%)**
F7292H	VP0°	5.116	6.060	0.944	18.45
	VP90°	5.295	6.025	0.730	13.79
	VSH	3.053	3.482	0.429	14.05
	VSV	2.886	3.500	0.614	21.28

F7290V	VP0°	4.797	5.950	1.153	24.04
	VP90°	5.070	6.171	1.100	21.70
	VSH	3.117	3.660	0.543	17.42
	VSV	2.829	3.590	0.761	26.90

Figure 5.2.8 shows a diagram illustrating the relationship between the Thomsen parameters ε (epsilon) and γ (gamma) for the analyzed carbonate samples under a confining pressure of 22.1 MPa, as measured in the laboratory, as described in other Reports of Rockphys 1. The dashed black line (y = x) indicates the condition where ε equals γ, representing isotropic or near-isotropic elastic behavior. The vertical green line (x = 0) and horizontal red line (y = 0) serve as reference axes, dividing the diagram into quadrants that classify the relative strength and direction of anisotropy.

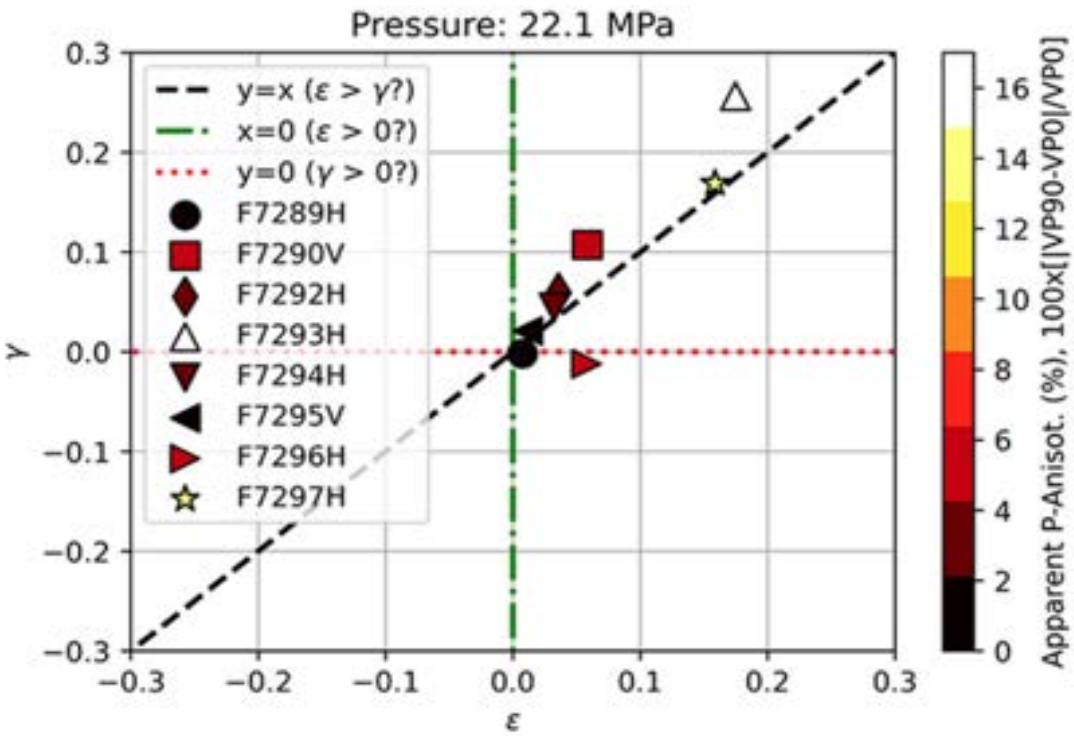


Figure 5.2.8. Cross-plot of ε vs. γ highlighting the anisotropy domain of the samples and the relationship between Thomsen parameters (ε, δ, γ) for samples F7292H and F7290V under a confining pressure of 22.1 Mpa (Lima Neto et al., 2023).

Samples F7292H and F7290V are highlighted, showing distinct anisotropic characteristics. F7292H, positioned near the origin, presents very small ε and γ values, confirming its weak anisotropy. In contrast, F7290V lies farther from the origin, in the positive ε–γ quadrant, indicating weak-to-moderate anisotropy and a slightly stronger elastic response difference between vertical and horizontal directions. The color scale represents the apparent P-wave anisotropy percentage ($100 \times |Vp_{90} - Vp_0| / Vp_0$), where both samples exhibit low to intermediate values (<16%), consistent with the observed degree of anisotropy, according to Lima Neto et al. (2023). The anisotropy of the S-wave is more than P-wave in sample F7290H (γ >ε).

Table 5.2.5 summarizes the quantification of the Thomsen parameters (ε, δ, γ) and the non-ellipticity indicators (η, η′, η″) derived from stiffness tensor analysis. For F7292H, the parameters ε = –0.0058, δ = –0.0118, and γ = –0.0052 confirm a nearly isotropic elastic behavior with negligible directional dependence. The corresponding non-ellipticity values (η ≈ 0.006) suggest minimal deviation from elliptic anisotropy. Conversely, the sample F7290V exhibits higher magnitudes of anisotropy parameters (ε = 0.0320, δ = 0.0216, γ = 0.0191), consistent with its weak-to-moderate anisotropy classification. The slightly elevated η, η′, and η″ values (~0.010) indicate small but measurable non-elliptic behavior, typically associated with more complex pore geometries and textural alignment. These results agree with previous findings for pre-salt carbonate reservoirs (Vasquez et al., 2019), confirming that anisotropy is generally weak to moderate in such lithologies, yet sufficient to influence seismic and elastic responses.

Table 5.2.5. Summary table of computed parameters and anisotropy classification, in relationship between non-ellipticity indicators (η, η′, η″) for samples F7292H and F7290V.

Parameter	F7292H	F7290V
Epsilon (ε)	-0.005802	0.032005
Delta (δ)	-0.011775	0.021581
Gamma (γ)	-0.005186	0.019114
Eta (η)	0.006117	0.009993
Eta' (η')	0.005973	0.010424
Eta'' (η'')	0.006098	0.009670

Anisotropy classification	Weak	Weak to Moderate

Figure 5.2.9 displays the error analysis for the predicted anisotropy parameters, Epsilon (ε) and Gamma (γ). This analysis reveals a consistent model bias for both carbonate samples, F7292H and F7290V. The error distributions, visualized through frequency histograms and boxplots, demonstrate a systematic negative shift, indicating a slight but consistent underestimation of both parameters.

For sample F7292H, the mean error for Epsilon is -0.0416, while the underestimation is more pronounced for Gamma, with a mean error of -0.0649. In contrast, sample F7290V shows smaller systematic errors, with mean values of -0.0262 for Epsilon and -0.0237 for Gamma. The boxplots further illustrate that the variability and central tendency of errors differ between the two parameters within each sample. Specifically, F7292H exhibits a greater discrepancy between the errors for Epsilon and Gamma compared to F7290V.

This pattern of small, negative errors contrasts sharply with the previously observed significant overestimation of wave velocities. This suggests that the model's representation of the relative anisotropic rock fabric is more accurate than its prediction of the absolute elastic moduli influencing wave speed, as indicated in Figure 5.2.9, which outlines the calculations for Thomsen's parameters for each subvolume of the samples.

Figure 5.2.9 shows the error distribution and summary statistics for predicted anisotropy parameters Epsilon (ε) and Gamma (γ) in carbonate samples F7292H and F7290V. The histograms show the frequency of error values, and the boxplots illustrate the data spread; all indicate a consistent slight underestimation by the model.

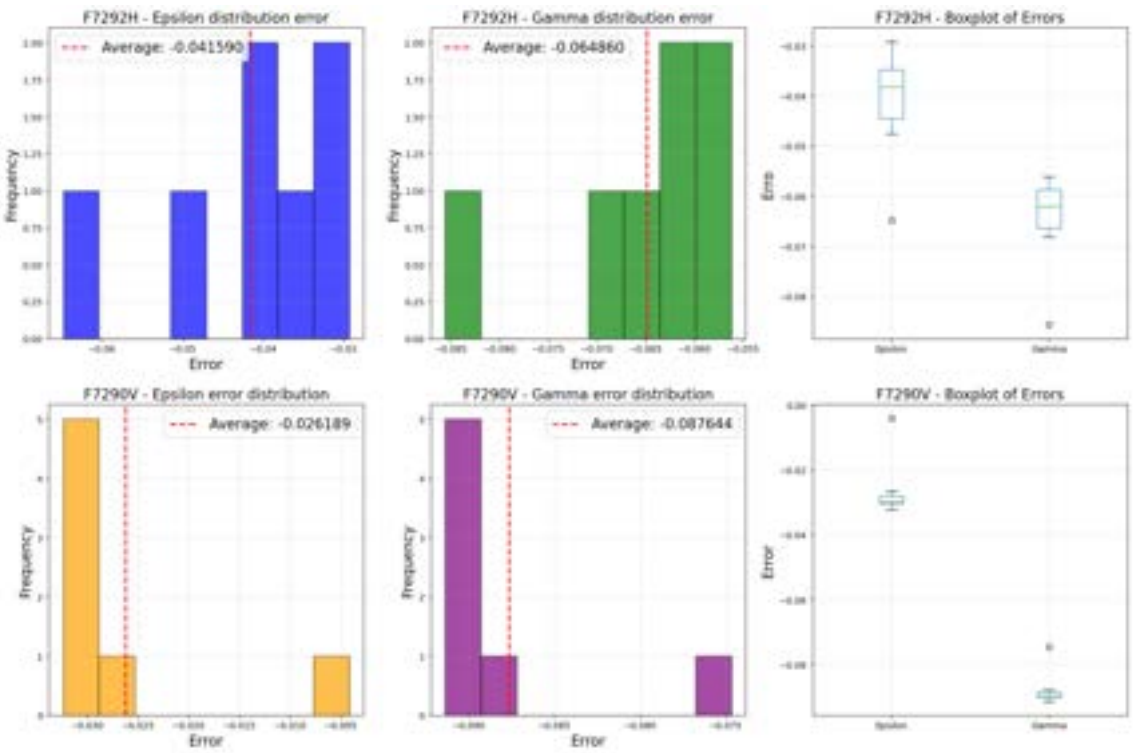


Figure 5.2.9. Statistical analysis of prediction errors for Thomsen anisotropy parameters. Panels show the error distributions for Epsilon (ε) and Gamma (γ) in samples F7292H and F7290V, with mean error values and boxplots summarizing the data variability.

Figure 5.2.10 displays the plot of the fundamental Thomsen parameters for P-wave anisotropy (ε) and S-wave anisotropy (γ). Data points representing all subvolumes of both samples are plotted, revealing a strong positive correlation where ε and γ increase concurrently for the F7292H sample. The two samples, F7292H and F7290V, are differentiated by color and marker, facilitating a comparison between them. The groups of points labeled "F7290V_total" and "F7292H_total" represent the overall average values for each complete sample.

This plot highlights a key finding: it relates the anisotropy to the non-ellipticity parameter (η), which quantifies the deviation from an elliptical anisotropic model. The data points demonstrate a direct correlation, indicating that as the values of the Thomsen parameters (a specific function or combination of ε and γ) increase, so too does the non-ellipticity (η). This confirms that the anisotropic behavior of these carbonate samples cannot be fully described by an elliptical model; instead, it becomes more pronounced as the samples display weak to strong anisotropies.

The central finding is that the complex pore structures within the subvolumes generate anisotropic properties that systematically deviate from simple elliptical anisotropy. The degree of this non-elliptic character can be predicted directly from the conventional Thomsen parameters, establishing a quantitative link between basic anisotropic measurements and more complex wave propagation behavior.

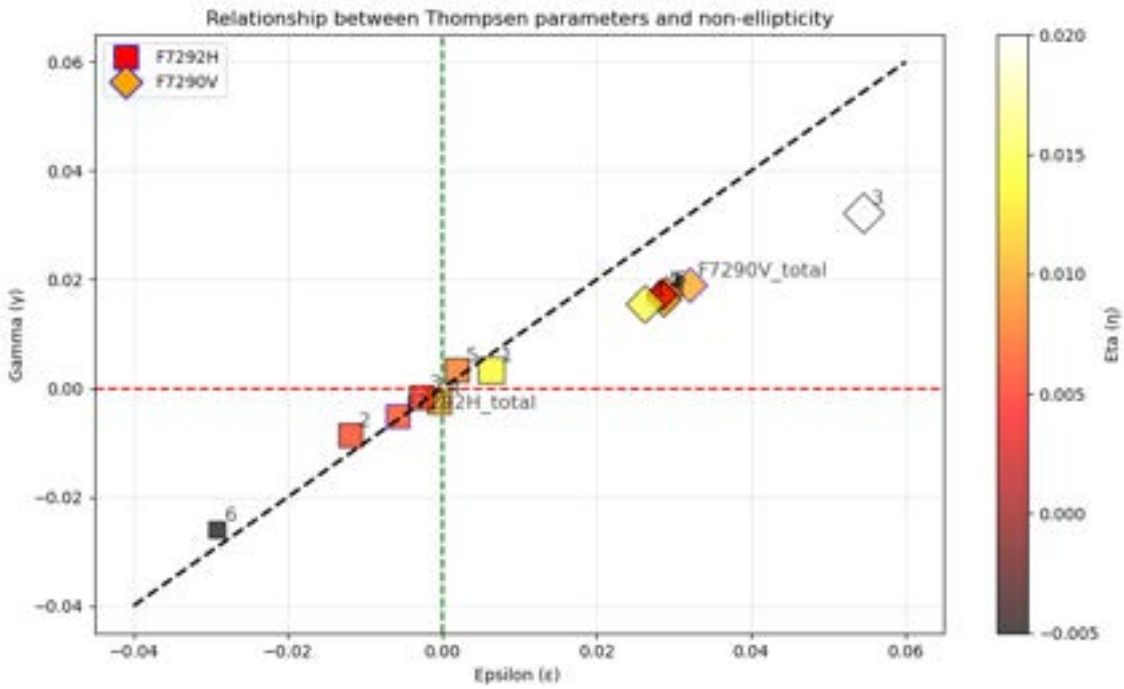


Figure 5.2.10. Cross-plot analysis of anisotropic properties. Thomsen parameters Epsilon (ε) versus Gamma (γ) for all subvolumes, showing their interrelationship. The resulting non-ellipticity parameter (η) illustrates the deviation from elliptical anisotropy as a function of the anisotropic strength.

5.2.4 Conclusions

The numerical simulations effectively computed the effective stiffness matrix and the resultant wave velocities. While these simulations did not achieve a perfect match with the absolute laboratory values of Vp and Vs, the models exhibited strong predictive capabilities by accurately replicating the relative anisotropic trends and directional velocity patterns observed in the laboratory data.

Analysis of the Thomsen parameters (ε, δ, γ) confirmed the anisotropic nature of the samples, classifying F7292H as weakly anisotropic and F7290V as weak to moderately anisotropic, consistent with laboratory classifications. Importantly, the investigation of non-ellipticity parameters (η, η′, η″) provided critical insights beyond conventional anisotropy, revealing that the more anisotropic sample (F7290V) also demonstrated more pronounced non-elliptical behavior. This finding supports the applicability of specific symmetry models (VTI/HTI) for these carbonates and highlights that anisotropy in these rocks is not strictly elliptical.

The integration of micro-CT data was essential for this advancement, as it supplied the detailed three-dimensional structural information necessary to quantify non-ellipticity. This established a direct and quantitative link between the complex microstructure of the rock and the resulting macroscopic elastic anisotropy, a relationship that cannot be derived from laboratory wave velocities alone.

Despite the discrepancies in absolute velocity values, the digital rock physics approach demonstrated here proves to be a robust and reliable tool for comparative analysis, offering unparalleled insights into the intrinsic anisotropic and non-elliptical behavior of carbonate rocks based on their true pore-scale architecture.

5.2.5 Current activities and next steps

The sample F7293H illustrates the consistent application of the digital rock physics methodology established for samples F7292H and F7290V. The results confirm the successful calculation of a complete set of parameters, including the effective stiffness matrix (see Figure 5.2.11), Thomsen anisotropy parameters, and non-ellipticity measurements. This is visually supported by polar diagrams representing the deformation planes and detailed displacement fields along the X, Y, and Z axes, respectively (see Figure 5.2.12 in top, middle, and base). These illustrations effectively demonstrate the anisotropic mechanical response of the rock microstructure under stress.

A key observation from this figure is the variation in the magnitude of the maximum displacement as a function of the different loading directions, a direct manifestation of the anisotropic elasticity of the material. It is important to note, however, that the analysis for this sample remains incomplete. Processing of subvolume 1 is ongoing, as indicated by its absence, due to the exceptionally high computational cost associated with numerical simulations. Consequently, although the methodology is validated, the results presented for F7293H should be considered preliminary until data from all subvolumes are integrated.

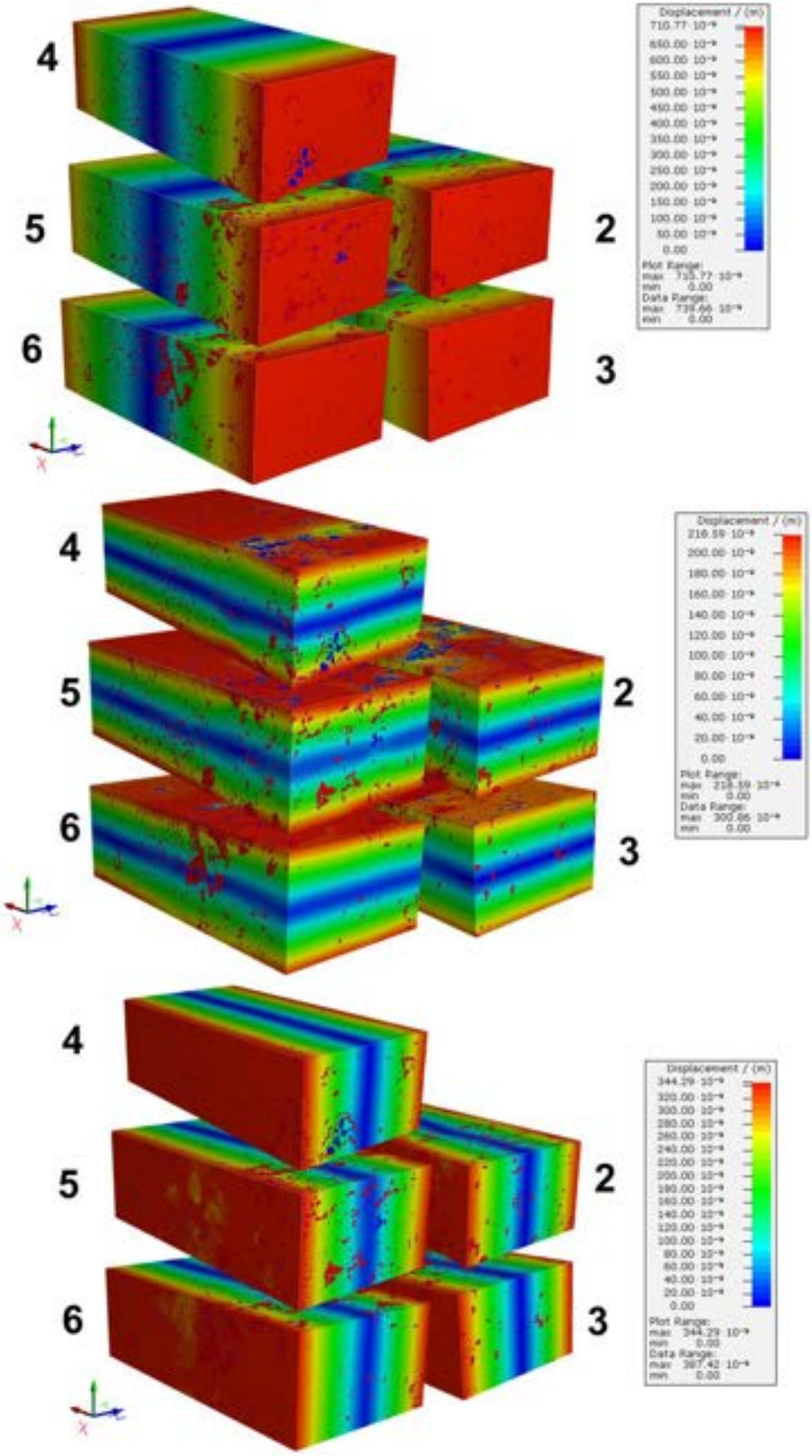


Figure 5.2.11. Numerical simulation results for sample F7293H showing directional displacement fields (preliminary, missing subvolume 1).

The polar diagrams of the stiffness surfaces for the subvolumes of Sample F7293H (excluding subvolume 1, which is unavailable) illustrate the angular behavior of the elastic response obtained from Digital Deformation (ElastoDict) simulations performed in GeoDict software. These diagrams show the magnitude of directional stiffness (in GPa) as a function of the propagation direction in three principal planes: XY, XZ, and YZ. This provides a graphical representation of the anisotropic behavior of each subvolume of the carbonate rock.

A visual analysis of the polar diagrams for subvolumes 2, 3, 4, 5, and 6 of Sample F7293H reveals a consistent pattern. Near-circular contours are observed in subvolumes 4, 5, and 6, and the directional stiffness curves in the XY, XZ, and YZ planes are similar. The proximity of these curves to a perfectly circular shape indicates a weakly anisotropic or nearly isotropic response. This characteristic is associated with a more homogeneous pore distribution and a less oriented microstructure within these subvolumes. The average stiffness for subvolumes 4, 5, and 6 is approximately 80 GPa, with subvolume 5 exhibiting slightly higher stiffness while maintaining the same pattern of weak directional anisotropy.

In contrast, subvolumes 2 and 3 display a different behavior, as their contours exhibit slightly elongated lobes in the three planes (XY, XZ, and YZ), indicating a more pronounced directional anisotropy. This is consistent with a more heterogeneous pore distribution and associated preferential pore alignment and textural heterogeneity. The elongation suggests that stiffness varies more significantly with the direction of propagation, being greater in certain orientations and less in others.

Figure 5.2.12 presents a comparative analysis of the non-ellipticity parameters (η, η', and η'') for the carbonate samples F7292H, F7290V, and F7293H, including both the total samples and their respective subvolumes. These parameters, derived from the stiffness tensor analysis, quantify the deviations from elliptic anisotropy and provide information on the degree of complexity of the elastic wavefront of each sample.

In the left section of the graph, the total values for F7292H and the six subvolumes show consistently low magnitudes for η, η', and η'' (generally between 0.004 and 0.012). This behavior reflects weak non-ellipticity, indicating that F7292H behaves similarly to a weakly anisotropic medium with minimal directional variation in wave propagation.

Toward the center of the figure, the results for F7290V demonstrate slightly higher non-ellipticity values in both the total sample and most subvolumes (typically between 0.008 and 0.020). This increase is highlighted by the downward red arrow and indicates a more pronounced deviation from elliptic anisotropy compared to F7292H. The greater dispersion among the subvolumes also reveals greater internal heterogeneity.

In the right portion of the graph, the subvolumes of F7293H exhibit negative values of η, η′, and η″, ranging approximately from -0.005 to -0.030. These negative values mean that the sample is more influenced by the gamma parameter, which is directly related to S-wave anisotropy (γ > ε), suggesting a distinct anisotropic structure compared to the other samples. The magnitude and sign inversion emphasize that F7293H exhibits the highest degree of non-ellipticity and structural complexity within this dataset.

These trends are aligned with microstructural observations and anisotropy classifications derived from Thomsen parameters, reinforcing the link between pore structure, elastic symmetry, and wavefront geometry, but further analysis is needed to better characterize the anisotropy of sample F7293H after processing all of its subvolumes.

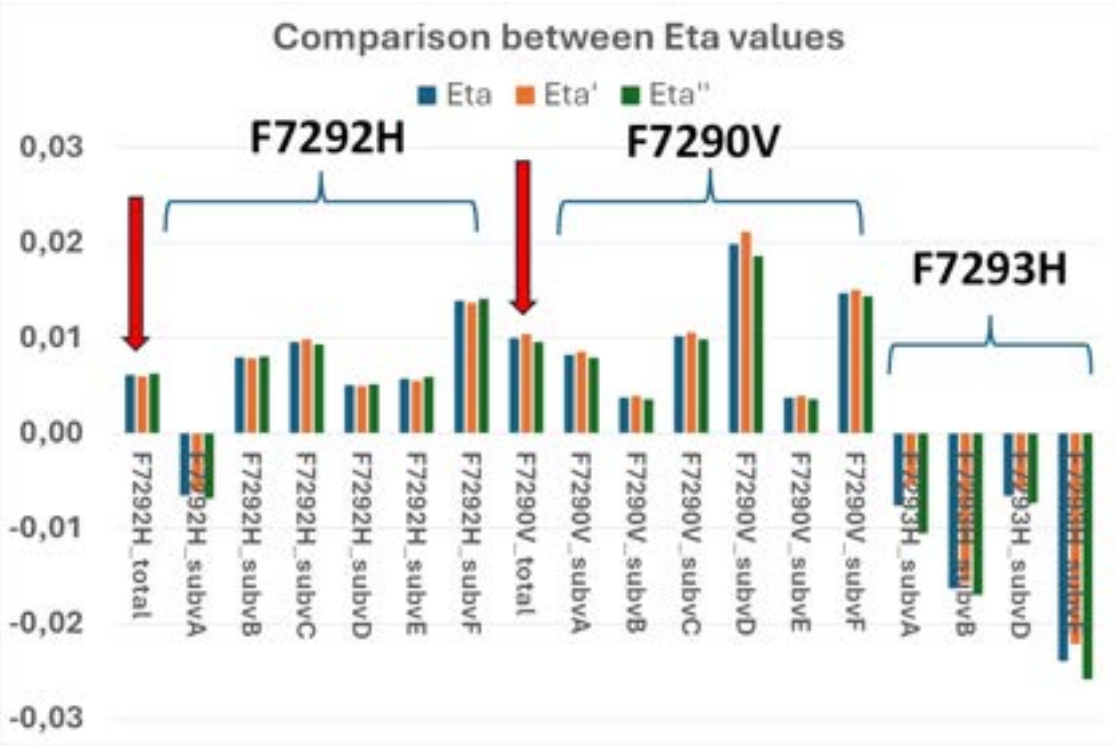


Figure 5.2.12. Partial comparison of non-ellipticity parameters (η, η′, η″) for samples F7292H, F7290V, and F7293H, including their respective sub-volumes.

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6. INTEGRATED DATA ANALYSIS

In this technical report, we present significant advances in petroelastic modeling of pre-salt carbonates, focusing on the development and refinement of the T-matrix model. A new Incremental T-Matrix scheme was introduced to improve sensitivity to variations in pore aspect ratio, outperforming traditional models such as DEM, Vernik-Kachanov, and Keys-Xu, due to its unique capacity to model frequency-dependent effects. The model was validated using well and laboratory data, showing satisfactory agreement with theoretical bounds and measured elastic properties.

The report also details a comprehensive petrophysical analysis of the 3-EQNR-3-SPS well in the Bacalhau Field, using the Pyrockphys Python package. This included data quality control, mineral inversion, facies classification, and estimation of porosity, permeability, water saturation, and pressure. The analysis identified a reservoir interval between 6050 and 6320 meters, characterized by high porosity, low water saturation, and good permeability, followed by an igneous rock zone of no economic interest.

Pressure estimation revealed a sharp transition across the salt–reservoir boundary, with overpressure conditions confirmed by formation test data. Hydraulic Flow Unit analysis highlighting the zones with excellent flow potential in the upper interval.

Overall, integrating T-matrix modeling and petrophysical workflows can yield robust reservoir characterization, enhancing understanding of carbonate systems and supporting geophysical monitoring strategies.

6.1. T-matrix modeling

In the first technical report, we presented an initial review of our implementation of the T-matrix model, showing some modeling results for elastic moduli as a function of porosity for different pore aspect ratios. These results were compared with the DEM model, with good agreement only for the bulk modulus at an aspect ratio of unity.

Now we were able to improve the results considerably, also benefiting from the release of Equinor's T-matrix implementation in Matlab, which prompted some adjustments to the extraction of the bulk and shear moduli from the stiffness tensor. A significant advance, however, came with the development of an alternative computational scheme for the differential version of the T-matrix model. This development was motivated by a comparative analysis of T-matrix modeling with other inclusion models, including the differential effective medium - DEM (Berryman, 1992), Vernik-Kachanov model (Vernik and Kachanov, 2010),

and Keys-Xu model (Keys and Xu, 2002). We identified a need to increase the T-matrix's sensitivity to variations in pore aspect ratio, which was achieved by implementing an incremental T-matrix computation with respect to porosity. This new computational scheme was successfully tested using a single inclusion model and is now being extended to model a family of multiple inclusions.

For initial model validation, the theoretical results are compared with well and laboratory data. This study includes modeling of the well-log data from 1-BRSA-1045 and 3-EQNR-3.

6.1. T-Matrix Modeling Strategies

Several modeling strategies are available within the T-matrix framework (Jakobsen et al., 2003a) to estimate the effective stiffness tensor $C^{\star}$ of the composite, that is

$$C^{\star} = C^{(0)} + C_1\left[I + C_1^{-1}C_2\right]^{-1}, \qquad (6.1.1)$$

where $C^{(0)}$ is the background stiffness tensor, I is the identity matrix for the fourth-rank tensors, and C_1 is the fourth-rank tensor representing first-order corrections to incorporate the effects of isolated pores. The second-order correction C_2 incorporates the interaction effect between cavities or pores. The tensors were defined in the previous report. The following subsections summarize the strategies for resolving equation 6.1.1.

6.1.1. Self-Consistent Approximation (SCTM)

The first-order approach imposes the condition that the effective medium must, on average, behave like an inclusion embedded within itself. The Coherent Potential Approximation (CPA) is more suitable for materials with high inclusion concentration or for composite rocks with no dominant host phase, i.e., with a well-defined initial stiffness tensor $C^{(0)}$. It leads to an implicit nonlinear equation for $C^{\star}$:

$$\sum_{r=1}^{N} v^{(r)}\left[C^{(r)} - C_{SC}^{\star}\right]\left[I - G_{SC}\left(C^{(r)} - C_{SC}^{\star}\right)\right]^{-1} = 0. \qquad (6.1.2)$$

For this approximation, the results of bulk and shear moduli as a function of porosity are shown in Figure 6.1.1. In this figure, we compare with the original T-Matrix formulation.

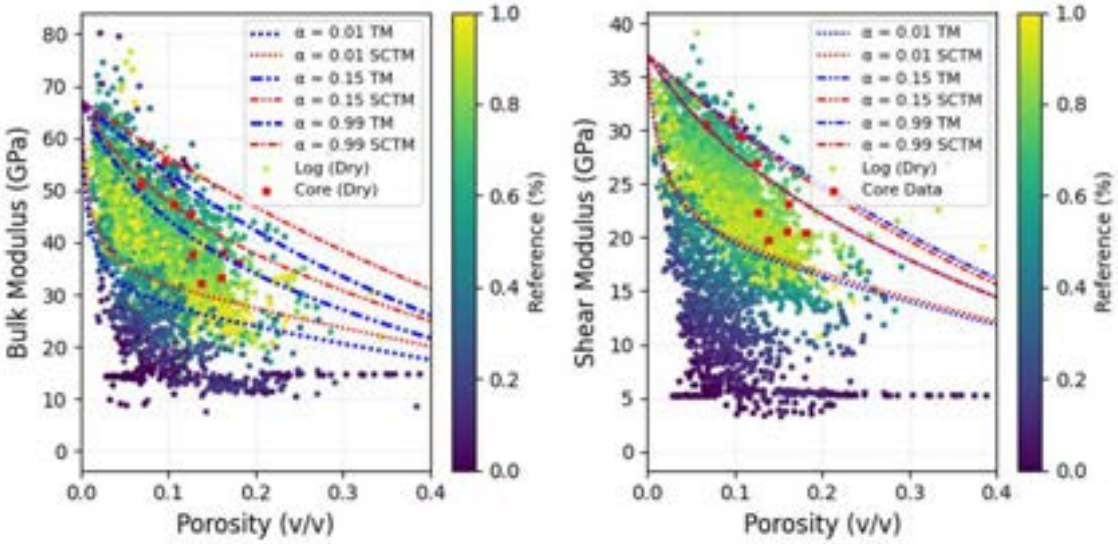

Figure 6.1.1: SCTM (red) bulk and shear Moduli in relation to porosity, showing low aspect ratio sensitivity as T-Matrix (blue). The points correspond to measured data from 1-BRSA-1045-SPS.

We can see that the T-matrix model exhibits substantially lower sensitivity to the pore aspect ratio (three aspect ratios: 0.01, 0.15, and 0.99 representing crack, reference, and spherical pores, respectively) than the DEM modeling, as shown in Figure 6.1.2.

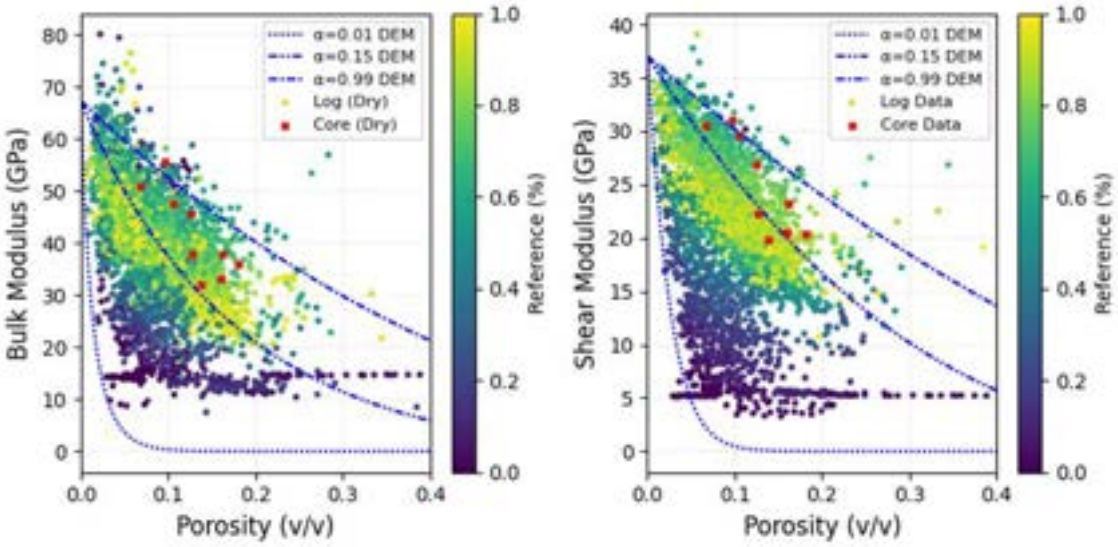

Figure 6.1.2: Differential Effective Medium (blue) bulk and shear Moduli in relation to porosity, showing higher aspect ratio sensitivity. Keys-Xu model (red) and Vernik-Kachanov (orange)

This comparison is relevant, as the DEM model is a standard in petroelastic modeling. Based on the results shown in the figure above, we will analyze the generalization of DEM to full tensor notation in the next subsection.

6.1.2. Differential T-Matrix Scheme (DTM)

The second-order, non-self-consistent T-Matrix approximation with a well-defined initial stiffness tensor $C^{(0)}$ can be formulated using the Generalized Optical Potential Approximation (OPA). When applied incrementally, this leads to the differential effective medium (DEM) formulation, which is expressed as a system of ordinary differential equations for the effective stiffness tensor

$$\left[1 - v^{(r)}\right]\frac{dC^{\star}{}_{DEM}}{dv^{(r)}} = \left[C^{(r)} - C^{\star}_{DEM}\right]\left[I - G_{DEM}\left(C^{(r)} - C^{\star}_{DEM}\right)\right]^{-1}, \qquad (6.1.3)$$

with the initial condition $C^{\star}_{DEM}(v^{(r)} = 0) = C^{(0)}$. So, this EDO is solved numerically. The results from this approximation are shown in Figure 6.1.3, which compares the T-matrix (represented by blue curves) and DEM (red) predictions with measured bulk and shear moduli, as a function of porosity.

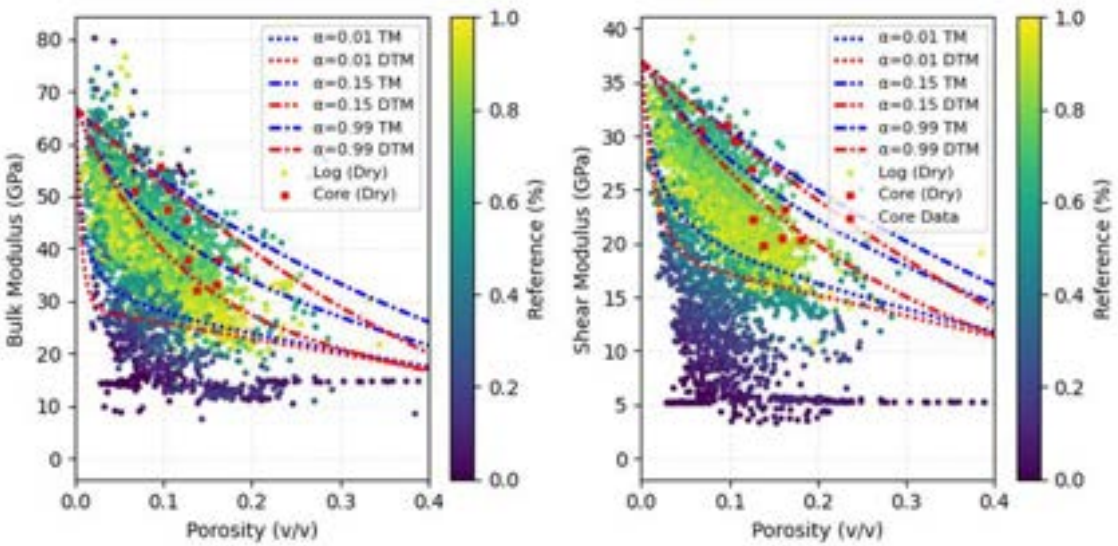

Figure 6.1.3: DTM (red) bulk and shear moduli in relation to porosity, showing low aspect ratio sensitivity as T-Matrix (blue). The points correspond to laboratory measured and well-log data from 1-BRSA-1045-SPS.

6.1.3. Incremental T-Matrix (Inc-TM)

DTM improves sensitivity with respect to the aspect ratio. A possible and accurate solution is achieved using the forward Euler method, which we refer to as the Incremental T-Matrix.

The Incremental T-Matrix Method is proving to be a robust approach for calculating the effective elastic properties of composite materials, particularly rocks with embedded inclusions such as pores or microcracks. This method extends the classical T-matrix formulation by updating the effective stiffness tensor step-by-step as inclusions are progressively added into the host matrix. It offers improved accuracy over classical mixing laws, especially at high inclusion concentrations or geometrically complex pore systems.

In each iteration, the composite's elastic response is updated based on a small increment in the inclusion volume fraction. The updated stiffness tensor incorporates the T-matrix contribution from the new inclusions, adjusted for the changing mechanical properties of the evolving effective medium. This formulation ensures that the medium's response evolves realistically as the inclusion phase increases.

By updating the background medium at each step, the method captures interactions between pores and the evolving matrix, which leads to nonlinear changes in effective properties (e.g., elastic moduli decrease faster than linearly with porosity).

The Incremental T-Matrix method is a numerical approach for modeling the effective properties of composite materials with inclusions. It is based on the mathematical foundation of the T-Matrix (Transition Matrix) formalism and implements a discrete approximation to the Differential Effective Medium (DEM) theory. In this framework, the effective stiffness tensor $C^{\star}$ of a composite material with inclusions can be described by:

$$C^{\star} = C^{(0)} + \sum_{r=1}^{N} v^{(r)} t^{(r)} + O\left(\left(v^{(r)}\right)^2\right) \qquad (6.1.4)$$

For a small inclusion volume increment of $\Delta v^{(r)}$, a forward Euler discretization or first-order Euler integration of the DEM differential equation 6.1.3, equation 6.1.4 becomes

$$C_{k+1}^{\star} = C_k^{\star} + \frac{\Delta v}{1 - v_k} t^{(k)}, \qquad (6.1.5)$$

where the effective stiffness tensor $C_{k+1}^{\star}$ at step $k+1$ is calculated from the previous step k. $C_k^{\star}$ is the effective stiffness at step k. Δv is the inclusion volume fraction increment ($\Delta v = v_{total}/N_{steps}$). The cumulative volume fraction at step k is $v_k = k\Delta v$. $t^{(k)}$ is the T-matrix computed using the current effective properties $C_k^{\star}$. Initial conditions start with the pure matrix properties $C_{k=0}^{\star} = C^{(0)}$, the background matrix stiffness, when $v^{(r=0)} = v_{matrix}$, and $v_{k=0} = 0$. The process continues until the desired total porosity v_{total} is reached.

At each step k, the T-matrix is computed as:

$$t^{(k)} = \left[C^{(r)} - C_k^{\star}\right]\left[I - G^{(k)}\left(C^{(r)} - C_k^{\star}\right)\right]^{-1}, \qquad (6.1.6)$$

where $G^{(k)}$ is the Eshelby tensor appropriate for the inclusion shape and the current effective medium $C_k^{\star}$.

For dry inclusions of ellipsoidal, spherical, penny-shaped, and needle types, the results are shown in Figure 6.1.4. The method provides a good approximation and is consistent with rigorous Hashin-Shtrikman bounds.

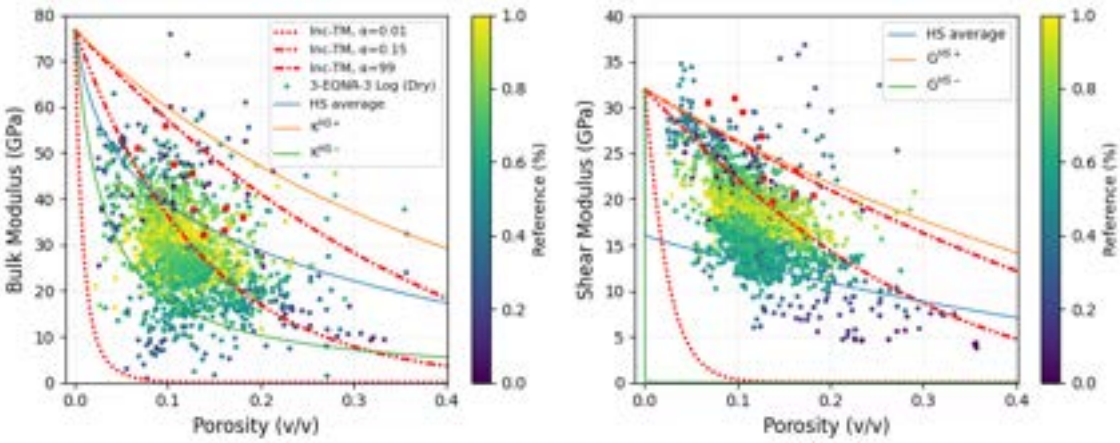

Figure 6.1.4: Incremental T-Matrix (red) bulk and shear moduli versus porosity plots, showing higher aspect ratio sensitivity. The results are consistent with rigorous Hashin-Shtrikman bounds (lower: green, upper: orange, and average: blue). The points correspond to measured data from 3-EQNR-3-SPS.

For multiple inclusion families, the method can be extended by cycling through different inclusion types at each step or by using a coupled approach:

$$C^{\star}_{k+1} = C^{\star}_{k} + \sum_{r=1}^{M} \frac{\Delta v_r}{1 - v_k} t^{(k,r)} , \qquad (6.1.7)$$

where Δv_r is the incremental volume fraction of the inclusion family r, $t^{(k,r)}$ is the T-matrix corresponding to the family r at step k, $v_k = \Sigma_r k_r \Delta v_r$ is the total inclusion volume fraction at iteration k.

This formulation allows simultaneous modeling of coexisting inclusion types, including mixtures of oblate microcracks, needle pores, and fluid-saturated inclusions. It provides a flexible framework for describing multiscale pore systems and anisotropic textures that are typical of natural rocks.

Figure 6.1.5 shows the Inc-TM modeling for multiple non-interacting pore inclusions in the right plot, using fractions of each pore fraction $v = v_1 + v_2 + v_3$, in the left plot. Additionally, there is a comparison that includes the saturated multiple inclusion family case using both Inc-TM and the Gassmann model.

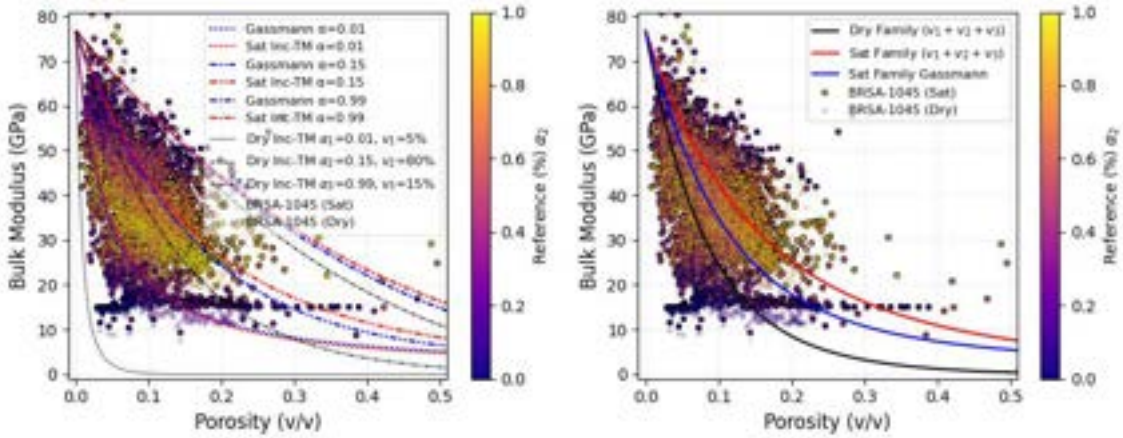

Figure 6.1.5: Left plot: Incremental T-Matrix (black) modeled bulk modulus as a function of porosity, considering a single inclusion in dry and saturated conditions. The plot compares the saturation effect computed directly by the Inc-TM (red) with that computed using Gassmann's equation (blue). Right plot: Modeled results, considering a combination of multiple inclusion families in dry (black), Inc-TM saturated (orange), and Gassmann saturated (blue). The points correspond to measured well-log data from 1-BRSA-1045-SPS.

The T-matrix analysis of frequency-dependent, complex-valued elastic moduli provides a direct link to the observed wave attenuation and dispersion in saturated porous media. In this framework, attenuation arises from poroelastic relaxation mechanisms that depend on wave frequency, rock permeability, relaxation time, and fluid viscosity (Jakobsen and Chapman, 2009). These factors control the diffusive equilibration of fluid pressure across pores of different aspect ratios and therefore govern the frequency-dependent moduli of the composite rock.

The initial results obtained using incremental T-matrix modeling are shown in Figure 6.1.6. These results demonstrate how the compressional and shear velocities evolve with frequency, reflecting the transition from the relaxed (low-frequency) to the unrelaxed (high-frequency) regime across different porosities, for a single family of saturated inclusions. In this case, the effective stiffness tensor given by

$$C^{\star} = C^{(0)} + C_1, \qquad (6.1.8)$$

with

$$C_1 = \sum_{r} \quad v^{(r)} t_{sat}^{(r)}, \qquad (6.1.9)$$

here, $t_{sat}^{(r)}$ it is given by Jakobsen et al. (2003b) using the permeability $\Gamma_{uv} = 0$.

In Figure 6.1.6, at lower frequencies, the fluid has time to flow and equilibrate pressure, so the rock appears softer with a lower velocity. At high frequencies,

the fluid is confined or unrelaxed, effectively stiffening the rock and leading to a higher velocity. The attenuation peak occurs precisely in the frequency band where the velocity is changing most rapidly. This is the point of a relaxation mechanism, where maximum energy is dissipated at this characteristic frequency (Agersborg et al. 2009).

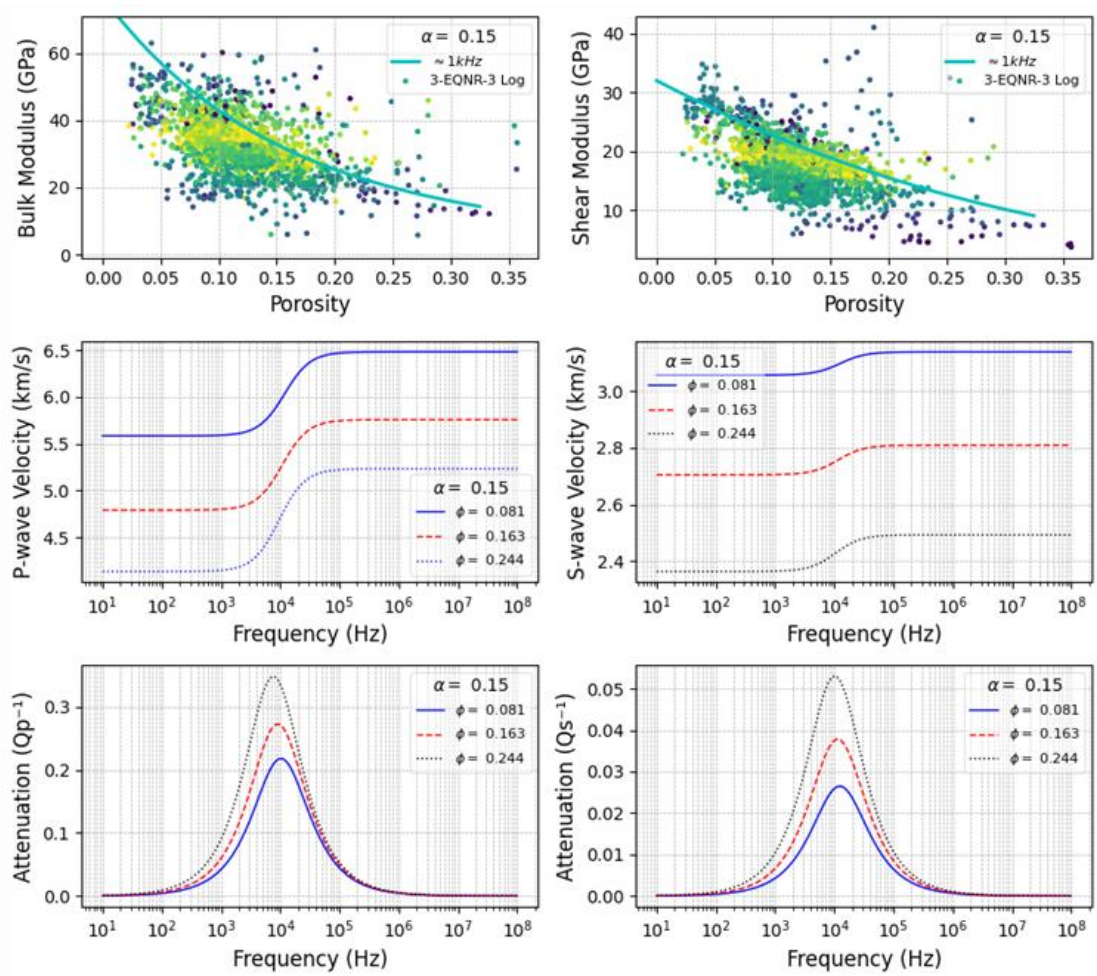

Figure 6.1.6: Predicting bulk and shear moduli for specific frequency response compared with well-log data from 3-EQNR-3 (top row), followed by the dispersion curves for P- and S-wave velocities (mid row), and the attenuation (the inverse quality factor) as functions of frequency.

The next steps of development are modeling the full-coupled interaction between the solid matrix and fluid. In this case, to simplify the analysis, we assume that $G^{(s)} = G^{(r)}$, indicating there is only one family type. Consequently, the effective stiffness is given by

$$C^{\star} = C^{(0)} + C_1[I + G_d C_1]^{-1}, \qquad (6.1.10)$$

where C_1 is given by equation 6.1.9, G_d is the Eshelby tensor for the effective aspect ratio of the correlation function, and T-matrix for dry inclusions is

$$t^{(r)} = -C^{(0)}\left[I + G^{(r)}C^{(0)}\right]^{-1} \qquad (6.1.11)$$

Thus, equation 6.1.10 includes fluid effects for fully saturated inclusions that can communicate with each other. This will be used to calculate the bulk and shear moduli for interacting pore inclusions, velocity dispersion, and attenuation.

The Incremental T-Matrix framework can be naturally extended to anisotropic host media, where the background stiffness tensor $C^{(0)}$ exhibits transverse isotropy (TI), orthotropy, or full anisotropy.

In such cases, the formulation must account for the direction-dependent elastic response of the host material. The Eshelby tensor $G^{(k)}$ must be computed for ellipsoidal inclusions embedded in an anisotropic background, following formulations such as those by Mura (1982). The update of the effective stiffness tensor $C^{\star}_{k+1}$ remains formally identical but requires full tensor operations that can be evaluated numerically via tensor rotation and coordinate transformations.

6.2. **Well data analysis**

In this section, we present the data analysis for the 3-EQNR-3-SPS well, using the Pyrockphys Python Package developed as part of the project.

6.2.1. Introduction

The 3-EQNR-3-SPS well, operated by Equinor, is located in the Bacalhau Field, within the Santos Basin, one of the most prominent sedimentary basins in Brazil. The Santos Basin is situated along the southeastern continental margin, in an offshore area between latitudes 23° and 28° South and longitudes 40° and 47° West, covering approximately 350,000 km².

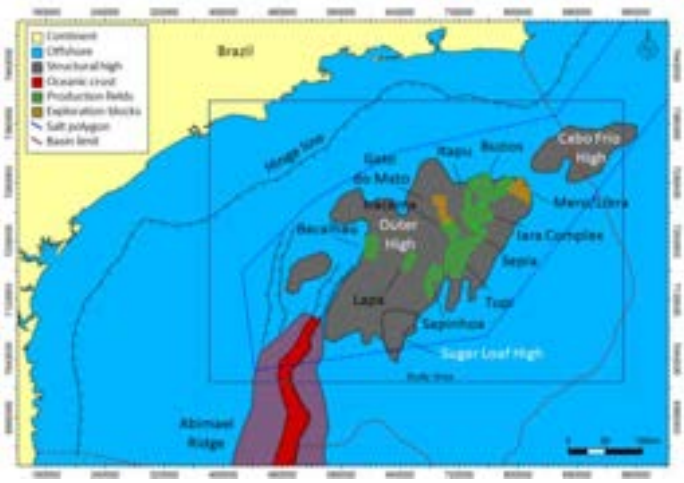

Figure 6.2.1: Location map of the main oil and gas fields in the Santos Basin, including the Bacalhau Field . Source: Coelho et al.(2022), adapted from Gamboa et al.(2021).

The Bacalhau Field is a geologically complex and overpressured field that hosts high-quality carbonate reservoirs. It is located in ultra-deep waters at a depth of around 2,100 meters and produces light, highly undersaturated oil with a high gas-to-oil ratio and no contaminants. The reservoirs correspond to two distinct intervals of the Aptian Barra Velha Formation, found at approximately 6,000 meters depth. These deposits developed on an isolated lacustrine platform, overlying a complex paleotopography composed of rifted Paraty volcanics and younger volcanic pedestal highs (Hansen, 2019; Hansen *et al.*, 2021; Hunt, 2021; Louback *et al.*, 2023; Nogueira *et al.*, 2022; Santos *et al.*, 2022).

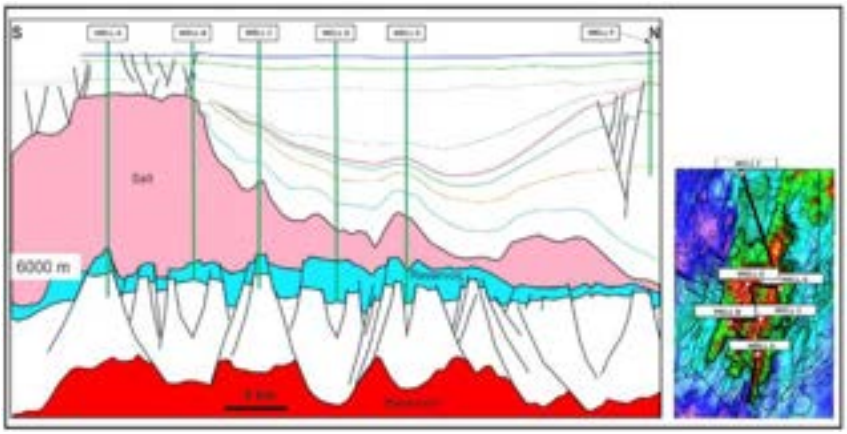

Figure 6.2.2: Geological cross-section along Bacalhau structure. Source: adapted from Santos et al.(2022)

The following sections present the progress of our Python package, Pyrockphys, through the application of a petrophysics workflow to the Pre-salt data from Bacalhau Field, i.e., the well 3-EQNR-3-SPS.

6.2.2. Rock Physics Modeling of Well 3-EQNR-3-SPS

In a study using well-log data from the Brazilian pre-salt in the Bacalhau Field, 3-EQNR-3-SPS, we implemented the Pyrockphys Python Package and applied the petrophysical workflow shown in Figure 6.2.3.

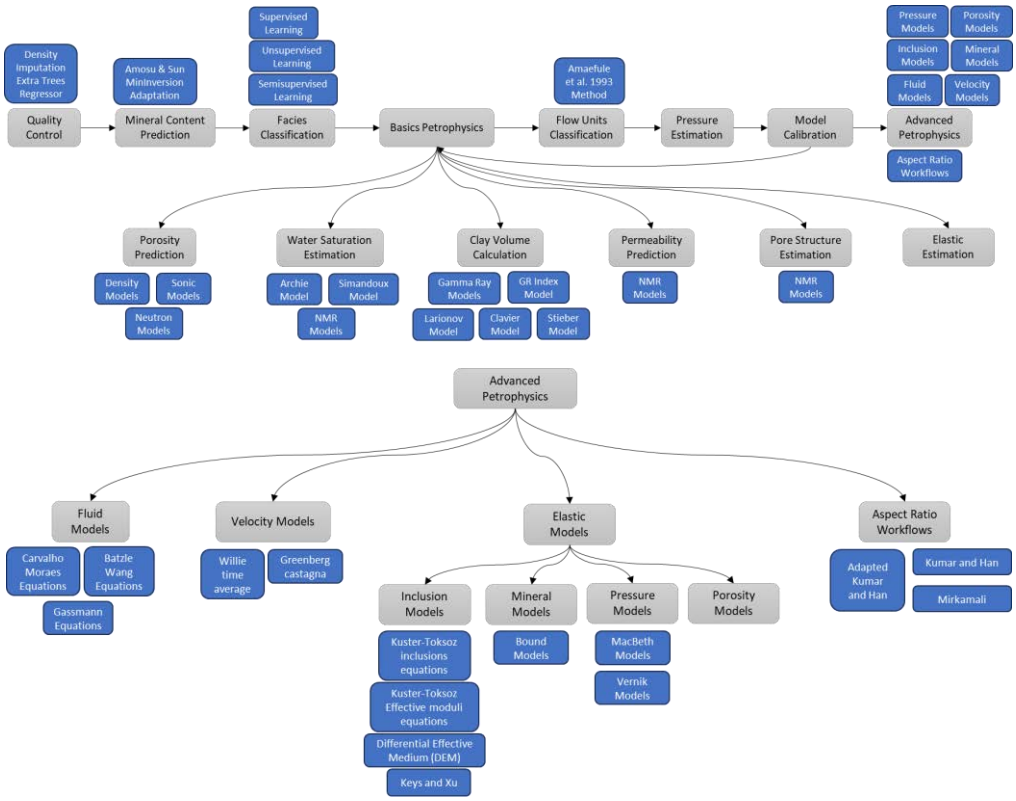

Figure 6.2.3: Petrophysical Workflow Analysis and Modeling from Pyrockphys Python Package.

The well-log data is composed of sonic logs (P and S-wave), density, gamma ray, total CMR porosity (from Nuclear Magnetic Resonance tool), thermal neutron porosity, resistivities, photoelectric factor, NMR logs (T2 distribution, bin porosities, free fluid, bound volume fluid, effective porosity and permeability), water saturation and clay volume from HILT (High Resolution Integrated Logging Tool), bulk and shear modulus, dry-weight elemental fraction logs.

6.2.2.1. Quality control

We used a tool from the package to visualize the distribution of each log curve along the well (Figure 6.2.4). From this visualization, it is evident that a significant portion of the data was recorded between 6000 and 6400 meters, mainly within the pre-salt zone. It is also noticeable that some curves contain gaps where data is missing, which need to be corrected before proceeding with further analyses.

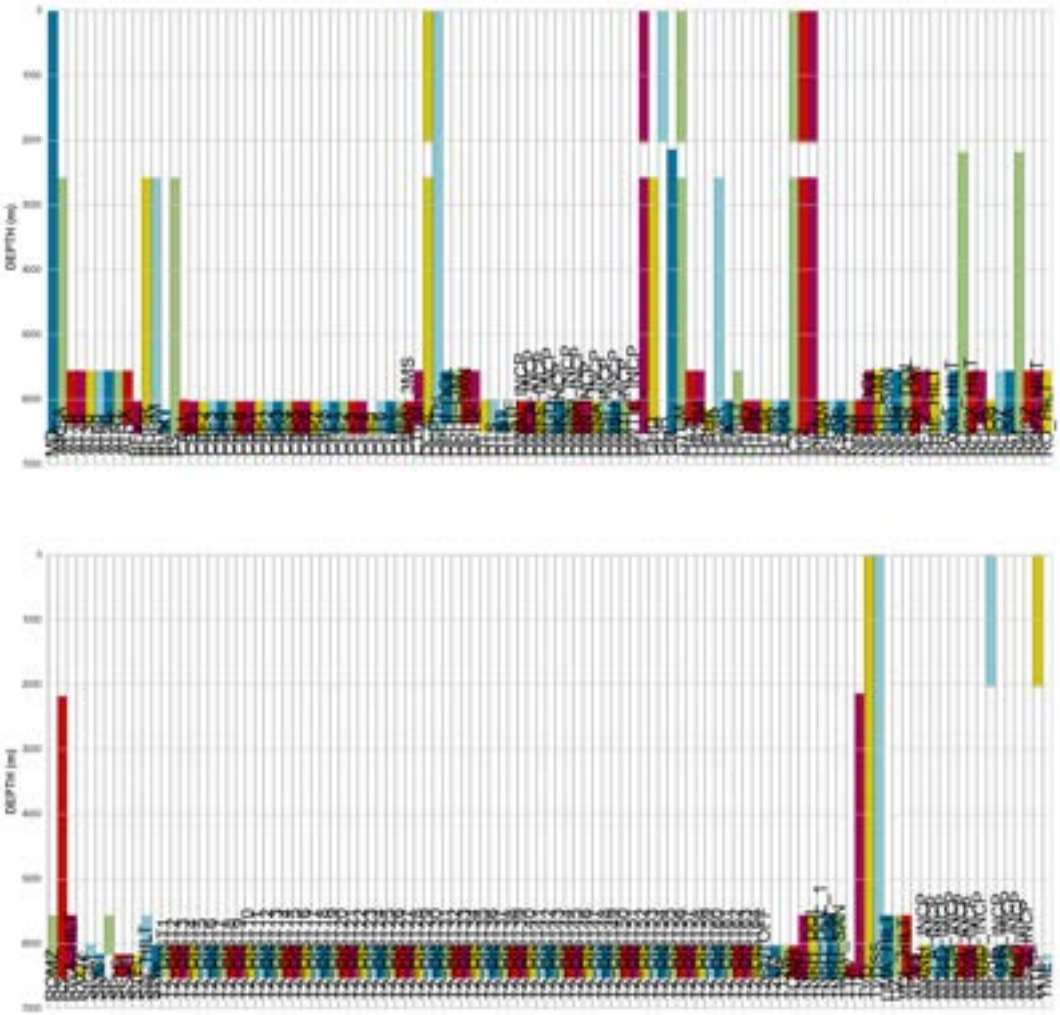

Figure 6.2.4: Available log curves from the 3-EQNR-3-SPS well plotted along depth.

Initially, it is important to define the post-salt, salt, and pre-salt zones of the well and identify the main lithologies in each zone. This zone identification can be done using the Composite Log, which describes lithology from the beginning of

the well. However, this file was not available for the 3-EQNR-3-SPS well. To perform this step, we used the Medium Resistivity (RM) log, recorded from the post-salt interval downward (Figure 6.2.5).

For the 3-EQNR-3-SPS well, no missing values were identified in the main logs within the pre-salt zone. However, to estimate pressure, a complete density (RHOZ) record is required from the water depth (~2105 m) to the total depth. In this dataset, the density log is available only from part of the salt interval to the end of the pre-salt section (Figure 6.2.5).

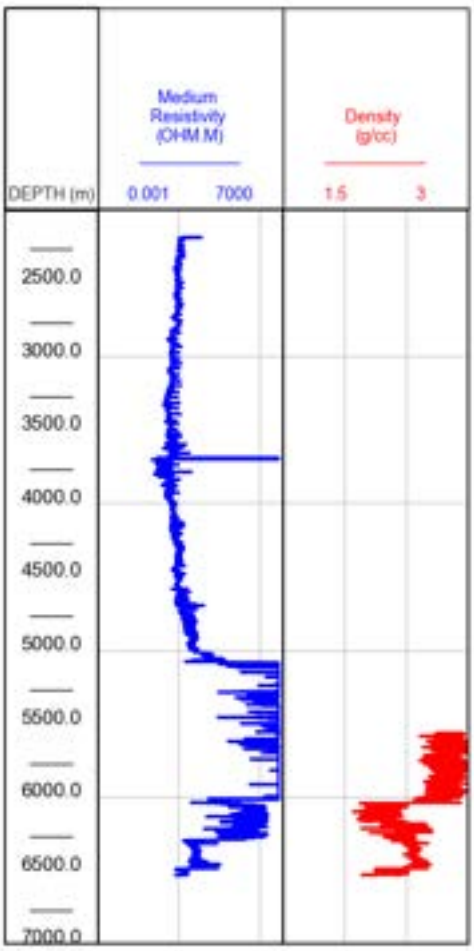

Figure 6.2.5: Medium Resistivity and Density Logs.

Based on the RM log, the post-salt zone extends from 2080 to 5080 m and is characterized by low resistivity (approximately 1–10 Ω·m) and lithologies composed mainly of shale and compacted sandstone. The salt zone, between 5080 and 6050 m, exhibits very high resistivity (around 5000 Ω·m). Well reports indicate the presence of anhydrite at 6044 m, and the density log begins within the salt interval with an average bulk density of 2.8 g/cm³, suggesting a higher proportion of anhydrite relative to halite. The pre-salt zone, from 6050 to 6400 m, exhibits variable resistivity, ranging from moderate to high, indicating a heterogeneous lithology.

Therefore, using the average density of which lithology as a reference, we imputed the missing portion of the density curve using the LogImputation class, which applies the Extremely Randomized Trees (ExtraTrees) algorithm (Geurts et al., 2006). This method estimates missing log values in unrecorded sections—such as salt or post-salt zones—based on correlations with fully available feature logs. The model is trained on intervals where both the target and feature logs are recorded, and the learned relationships are then applied to predict the missing values. Selecting the most geologically relevant input logs is crucial to ensure reliable imputation. Figure 6.2.6 shows the result of density imputation.

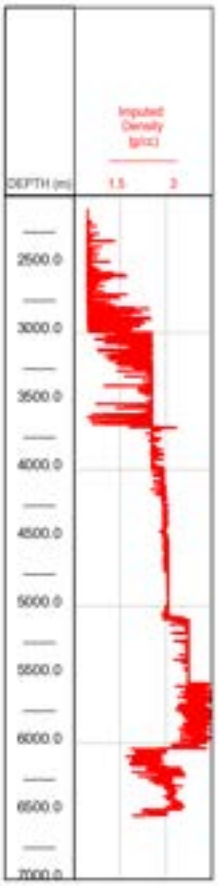

Figure 6.2.6: Result of density log imputation.

The feature logs used were Deep, Medium, and Shallow Resistivities (RD, RM, RS) and Gamma Ray (GR). The results presented a Mean Squared Error of 0.0098 and an R^2 of 0.93, demonstrating excellent model performance and consistency with the expected geological characteristics of the studied zones.

6.2.2.2. Mineral Modeling

The PyRockPhys package performs a mineral-inversion process to estimate the mineralogical composition from the well-log data. It applies an adapted Multiple Linear Regression formulation based on the approaches proposed by Amosu and

Sun (2018) and Cheng et al. (2022). The algorithm uses the scipy minimize function to determine the optimal mineral volume fractions.

Figure 6.2.7 presents the current results. We are currently investigating alternative methods to improve the fitting performance.

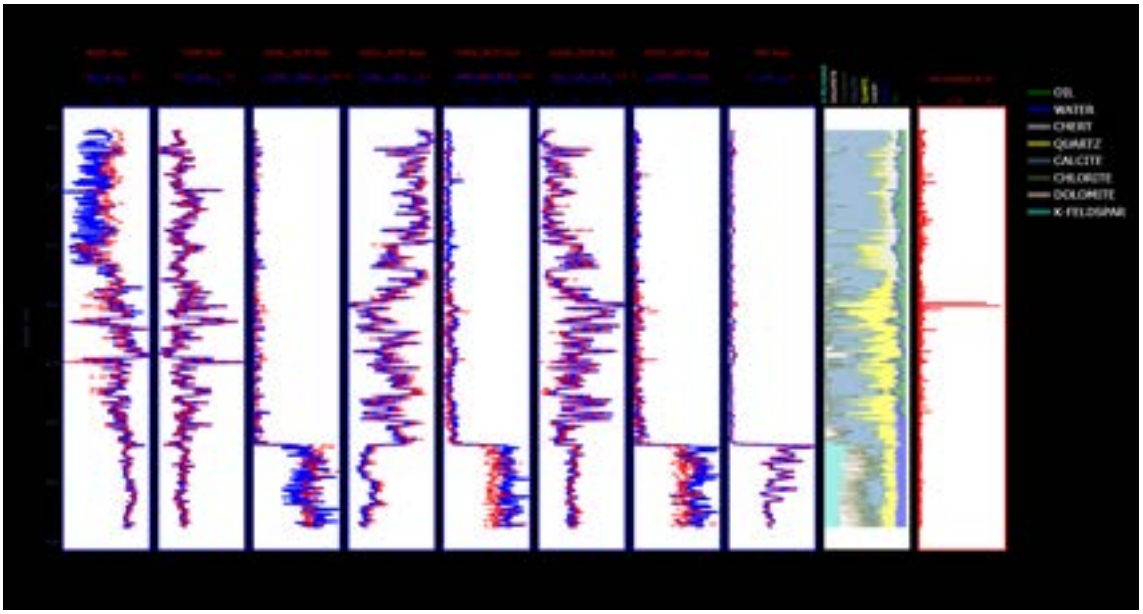

Figure 6.2.7: Multimineral Inversion result.

In Figure 6.2.7, the blue lines represent the well log curves used in the inversion, while the red dashed lines correspond to the fitted curves generated by the model using the estimated mineral volume fractions. The logs used for modeling were Track 1 – Bulk Density (RHOZ), Track 2 – Total Porosity (TCMR), Track 3 – Dry Weight Aluminium (DWAL), Track 4 – Dry Weight Calcium (DWCA), Track 5 – Dry Weight Iron (DWFE), Track 6 – Dry Weight Silicon (DWSI), Track 7 - Dry Weight Titanium, and Track8 - Formation Potassium Concentration. Track 9 displays the estimated components — Oil, Water, Quartz, Chlorite, Calcite, Dolomite, Chert and K-Feldspar — while Track 10 shows the normalized inversion error.

According to well reports, the interval below 6318 m, corresponds to a volcanic section. This interval was later excluded in a separate test of the mineral inversion, which did not yield significant changes in the outcome.

6.2.2.3. Facies Classification

The Pyrockphys Package includes three machine learning approaches for facies classification: supervised, unsupervised, and semi-supervised learning. The choice among them depends primarily on the availability and quality of training data. When labeled training data are available, supervised learning can be applied. In the absence of such information, unsupervised learning is used to

group data based on intrinsic similarities. The semi-supervised approach, in turn, is useful when existing classifications have different sampling intervals or contain numerous missing values.

For the 3-EQNR-3-SPS well, no labeled training data were available at this stage; therefore, unsupervised learning was employed. The package provides three algorithms for this purpose: K-Means, Spectral Clustering, and Hierarchical Clustering.

For the classification, the following logs and estimated mineral fractions were used: Density (RHOZ), P-Wave Sonic (DTCO), Gamma Ray (GR), Formation Photoelectric Factor (PEFZ), Resistivity (AT90), and Quartz, Chlorite, Calcite, Dolomite, Chert and K-Feldspar. Figure 6.2.8 presents the resulting facies classification.

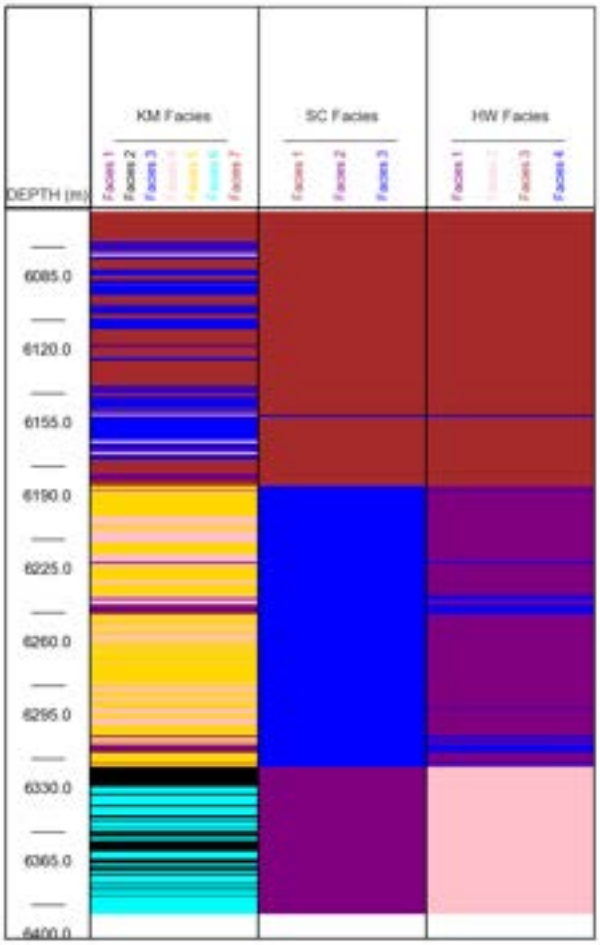

Figure 6.2.8: Facies classification using K-Mean (track 1), Spectral (track 2), and Hierarchical (track 3) Clustering.

Among the tested methods, K-Means Clustering produced well-differentiated clusters along the depth profile, whereas Spectral and Hierarchical Clustering yielded fewer facies and showed limited variability throughout the well.

6.2.2.4. Basic Petrophysics

This section presents the calculations of basic petrophysical parameters, including porosity, shale volume, water saturation, and NMR attributes.

6.2.2.4.1. Shale and Clay Volume Estimation

The clay volume estimation uses five techniques: Gamma Ray Index (IGR), Larionov, Stieber, Clavier, and NMR. These techniques combine information from radioactive and fine-grain materials, affecting Gamma-ray and NMR data. Figure 6.2.9 shows the results of clay volume estimation.

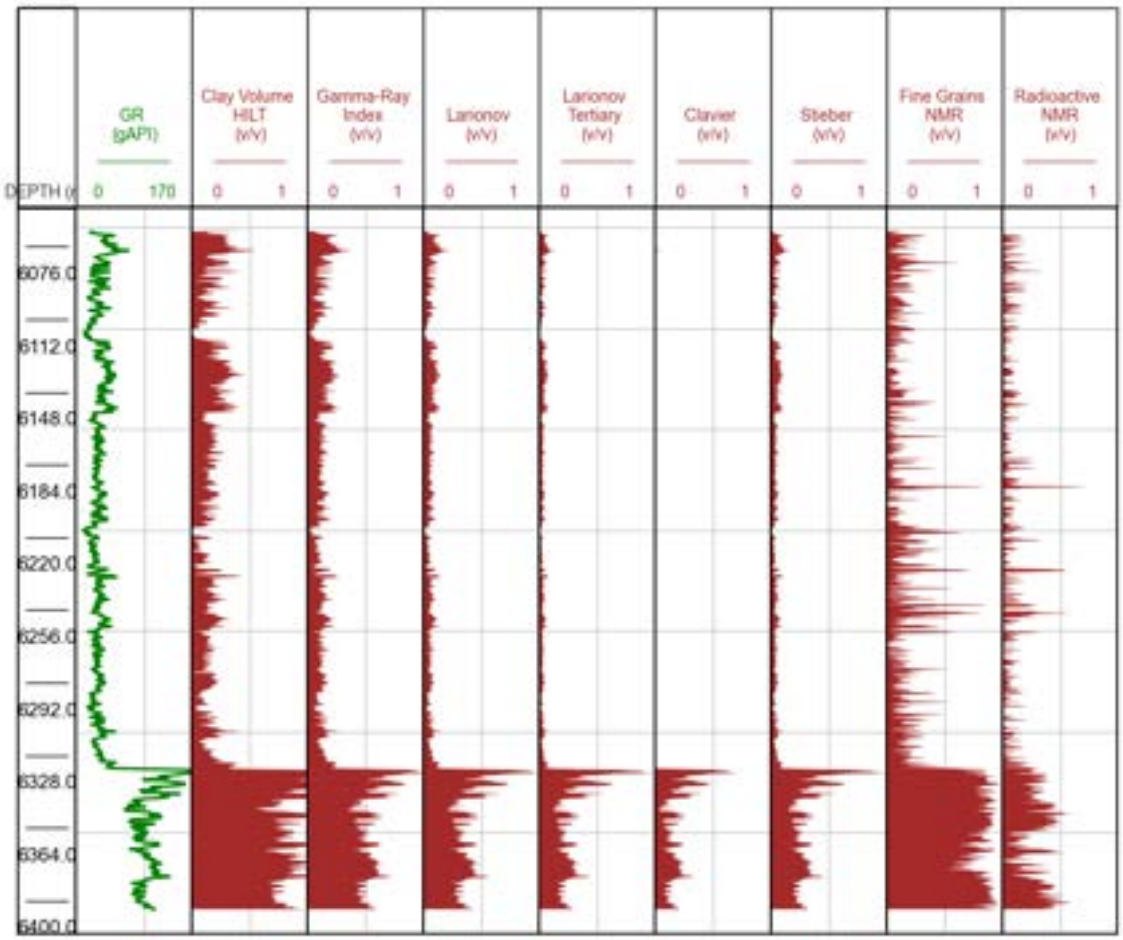

Figure 6.2.9: Clay volume estimation. Track 1 - GR curve, Track 2 - Clay Volume from HILT, Track 3 - Gamma ray Index, Track 4 - Larionov old rocks, Track 5 - Larionov tertiary rocks, Track 6 - Clavier, Track 7 - Stieber, Track 8 - NMR Fine Grain, Track 9 - NMR Radioactive.

The low clay volume in 6050 to 6310m indicates good permeability and the reservoir zone of this well. The high gamma-ray values below 6318 meters indicate the presence of igneous rocks.

6.2.2.4.2. Pore Structure Estimation

In the package, we use bin porosities (BP1 to BP8) to estimate the porosity structure following Rezaee (2022), Figure 6.2.10, where we can calculate Clay Bound Water (CBW), Bound Volume Irreducible (BVI), Free-Fluid Index (FFI), Effective Porosity (PHIE), and Total Porosity (PHIT).

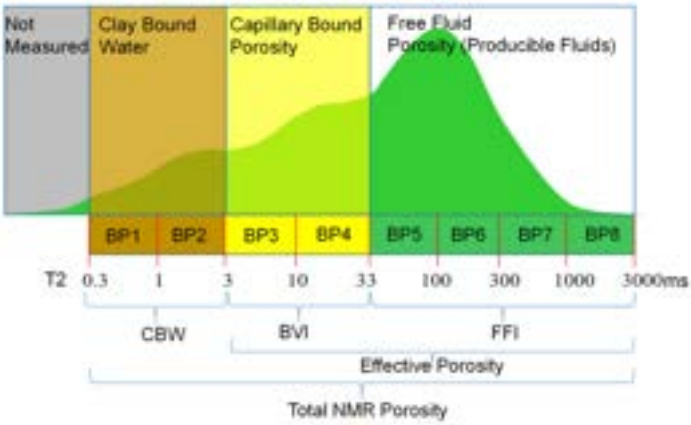

Figure 6.2.10: The T2 distribution cutoffs to determine porosity structure. Source: Rezaee (2022).

The package also uses bin porosities to calculate the fractions of Mesopores, Micropores, and Macropores. Figure 6.2.11 shows the results of pore structure estimation.

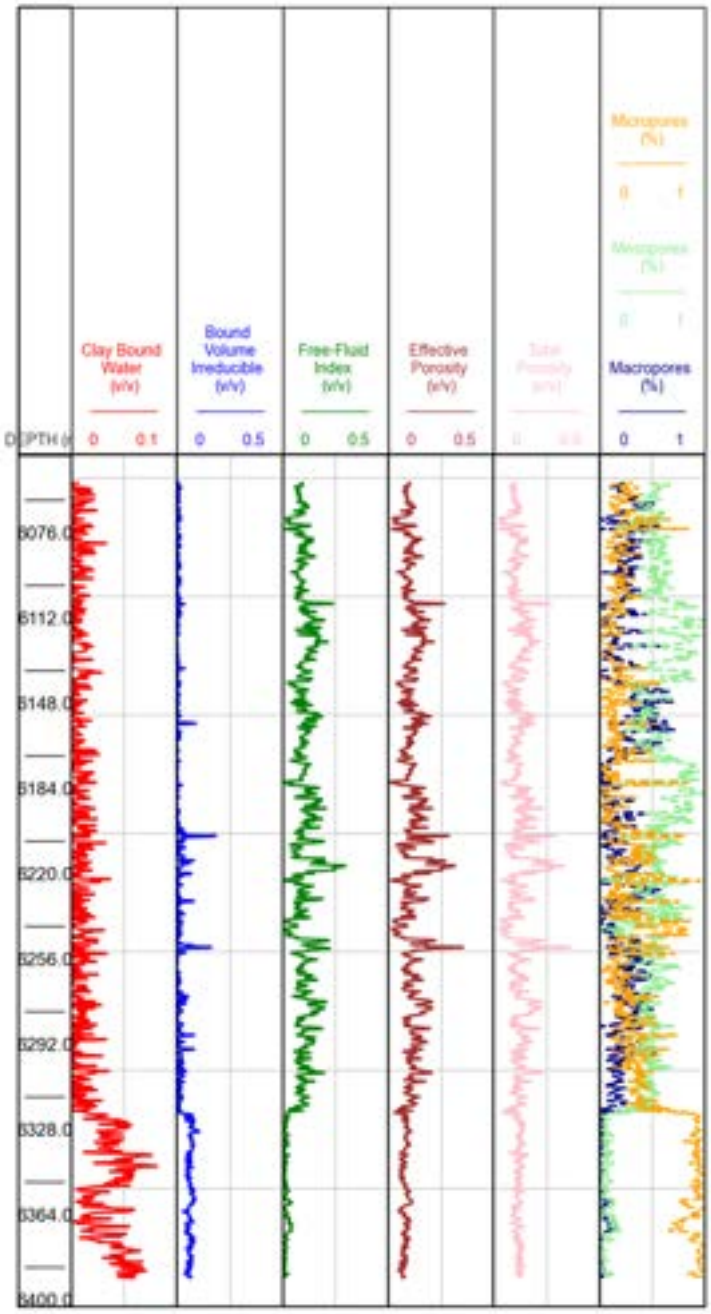

Figure 6.2.11: Pore Structure Estimation. Track 1 - CBW, track 2 - BVI, track 3 - FFI, track 4 - PHIE, track 5 - PHIT.

Between 6050 and 6320 m, the interval is characterized by low values of CBW, BVI, and Micropore fractions, combined with high FFI and Mesopore fractions. These properties indicate a zone with good storage capacity and efficient fluid flow, consistent with a reservoir interval.

From 6320 to 6400 m, higher CBW, BVI, and Micropore fractions are observed, along with lower FFI and Mesopore fractions. This combination suggests a zone of low permeability and predominantly immobile fluids, corresponding to the igneous rock interval.

6.2.2.4.3. Water Saturation Estimation

The package includes several methods for water saturation estimation, such as Archie, Simandoux, and NMR-based calculations (Irreducible Water and Fluid Partition).

These methods were applied across the entire well, and the results (Figure 6.2.12) show water saturation below 30% between 6050 and 6320 m, indicating a reservoir zone in the first 300 m of this interval. Between 6320 and 6400 m, high water saturation was observed, indicating an igneous rock zone with low permeability and predominantly immobile fluids.

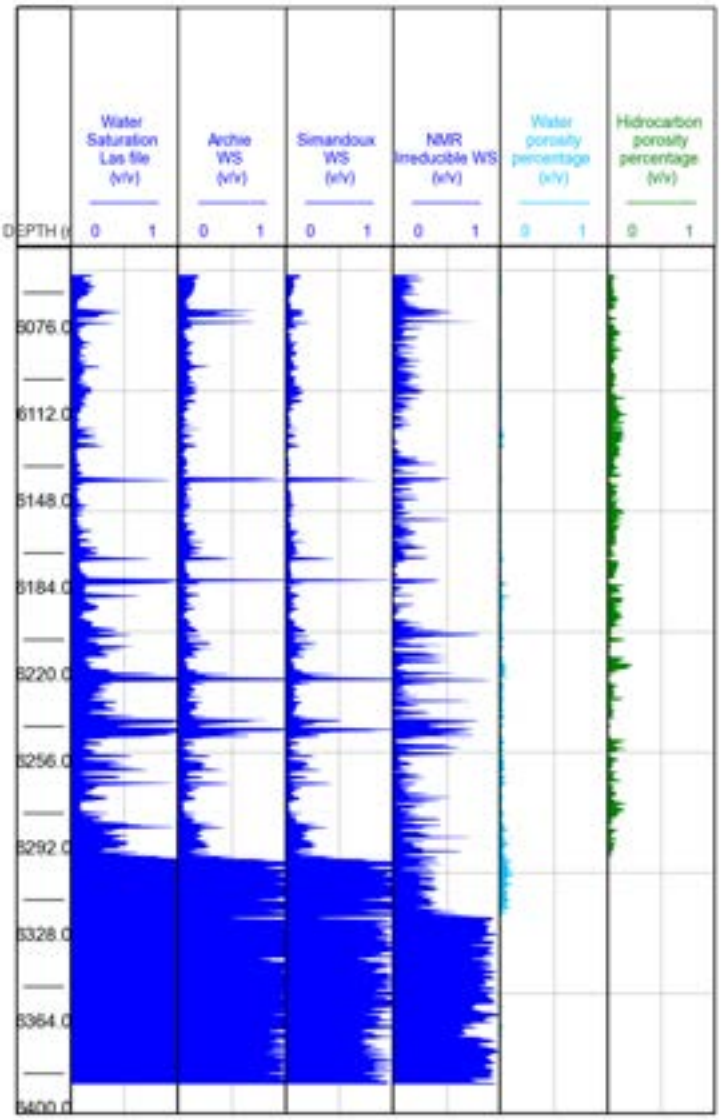

Figure 6.2.12: Water Saturation Estimation. Track 1 - Water Saturation from HILT, track 2 - Water Saturation from Archie, track 3 - Water Saturation from Simandoux, track 4 - Irreducible Water Saturation from NMR, track 5 - Water Porosity percentage from NMR, track 6 - Hydrocarbon Porosity percentage from NMR.

6.2.2.4.4. Porosity Estimation

The package provides several porosity estimation methods, including Density, Neutron, Sonic, and NMR-based models, allowing a comprehensive evaluation of pore systems and lithological variations throughout the well. The NMR-based model was previously discussed in the Pore Structure Estimation section.

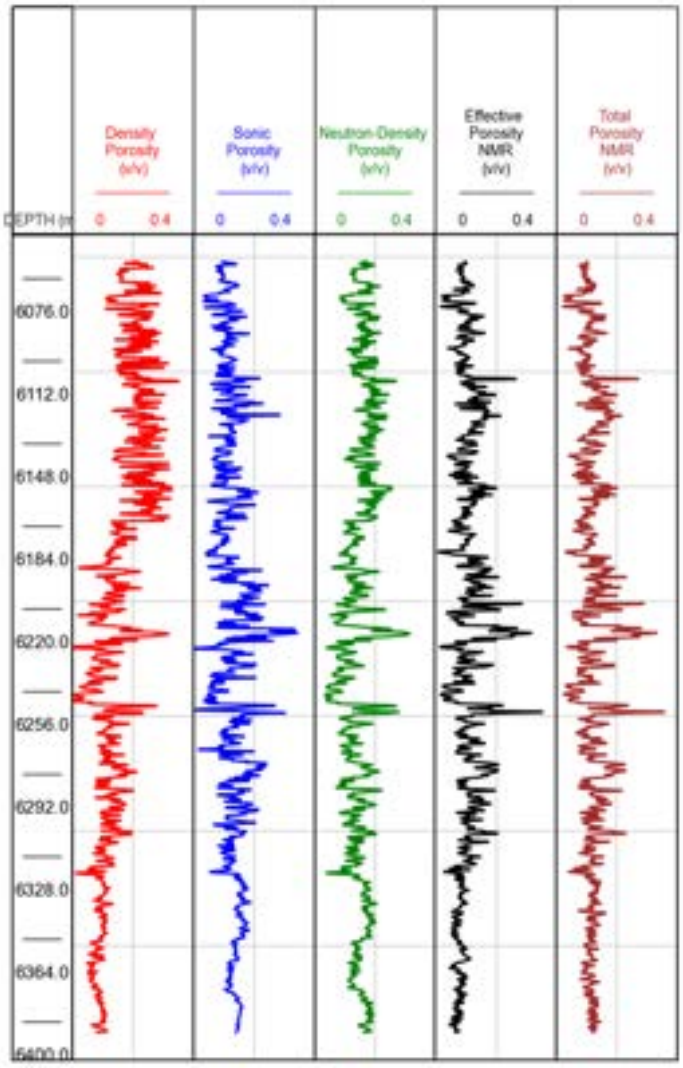

Figure 6.2.13: Porosity Estimation. Track 1 - Density Porosity, track 2 - Sonic Porosity, track 3 - Neutron-Density Porosity, track 4 - Effective Porosity from NMR, track 5 - Total Porosity from NMR.

Figure 6.2.13 presents the porosity results obtained from these methods. The results from density, sonic, neutron-density, and NMR effective and total porosities are very similar, with high, variable porosity values between 6050 and 6320 m, indicating a reservoir zone in the first 300 m of this interval. From 6320 to 6400 m, porosity decreases and becomes more uniform, marking the end of the reservoir section.

6.2.2.4.5. Permeability Estimation

The package also includes permeability estimation methods based on NMR-derived models, such as the Timur-Coates and Schlumberger Doll Research (SDR) equations. These approaches enable the evaluation of fluid flow potential and pore connectivity throughout the well.

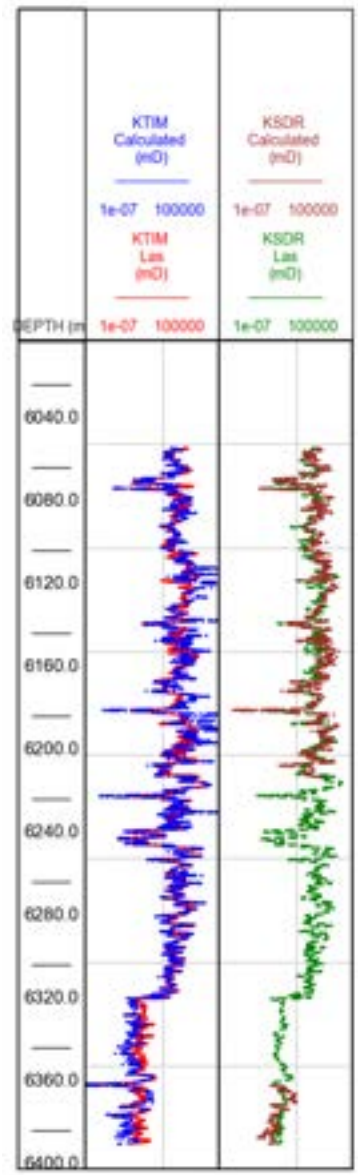

Figure 6.2.14: Permeability Estimation. Track 1 - Permeability from Timur Coates (blue is calculated and red is from the LAS file), track 2 - Permeability from SDR (green is calculated and brown is from the LAS file)

Figure 6.2.14 presents the permeability results obtained from these methods. Between 6050 and 6320 m, the results indicate high permeability, characterizing a reservoir zone in the first 300 m of this interval. Between 6320 and 6400 m, low permeability is observed, corresponding to an igneous rock zone with limited fluid mobility.

6.2.2.5. Pressure Estimation

The overburden pressure (OP) refers to the stress exerted on subsurface formations by the weight of the overlying rock layers (Karimi et al., 2013). In the package, the overburden pressure was calculated from the density log by using

$$OP = P_{OB} + \sum \quad (\rho_{bulk} * g * 10^{-6} * dz) \tag{6.2.1}$$

where OP is the overburden pressure, g is the gravitational acceleration, ρ_{bulk} is the bulk density log, dz is the sampling rate derived from the true vertical depth subsea (TVDSS) data, and P_{OB} is the ocean bottom pressure from the water column, given by

$$P_{OB} = \rho_{water} * g * 10^{-6} * dz_{wd} \tag{6.2.2}$$

where ρ_{water} is the water density (1023 km/m³) and z_{wd} is the true vertical depth subsea (TVDSS) data for water depth.

In porous formations, the pore pressure is defined by the pressure of fluids occupying the pore spaces (Dasgupta & Mukherjee, 2019). To estimate pore pressure (PP), we used hydrostatic pressure before the reservoir zone and static pressure data from formation tests in the reservoir zone. The hydrostatic pressure (HP) is calculated by

$$HP = P_{OB} + \sum \quad (\rho_{brine} * g * 10^{-6} * dz) \tag{6.2.3}$$

where ρ_{brine} is the brine density (1100 km/m³).

The pore pressure estimation was divided into two stages. The first corresponds to the hydrostatic pressure calculated to the depth of the first formation test, while the second involves interpolating pressure values from the available formation test data (Table 6.2.1).

Depth (m)	Pressure (MPA)
6149.9	91.86
6282.1	92.70
6301.0	92.87

Table 6.2.1: Formation Test from 3-EQNR-3-SPS

For well 3-EQNR-3-SPS, only three formation test points are available, starting at a depth of 6149.9 m. The lack of data near the top of the reservoir, combined with the well's overpressure, results in a significant pressure jump across the salt–reservoir transition zone. Since samples are available in this region, the pore pressure was calculated by adding the hydrostatic pressure to the pressure

difference between the formation test and the hydrostatic curve, ensuring a more secure estimation for this interval.

Alternative methods for pore pressure estimation in this transition zone are being studied to refine the results. Finally, the differential pressure (DP) is the difference between the overburden pressure and pore pressure. The results of the pressure estimations are shown in Figure 6.2.15.

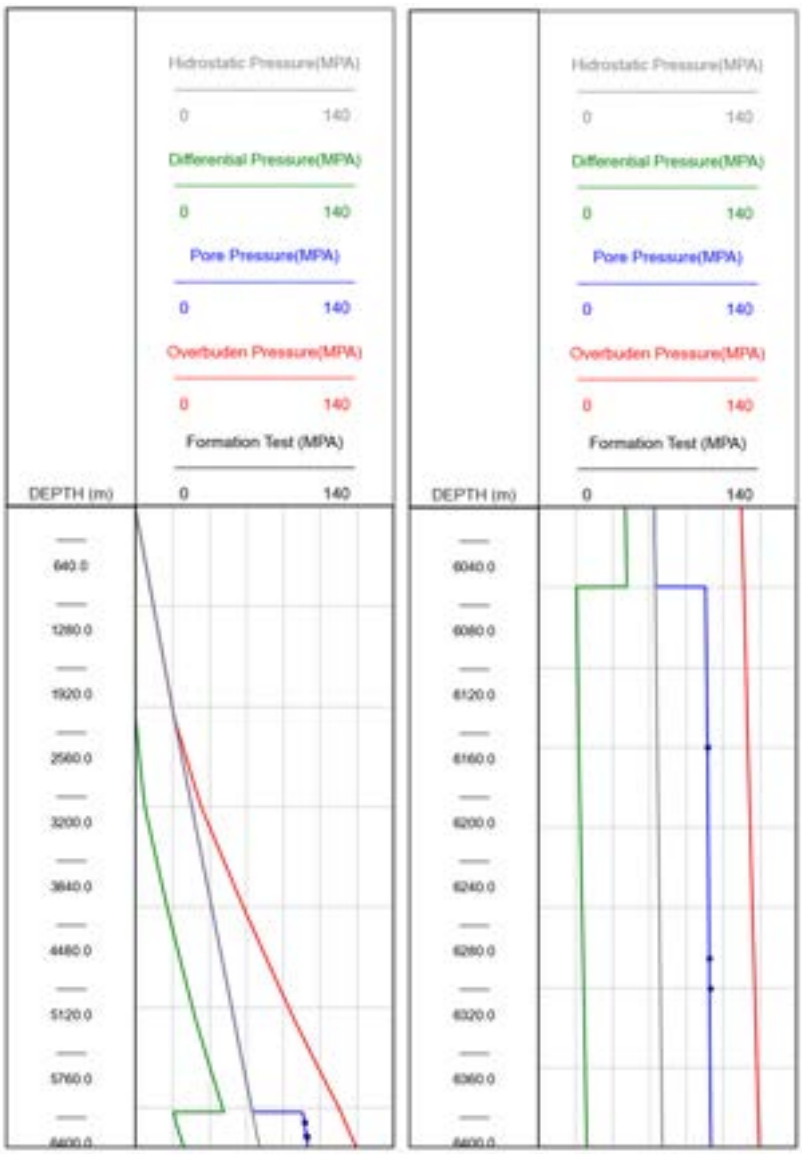

Figure 6.2.15: Track 1 – Pressure estimations along the entire well. Track 2 – Detailed pressure estimations within the reservoir zone.

The red line is the overburden pressure, the blue line is the pore pressure, the gray line is the hydrostatic pressure, the green line is the differential pressure, and the black dots are the formation test pressures.

6.2.2.6. Hydraulic Flow Units

Amaefule et al. (1993) proposed a practical, theoretically consistent method for identifying and characterizing Hydraulic Flow Units (HFUs). This approach is based on a modified Kozeny–Carman equation and the concept of mean hydraulic radius, grouping intervals with similar flow behavior into Flow Zone Indicators (FZI). This method is implemented in the package and used to perform the HFU analysis.

To calculate the Flow Zone Indicator (FZI), the procedure begins with determining the normalized porosity ϕ_z, given by

$$\phi_z = \frac{\phi}{1-\phi} \quad , \tag{6.2.4}$$

where ϕ is the porosity expressed as a volume fraction. Next, the Reservoir Quality Index (RQI) is calculated using

$$RQI = 0.0314\sqrt{\frac{k}{\phi}} \tag{6.2.5}$$

where k is the permeability in millidarcies. Finally, the FZI is obtained as the ratio between the RQI and the normalized porosity, as expressed in

$$FZI = \frac{RQI}{\phi_z} \tag{6.2.6}$$

This parameter represents the hydraulic quality of each flow unit within the reservoir. The result is shown in Figure 6.2.16.

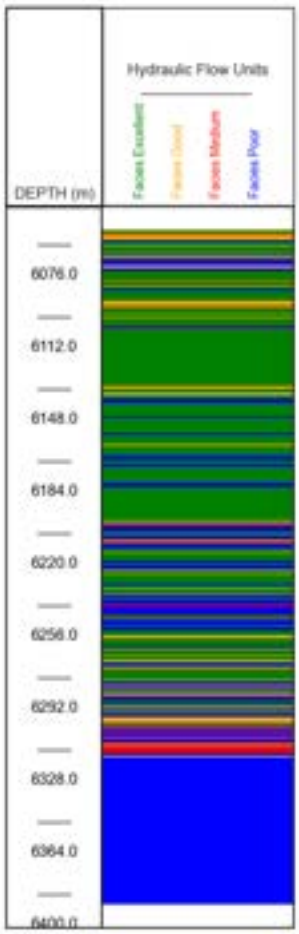

Figure 6.2.16: Hydraulic Flow Units for 3-EQNR-3-SPS

The Hydraulic Flow Unit (HFU) results indicate distinct flow characteristics along the well. Between 6050 and 6320 m, there is a high occurrence of Excellent, Good, and Medium flow units, corresponding to a reservoir zone with favorable permeability and fluid flow capacity. In contrast, from 6320 to 6400 m, the predominance of Poor flow units reflects the presence of an igneous rock zone, characterized by low permeability and limited fluid mobility.

6.2.3. Conclusion

The petrophysical analysis performed using the Pyrockphys Package provided an integrated characterization of the 3-EQNR-3-SPS well, combining log interpretation, mineral inversion, petrophysical property estimation, pressure estimation, and flow unit analysis. The results allowed the identification of a clear reservoir interval between 6050 and 6320 m, characterized by high porosity, low water saturation, and good permeability, followed by an igneous rock zone (6320–6400 m) with reduced flow potential.

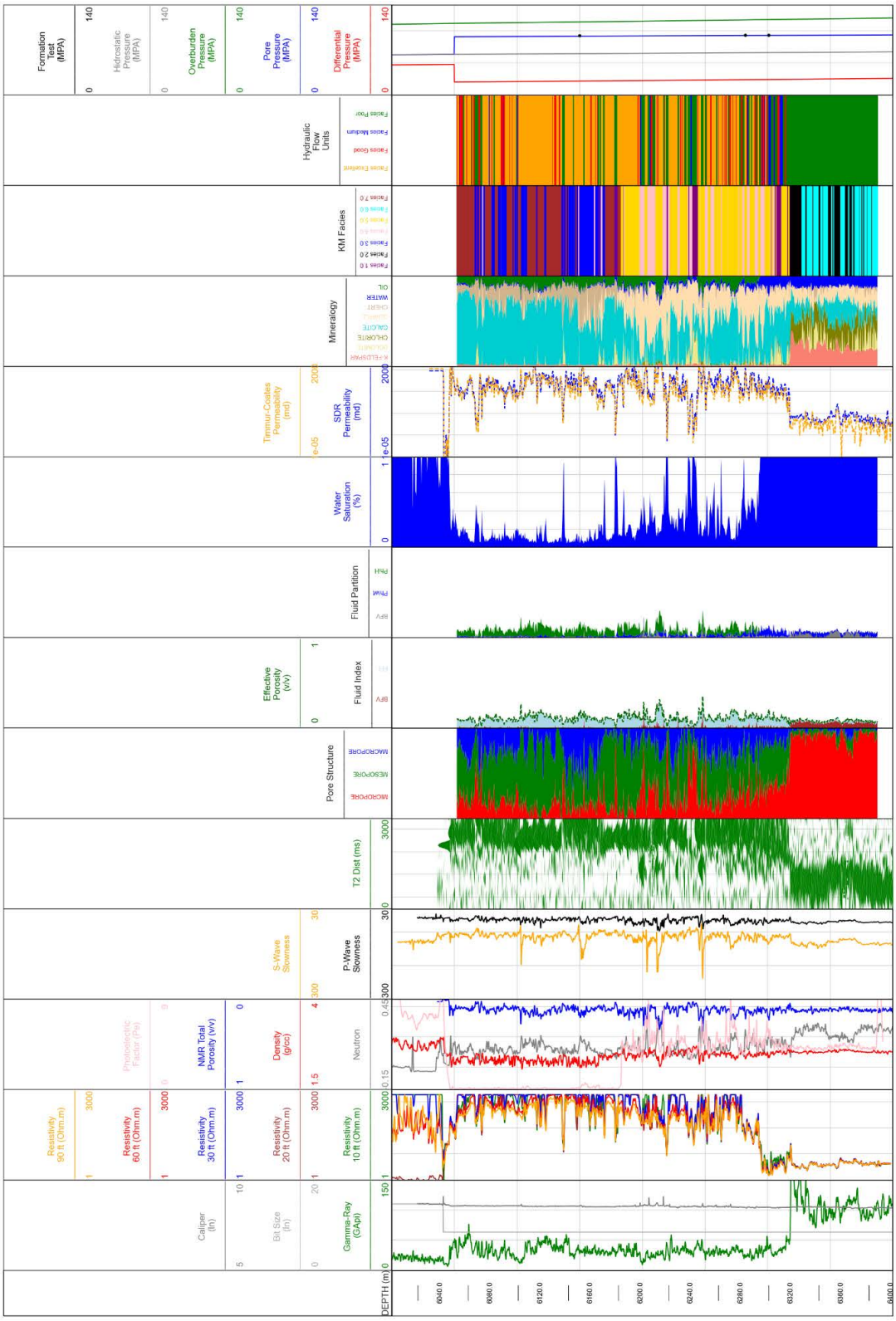

Figure 6.2.17: Measured and estimated logs from 3-EQNR-3-SPS. Track 1 - Bit size, Caliper and Gamma Ray; track 2 - Resistivities; track 3 - Photoelectric Factor, NMR Total Porosity, Density, Neutron; track 4 - S-Wave Slowness, P-Wave Slowness; track 5 - NMR T2 distribution; track 6 - Pore Structure (micro, meso and macropore); track 7 - Effective Porosity and Fluid Index (BVI and FFI); track 8 - Water Saturation form HILT; track 9 - Timur-Coates and SDR Permeability; track 10 - Mineralogy; track 10 - Facies Classification; track 11 - Hydraulic Flow Units; track 12 - Pressure Estimation.

Figure 6.2.17 presents an overview of the well, integrating measured and estimated logs to offer a comprehensive view of the well behavior and reservoir characteristics throughout the analyzed interval.

6.3. Software development

This section details the design, architecture, and implementation of the Pyrockphys Library, a tool developed to support advanced rock physics analysis within the Python ecosystem. Pyrockphys serves as a foundational component, enabling computations for characterizing reservoirs. Its development was driven by the need for a robust, flexible, and extensible framework capable of handling various rock physics models and data types essential for comprehensive petrophysical studies. The following subsections outline the library's developmental methodology, its core features, architectural principles, and practical application considerations.

6.3.1. Development Approach and Methodology

The creation of the Pyrockphys library followed a meticulous development approach, emphasizing modern software engineering practices to ensure its reliability and scalability. The foundational principles guiding its development include:

- **Implementations in the Python Ecosystem**: Python was selected as the primary programming language due to its prominence in scientific computing and extensive ecosystem of data analysis libraries. This choice facilitates integration and collaboration of the development environment.
- **Object-Oriented Modeling and Implementation**: A rigorous Object-Oriented Programming paradigm was adopted. This approach enables the abstraction of complex rock physics models and associated functionalities into modular, reusable objects. Such an approach enhances maintainability, promotes data integrity, and simplifies the management of interdependencies among internal components.
- **Conceptual Structures Supported by Design Patterns**: The library's architectural integrity and flexibility are significantly enhanced by the strategic application of established software design patterns. These patterns provide proven solutions to common software design challenges, contributing to a more robust, extensible, and adaptable framework. This structure ensures that the library can accommodate future extensions, new rock physics models, and evolving research requirements with minimal refactoring.

6.3.2. Key Features

The Pyrockphys library provides a comprehensive suite of functionalities critical for rock physics analysis. These include, but are not limited to, modules with methods for:

Data Handling and Pre-processing: LAS file importation, DLIS-to-LAS conversion, Spike removal, and log filtering.

Basics Petrophysics: Property models, porosity, pressure, permeability, shale volume, fluid saturation, pore structure, elastic templates, hydraulic flow units.

Advanced Petrophysics: Fluid models, velocity models, elastic moduli models, and aspect ratio workflow.

Machine Learning: Facies classification and log imputation.

Visualization: log view, interactive log view, crossplot view, interactive crossplot statistical plot, log correlation, mineralogy indicator, registered log, and confusion matrix.

Selector strategy: A search component that receives arguments and identifies all compatible models (classes/strategies) that require them.

Pipeline: A component that enables the construction of sequence models of interest.

6.3.3. Architectural Design

The architecture of the Pyrockphys Library, as shown in Figure 6.3.1, is organized to ensure modularity, extensibility, and usability. It integrates modern software engineering principles such as object-oriented programming and design patterns, providing a scalable framework to accommodate both basic and advanced rock physics workflows.

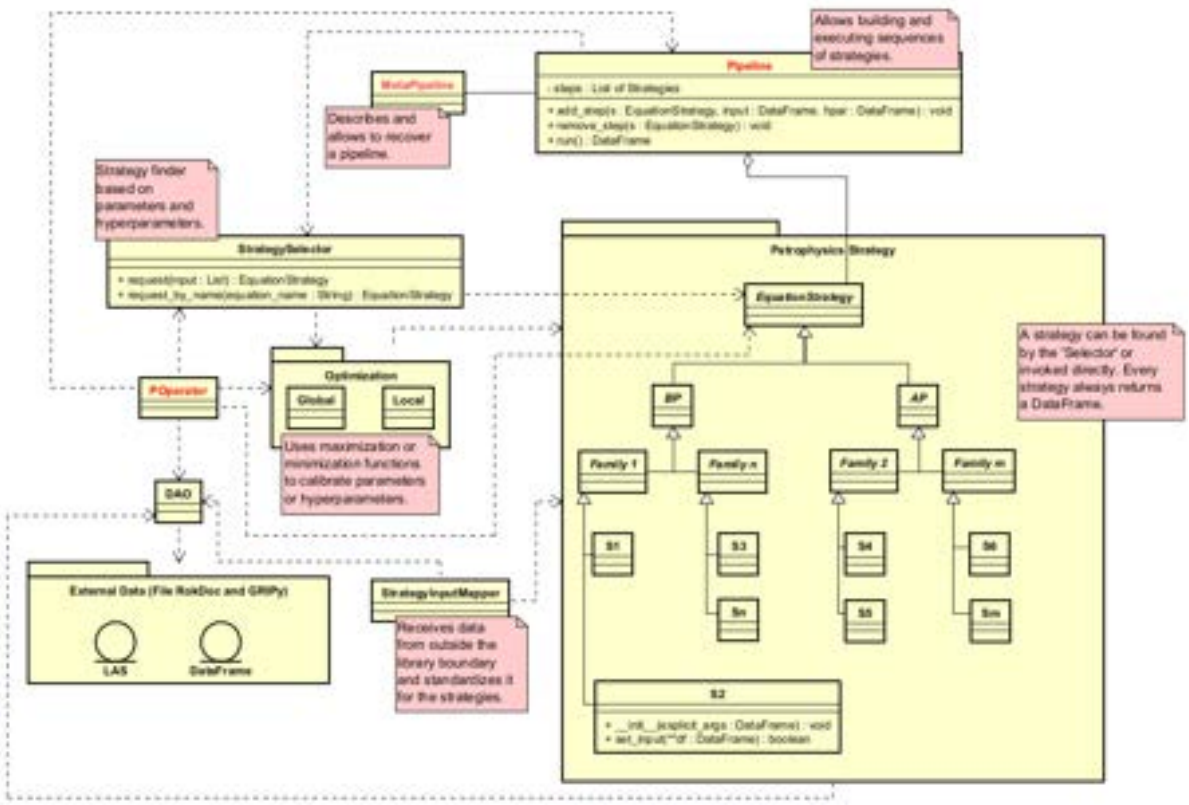

Figure 6.3.1: Pyrockphys Architecture.

At the center of the Pyrockphys architecture is the **Strategy Design Pattern**, which supports the flexibility and modularity of its computational workflows. This design enables dynamic selection and execution of computational models (strategies), ensuring the library can handle a variety of methods for both direct and inverse modeling. The architecture is divided into the following key components:

EquationStrategy: It is a class that serves as the abstract foundation for all calculation models in the library. It defines a common structure and ensures consistency across various implementations of rock physics models. Concrete strategies are grouped into two primary families (**BasicPetrophysics** - BP, **AdvancedPetrophysics** - AP). Each strategy operates on input data provided in a standardized format (typically as Pandas DataFrames) and produces outputs in the same format. This ensures seamless integration between models and facilitates data consistency.

StrategySelector: It is a search-and-selection mechanism that dynamically identifies and retrieves the appropriate computational strategy based on user-provided arguments. This capability enables high flexibility and supports diverse workflows (or the direct use of strategies) without requiring prior knowledge of the underlying model names.

Pipeline: A component that enables the construction, management, and execution of sequential strategies. Users can define a series of steps, each representing a specific calculation.

StrategyInputMapper: A component that addresses the diversity of data formats typically encountered in petrophysical studies. This component ensures that all incoming data, regardless of its source, is standardized into a compatible format for use by the library.

Optimization: This module introduces specialized tools for calibrating computational models. These tools include Global Solver (for large-scale parameter adjustments and general optimization) and Local Solvers (for refining parameters within a constrained domain).

POperator: It is a component that orchestrates interactions among the user, data management, and computational strategies. Additionally, the POperator works as a Facade (a design pattern that simplifies the use of strategies).

Data Access Object (DAO): This component handles the retrieval of persisted data (LAS file), ensuring that input data is efficiently stored and managed.

Then, the architectural design of Pyrockphys offers several key benefits. Among them are modularity, flexibility, reproducibility, and extensibility.

6.3.4. Some Illustration of Use

To demonstrate the practical application and operational mechanics of the Pyrockphys Library, this section presents a series of code examples. These illustrations highlight the library's modularity, adherence to object-oriented principles, and the ease of use of its implemented petrophysical calculation strategies. The examples showcase how users can interact with the library, from basic importation to the application of specific rock physics models, utilizing Pandas DataFrames as a standardized data interface.

The initial case for using the Pyrockphys Library involves importing it into the Python environment, a standard procedure that makes its functionalities accessible, as shown in Figure 6.3.2.

```
# To import the library
import pyrockphys
```

Figure 6.3.2: Importation of the Pyrockphys.

This fundamental command `import pyrockphys` makes the entire library available for subsequent operations.

Figure 6.3.3 illustrates a specific import fo a selected petrophysical class, `Rhob_PhiSwVsh`, which calculates density (`Rhob`) based on porosity (`Phi`),

water saturation (`Sw`), and shale volume (`Vsh`). This model is part of the `density_family` module within the basic rock physics functionalities, consistent with the modular design discussed in the previous section.

```
# To import a specific strategy
from pyrockphys.rockphysics.basic.density_family import
Rhob_PhiSwVsh, Phi_RhoSwVsh
```

Figure 6.3.3: Importation of a Specific Strategy.

This code segment explicitly imports `Rhob_PhiSwVsh` (and `Phi_RhoSwVsh` for later use), demonstrating how individual models are selectively accessed from their hierarchical module structure within the library.

Next, as Figure 6.3.4 shows, a hypothetical input dataset is constructed as a Pandas DataFrame, representing typical petrophysical parameters (with simple test values):

```
# Consider a hypothetical DataFrame as input
df_input = pd.DataFrame({
   'saturation': [0.2, 0.8],
   'porosity': [0.25, 0.75],
   'vshale': [0.1, 0.5],
   'rhohc': [0.8, 2.8],
   'rhograin': [2.65, 3.65],
   'rhomatrix': [2, 4],
   'rhobrine': [1.1, 3.1]
})
```

Figure 6.3.4: Hypothetical DataFrame.

This `df_input` DataFrame serves as the standardized input for the chosen rock physics model, with column names corresponding to the expected parameters of the `Rhob_PhiSwVsh` strategy.

The calculation of density using the `Rhob_PhiSwVsh` model can then be performed directly, as shown in Figure 6.3.5.

```
# Calculates density from PhiSwVsh model
df_inf = Rhob_PhiSwVsh(**df_input).calculate()
```

Figure 6.3.5: Calculates Density.

Here, the `Rhob_PhiSwVsh` class is instantiated and its `calculate()` method is immediately invoked. The `**df_input` syntax unpacks the DataFrame's

columns as keyword arguments, efficiently mapping the input data to the model's required parameters. The result, `df_inf`, is a DataFrame containing the inferred density values.

Alternatively, the model can be instantiated and its input set in separate steps, which offers greater flexibility for dynamic parameter assignment or sequential processing, as shown in Figure 6.3.6.

```
# Set input and calculates
model = Rhob_PhiSwVsh()
model.set_input(**df_input)
df_inf1 = model.calculate()
```

Figure 6.3.6: Set the input arguments and calculate the target property.

This approach, where the model object is first created `(model = Rhob_PhiSwVsh())` and then populated with input data via `model.set_input(**df_input)` before calculation, exemplifies the object-oriented design, allowing for greater control over the model's lifecycle and parameterization.

This example extends the illustration by applying a different strategy, `Phi_RhoSwVsh`, to infer porosity from density and other petrophysical parameters. This demonstrates the library's versatility in providing formulations for interconnected petrophysical properties, potentially serving as an inverse model or an alternative to direct calculations.

The input data for this case simulates actual well log measurements, representing a more realistic application scenario. The `df_log1045` (an assumed existing DataFrame representing well log data) is used to construct `df_input2`, as shown in Figure 6.3.7.

```
# Using hypothetical log1045 DataFrame as input
df_input2 = pd.DataFrame({'density': df_log1045['RHOZ'],
'saturation': df_log1045['SW'], 'vshale':
df_log1045['VCL'], 'rhohc': 0.89, 'rhograin': 2.7,
'rhomatrix': 2.5, 'rhobrine': 1.1})
```

Figure 6.3.7: Hypothetical log1045 DataFrame.

This `df_input2` DataFrame is specifically prepared to match the required parameters of the `Phi_RhoSwVsh` strategy, which calculates porosity (`Phi`)

based on density (RHOZ), water saturation (SW), and shale volume (VCL), among other parameters.

The calculation of porosity using this strategy is performed in the same way as in the previous case, as shown in Figure 6.3.8.

```
# Calculates porosity from RhoSwVsh model
df_inf2 = Phi_RhoSwVsh(**df_input2).calculate()
```

Figure 6.3.8: Calculates porosity.

Here, the `Phi_RhoSwVsh` strategy is instantiated with `df_input2 and its calculate()` method is executed, yielding `df_inf2,` which contains the inferred porosity values. This highlights Pyrockphys's capacity to facilitate diverse calculations, demonstrating its utility for both forward and potential inverse modeling within a consistent, structured framework.

These illustrative examples collectively demonstrate the user-friendly interface and the robust, modular design of the Pyrockphys Library, affirming its capability to support complex petrophysical analysis efficiently and reproducibly.

6.3.5. Access and Installation

The Pyrockphys Library is designed for ease of access and deployment, facilitating its integration into research workflows. The installation process generally follows standard Python package management practices, ensuring a straightforward setup for users.

The project's source code, along with detailed installation guidelines and documentation, is hosted in a private GitHub repository (https://github.com/giruenf/Pyrockphys-Delivery).

Users can clone the repository to their local environment and follow the instructions in the README file to set up dependencies and install the library (typically using pip). This open-source approach, coupled with accessible documentation, encourages community contributions, collaborative development, and widespread adoption, ensuring that Pyrockphys remains a dynamic and evolving tool for the research community.

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7. **Next steps**

We recently received pre-salt core plugs from two sources, which will be used in further tests:

- ANP direct request (SAA 31.25): 10 plugs from well 3-EQNR-3 and 10 sidewall core plugs from well 4-BRSA-971-SPS. Both wells from the Bacalhau field.
- UENF/UNICAMP agreement: 11 plugs from well 9-BRSA-1212 (Sururu) and 7 plugs from well 3-BRSA-1132 (Berbigão).

Other steps:

- Selection and characterization of new fluids to simulate oil in fluid substitution experiments. Research on ISOPAR and Lamp oil.

- Fluid substitution experiments - Measurement and data analysis of outcrop carbonate samples (Silurian Dolomite, Indiana Limestone).
- Porosity and Permeability measurements of the pre-salt rocks.
- Compressibility of the pre-salt rocks.
- Measurements of Vp and Vs of the pre-salt rocks. Dry condition.
- Measurements of Vp and Vs of a selection of pre-salt rocks. BFW saturated.
- Attenuation evaluation using the spectral ratio technique. Pre-salt rocks.
- Attenuation evaluation using the spectral ratio technique. Outcrop analog rocks. Dry, BFW-saturated, and pore pressure experiments.
- Characterization of Bacalhau oil properties (density, viscosity, P-wave velocity). This can be done using some oil samples received on the SAA 31.25.
- Pre-salt thin section evaluation.
- Micro-CT images of a selection of pre-salt core plugs.
- Continuation of the anisotropy evaluation based on the micro-CT data.
- Fluid substitution simulations based on the micro-CT data.
- Continuation of the T-matrix implementation for Rock Physics modeling.
- Continuation of the Reservoir Characterization based on well-log analysis and formation tests.

Appendix A – GESTI – Lab sample management software

GESTI is a web-based software for managing and tracking laboratory rock samples and correlated items. It centralizes type-based records derived from rocks (plugs, thin sections, blocks, discs, fragments, etc.), as well as complementary characteristic information, such as location, technician responsibilities, project associations, and management of lab services & measurements workflows, ensuring full auditing and operational indicators. The main view displays a dashboard of the dataset and general information (see Figure A.1). A menu on the left side of the webpage provides access to system functionalities, some of which are illustrated in Figures D.2 and D.3.

The software layered architecture can be summarized by the following technologies:

- Frontend (React + Vite + TypeScript): A responsive SPA with fast search/filters, tables, and modals; data caching and smooth UX for large lists.
- Backend (NestJS + Prisma): Modular API (controllers/services/DTOs), domain-based business rules, validation, and route-level security.
- Database (PostgreSQL): Relational model with Item as the core and 1:1 subtypes (type-specific attributes), plus indexes and uniqueness for consistency.
- Deployment & Ops (Docker + Nginx): The stack runs in containers (frontend, backend, PostgreSQL) orchestrated with Docker Compose, with Nginx in front as a reverse proxy and static asset server. Benefits: reproducible environments, easy updates/rollback, network isolation, persistent volumes for databases and attachments, and centralized logging.
- Operations/Integrations: Google Drive for per-item attachments; Services/Tasks with ordering and status; scheduled backups (e.g., pg_dump) with retention and periodic restore tests.

Some basic operations can be highlighted:

Working with “Item and types” (a general view in Figure A.2):

The type “Plugs” list illustrates the standard pattern: search by name/attributes, filters (e.g., conservation), pagination, and quick actions (view, history, update location, delete).

Data rules: each type of item keeps its own specific attributes (e.g., Block dimensions, Plug inch diameter, etc.) in a 1:1 relationship with Item. It simplifies data entry and preserves traceability.

Typical flow to register an “Item”:

Select Type → add the item + type-specific attributes;

Set Location (required for traceability);

Attach files (images/reports);

Use the item in daily routines (services, projects).

Working with “Services, tasks, and histories” (a general view in Figure A.3):

“Service”: It is a set of actions over one or more items at a given location, with status (not started, in progress, completed, canceled), responsibles (ordered; 0 = lead), tasks, and timestamps (start/finish).

“Tasks”: It is an order for executing a lab procedure.

Service and tasks properties: status, start/finish, and order (sequencing). The management of services and tasks is fundamental to controlling the scheduled time, responsibilities, location of tests, and the control of the rock item repository.

Attachments (File Manager): the File Manager supports local uploads (images/PDFs/spreadsheets) and links to existing Google Drive files. Each file is tied to the corresponding item, keeping technical evidence together with the record. Recommended practices:

Naming convention: use a consistent pattern (e.g., ITEMCODE_TYPE_YYYYMMDD.ext).

Service evidence: attach before/after photos or reports to the relevant service for auditability.

Versioning: prefer adding a new file rather than overwriting an older one; keep prior versions when they document a process step.

Storage: metadata (title/URL/Drive ID) stays in the database; binaries live in Drive (for linked files) or the configured storage for uploads.

Backups (containerized deployment): in the Docker + Nginx setup, backups are performed from the PostgreSQL container using pg_dump, and written to a mounted backup volume. A scheduler (host cron or a small utility container) runs the routine and enforces retention, keeping a database in backup.

Restore drills and governance: it keeps access to the backup folder restricted, documents who can restore, and logs successful/failed jobs. This combination of

scheduled dumps, clear retention, and restore drills ensures the database can be recovered quickly and safely when needed.

The GESTI development is in progress for improvements and eventual bug correction, according to laboratory and information management necessities.

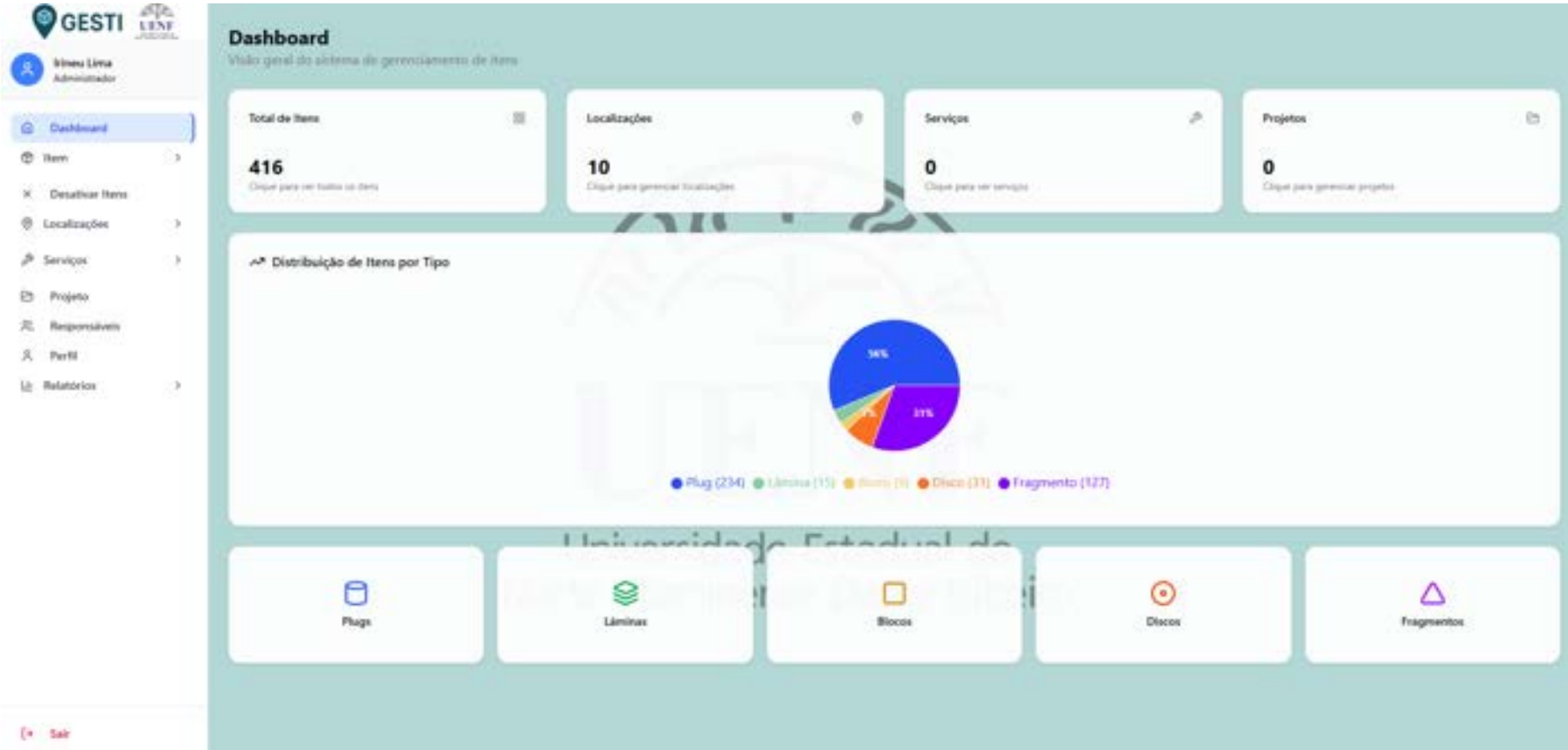

Figure A.1 – GESTI web-based main view: a dash board overviewing the data base & general information (in Portuguese language only).

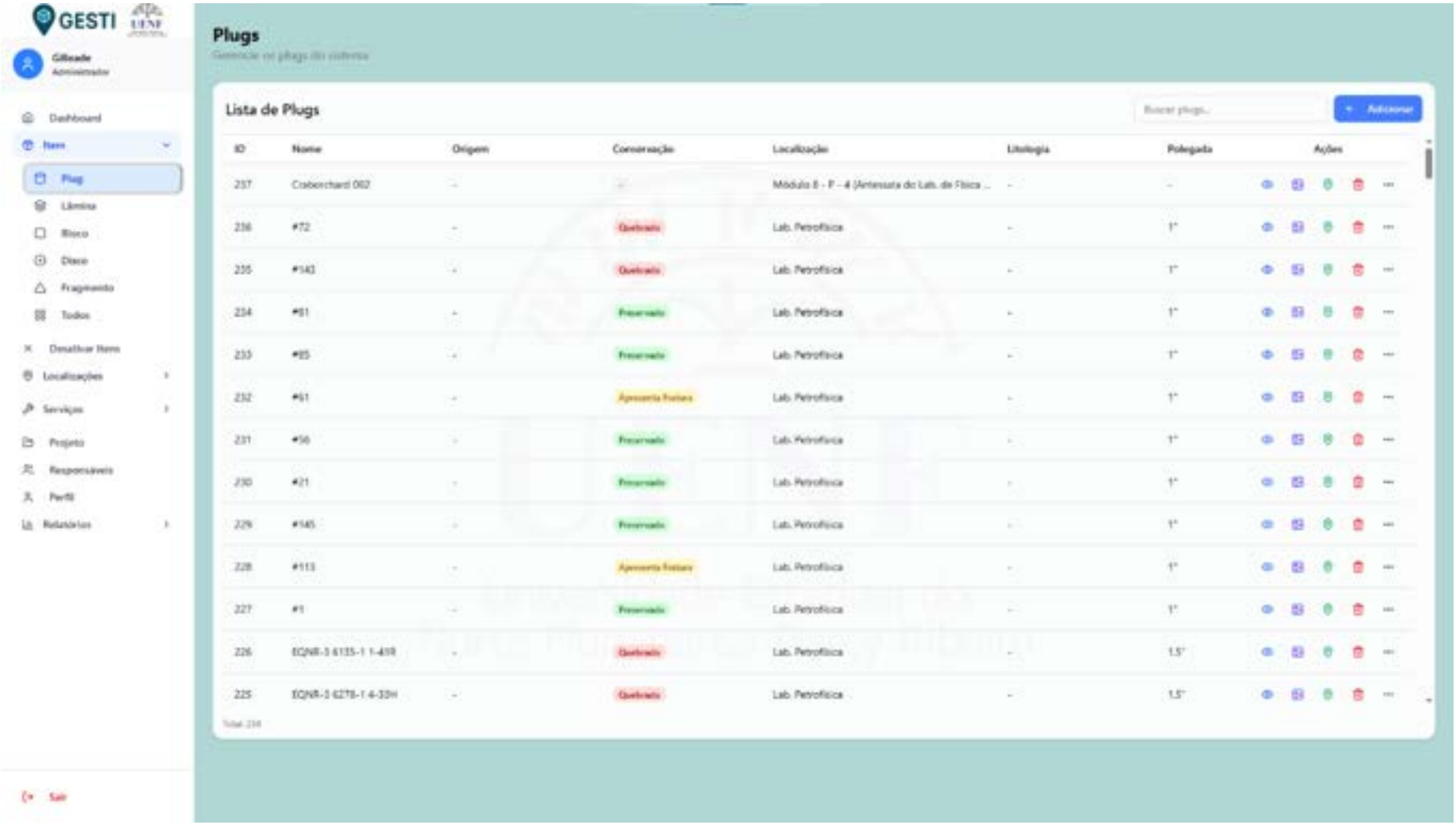

Figure A.2 – GESTI web-based view of the menu “Item > Plug”: exhibiting a searchable and filterable table list with basic info of the status, location, and actions for services, etc.

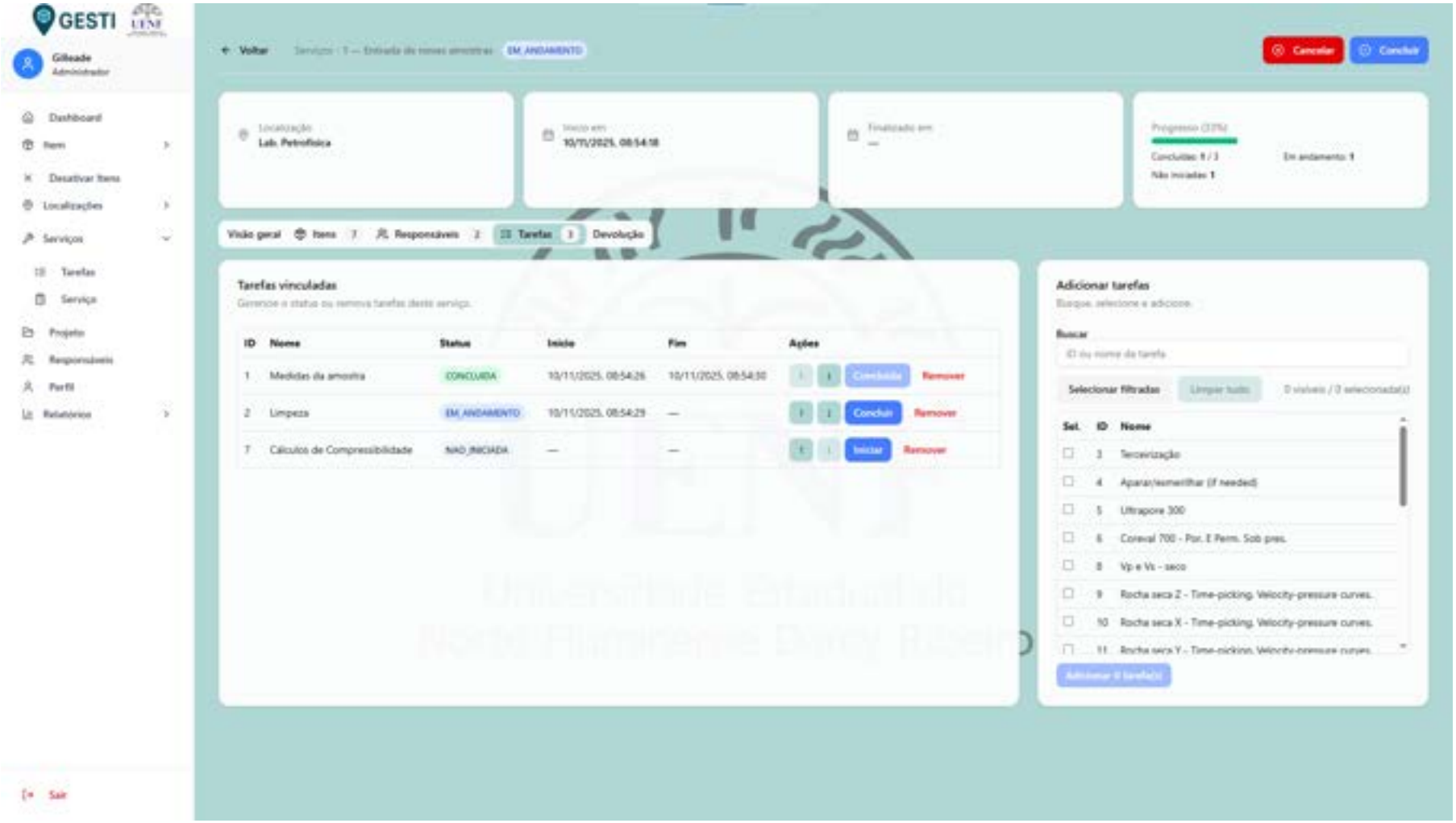

Figure A.3 – GESTI web-based view of the menu “Service > Task”: planning a service based on a sequence of tasks, ordering the execution by date/time, status, and details related to laboratory procedures.

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React — The library for web and native user interfaces. Available at: https://react.dev/.

Vite — Next-generation frontend tooling. Available at: https://vitejs.dev/.

Prisma — Next-generation Node.js and TypeScript ORM. Available at: https://www.prisma.io/docs.

PostgreSQL — The world's most advanced open source database. Available at: https://www.postgresql.org/docs/.

Google Drive API — Manage Drive resources programmatically. Available at: https://developers.google.com/drive.

Docker / Nginx. Available at: https://docs.docker.com/ and https://nginx.org/en/docs/.

Appendix B – Wave Time Analysis software

The **Wave Time Analysis** software is a Python-based application developed for the visualization, processing, and analysis of waveforms obtained from laboratory rock physics experiments. Its main purpose is to streamline and integrate the processing of experimental data, offering a complete platform that gathers, within a single environment, all steps of the workflow — from data loading and organization to filtering, spectral analysis, and waveform comparison.

The application was designed to be user-friendly, intuitive, modular, and interactive, allowing users to perform waveform analyses in time and frequency domains in an efficient way, as required by the lab procedures analysis of the data, which are stated by the Technicians from Rock Physics Lab in the LENEP/UENF, avoiding the need to switch between different programs as employed before. Moreover, the application is built on an independent architecture, enabling it to run as a stand-alone executable, without requiring any additional library installation or Python environment setup.

The main interface is divided into four primary sections (menu items), which follow the natural sequence of the analysis workflow:

1. “Input Data” (Figure B.1): the user loads the waveform files to be analyzed using the “Open” button, the input files are selected and automatically displayed in the main table, which contains key information such as “Editable Tag” a possibility to inform the wave data name, “Wave Type” as P- and S-wave, “Sample Length (mm)”, and “Pressure” value during acquisitions. This tab also includes a “Wave Management panel” that allows users to check or uncheck all waves, clear selected entries, and quickly apply parameters such as sample length and wave type (for example, P-wave or S-wave). This section functions as the entry point of the system, where the dataset is organized before any processing or analysis.

2. “Filters” (Figure B.2): This option in the menu provides several pre-processing options for signal conditioning. The user can remove amplitude noise upon data loading, apply a moving average filter to smooth the waveform, and remove the blank time from the signal. At the bottom of the interface, there are specific fields for setting blank times for each wave type (P, S1, and S2), allowing precise adjustment of the adequate starting time according to the experimental setup. These filters are essential for improving data quality before proceeding to the more advanced stages of analysis.

3. “Processing” (Figure B.3): it concentrates the time-domain visualization tools. In this section, the user defines axis limits (time and amplitude), adds an optional title to the plot, and chooses how to visualize the data using the “Plot”, “Subplot”, and “Wiggle Plot” buttons. The “Plot” button generates a standard

combined graph with all selected waves; “Subplot” organizes each waveform in separate panels for easier inspection; and “Wiggle Plot” creates the traditional seismogram-style visualization, emphasizing amplitude variations over time. Additionally, the “Plot Diff.” button allows users to compare two curves — one as a reference and one as a target — displaying the difference between them. This functionality is particularly useful for detecting subtle variations between signals acquired under different pressure conditions or experimental settings.

4. “FFT” (Figure B.4): this option menu brings the most advanced analytical modules for analysis, described as:
    - “Spectral Analysis”: it computes the Fast Fourier Transform (FFT) and displays the power spectrum of the signals, allowing users to visualize the energy distribution across frequencies and identify noisy or dominant frequency bands.
    - “Multiplotter”: this module provides an interactive environment for waveform comparison and time picking, enabling users to visualize multiple waves with vertical offsets, perform manual or automatic picking of P- and S-wave arrivals, apply curve fitting (Power and Bowers models), and generate Velocity vs. Pressure plots, with the option to export results directly to Excel.
    - “Butterworth Bandpass Filt”: it performs interactive band pass filtering, users select the cutoff frequencies (fc1 and fc2) directly from the spectral plot, and the system applies a Butterworth filter, displaying side by side the original and filtered signals in both time and frequency domains.

In summary, these four sections define the entire workflow of the Wave Time Analysis software, enabling users to progress seamlessly from data loading and preparation to complete time- and frequency-domain analysis. The system stands out for its intuitive interface, robust design, and ability to integrate multiple analytical stages in a single environment, representing a significant improvement in productivity and standardization for laboratory waveform analysis. The development is in progress, particularly for testing and bug detection & correction, as well as designing new future functionalities for automatic wave picking analysis and velocity computation, aiming to improve analysis quality as quickly as possible for users.

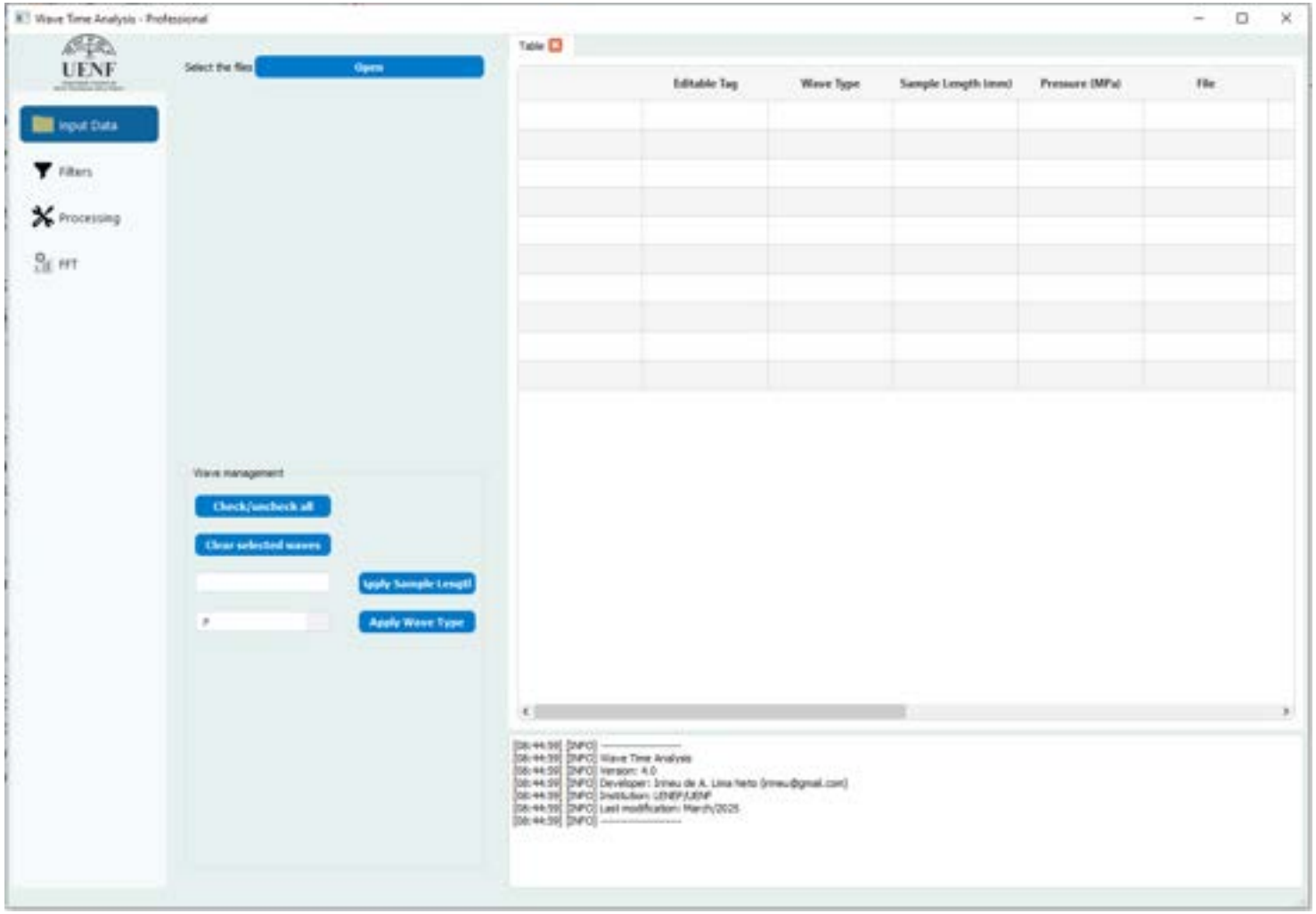

Figure B.1 – Wave Time Analysis, view of the menu: “Input Data” - the user loads the waveform files and set information as necessary in the input table view.

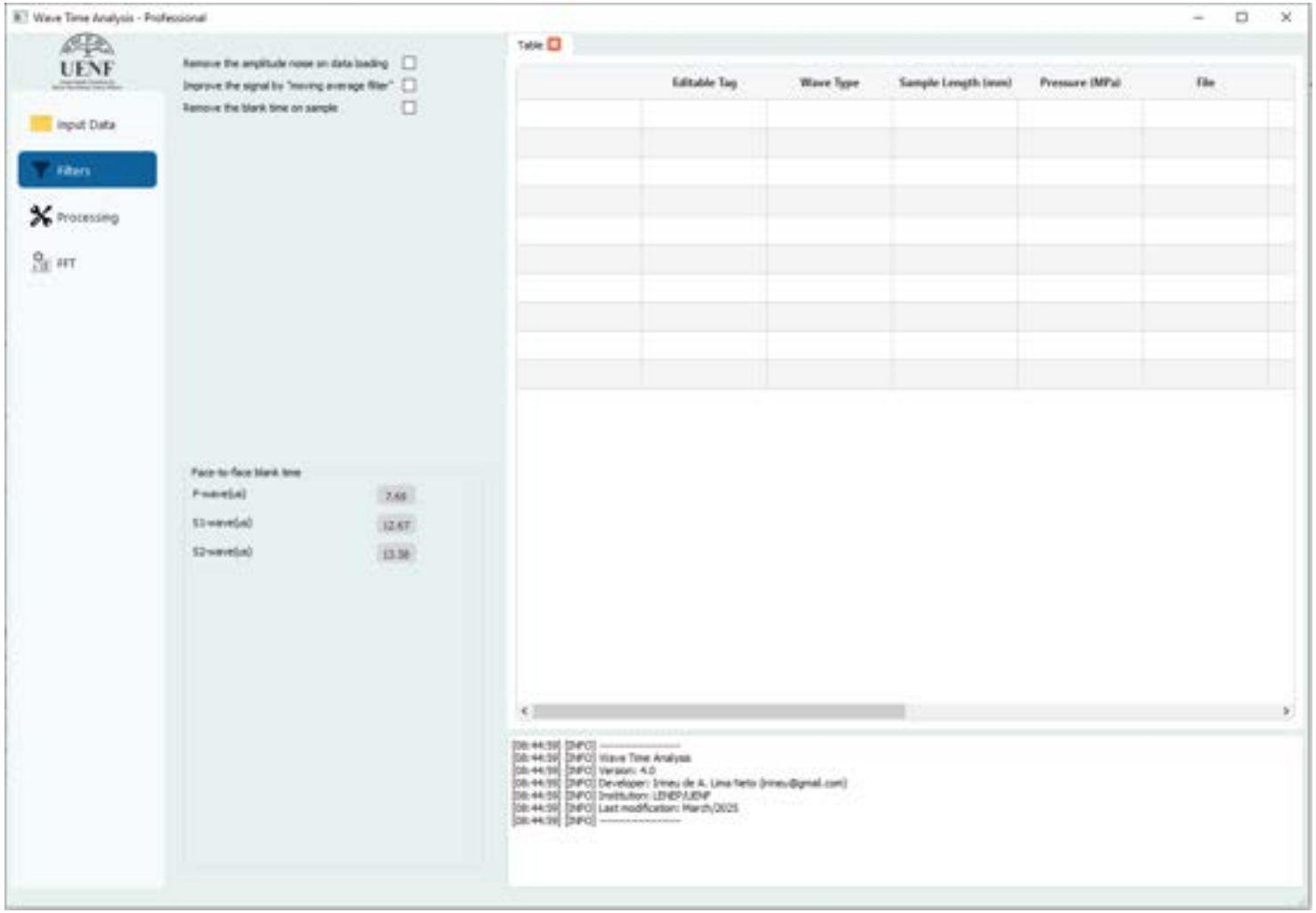

Figure B.2 – Wave Time Analysis, view of the menu: “Filters” – it provides several pre-processing options for signal conditioning.

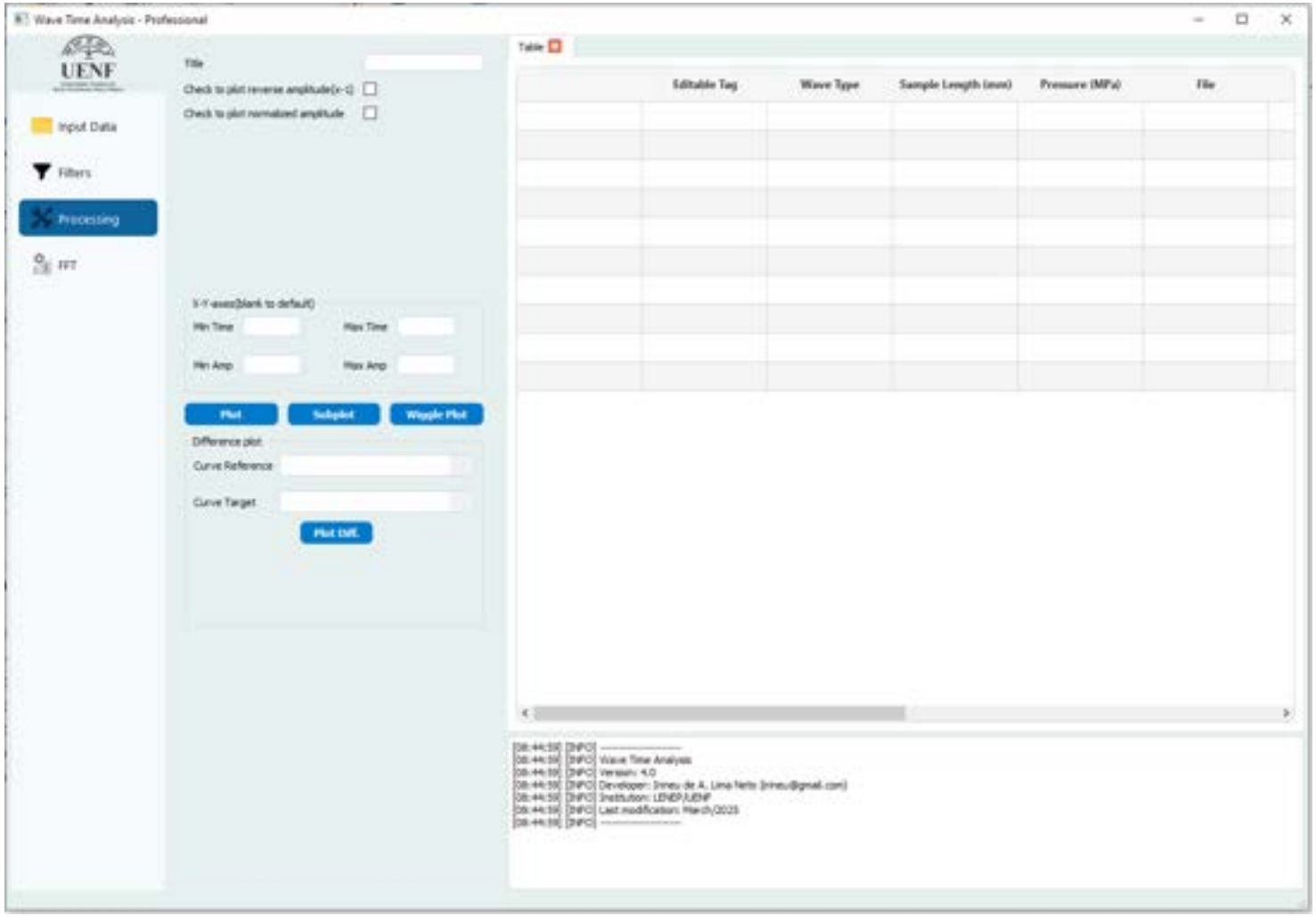

Figure B.3 – Wave Time Analysis, view of the menu: “Processing” – it concentrates the time-domain visualization tools for “Plot”, “Subplot”, and “Wiggle Plot” functionalities.

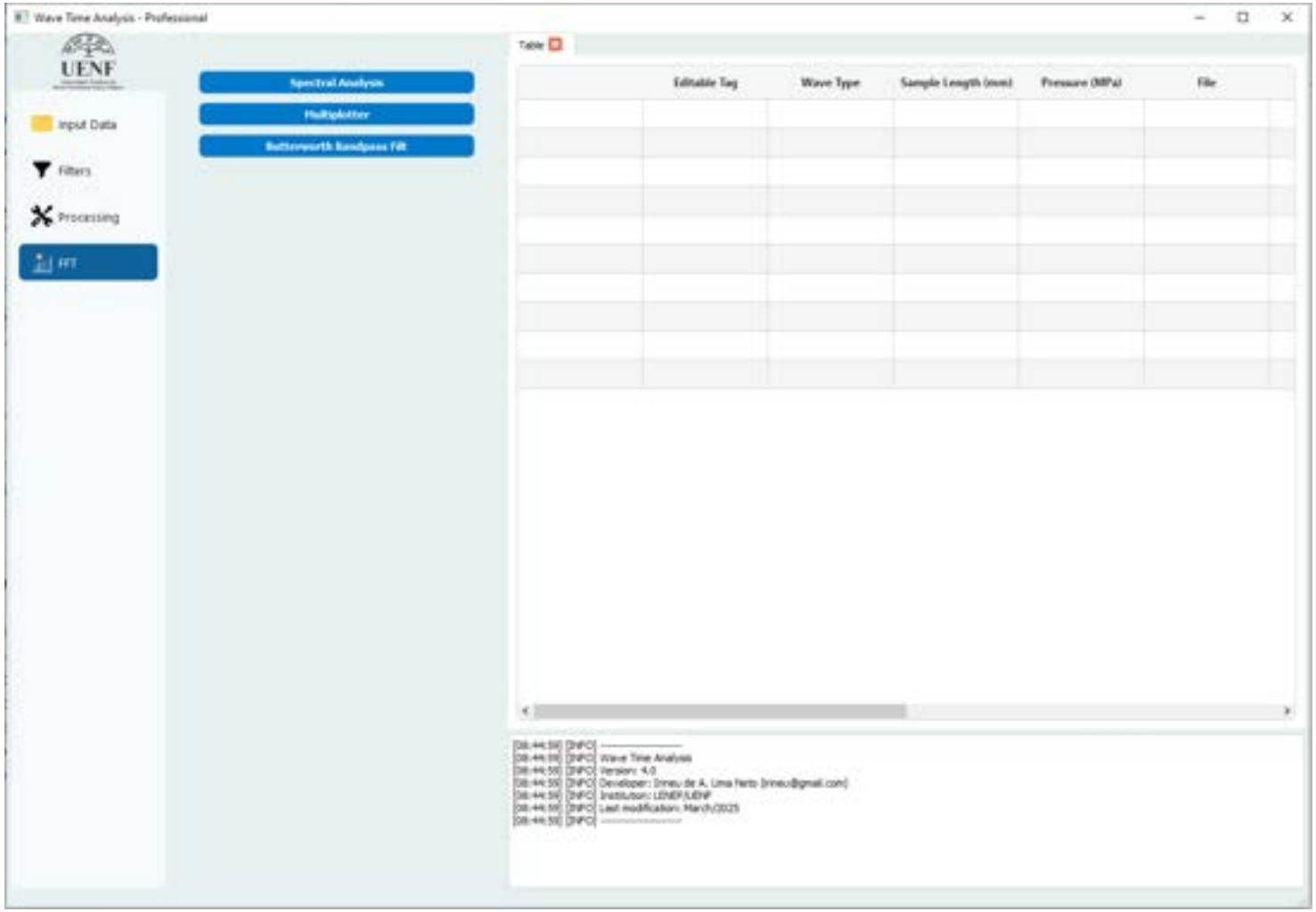

Figure B.4 – Wave Time Analysis, view of the menu: “FFT” – advanced analytical modules for analysis: “Spectral Analysis” based on FFT, “Multiplotter” for waveform comparison and time picking, and “Butterworth” band pass frequency filter